

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

**Before the Honorable Doris Johnson Hines
Administrative Law Judge**

In the Matter of:

CERTAIN WEARABLE DEVICES

Investigation No. 337-TA-1478

COMPLAINANTS' OPENING CLAIM CONSTRUCTION BRIEF

TABLE OF CONTENTS

	Page
I. INTRODUCTION	1
II. PATENT BACKGROUND	2
III. PERSON OF ORDINARY SKILL IN THE ART (“POSITA”)	4
IV. DISPUTED CLAIM CONSTRUCTIONS	5
A. U.S. Patent No. 10,231,675 (“’675 Patent”)	5
1. “acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user's body” (’675 Patent, Claim 13)	5
2. “acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes; and activating the at least one second sensor based on at least a part of the additional information” (’675 Patent, Claim 15)	10
B. U.S. Patent No. 10,642,359 (“’359 Patent”):	13
1. “[the] biometric signal” (’359 Patent, Claim 14)	13
V. CONCLUSION	19

TABLE OF AUTHORITIES

Page(s)

Cases

<i>Biosig Instruments, Inc. v. Nautilus, Inc.</i> , 783 F.3d 1374 (Fed. Cir. 2015)	6
<i>Dow Chem. Co. v. Sumitomo Chem. Co.</i> , 257 F.3d 1364 (Fed. Cir. 2001)	18
<i>Enzo Biochem, Inc. v. Applera Corp.</i> , 599 F.3d 1325 (Fed. Cir. 2010)	9, 10
<i>Guangdong Alison Hi-Tech Co. v. Int'l Trade Comm'n</i> , 936 F.3d 1353 (Fed. Cir. 2019)	6
<i>Nautilus, Inc. v. Biosig Instruments, Inc.</i> , 572 U.S. 898 (2014).....	6
<i>Tuna Processors, Inc. v. Haw. Int'l Seafood, Inc.</i> , 327 F. App'x 204 (Fed. Cir. 2009).....	14

Other Authorities

<i>Biometrics</i> , Dictionary of Computer Vision and Image Processing (2005)	18
<i>Biometrics</i> , Newton's Telecom Dictionary 27th ed. (2013).....	18

TABLE OF EXHIBITS

Exhibits 1–7 are being filed concurrently with this opening brief.

Ex. No.	Description
1	U.S. Patent No. 10,231,675 (SAMS382-0003655)
2	U.S. Patent No. 10,642,359 (SAMS382-0003704)
3	Oura's Invalidity Contentions [excerpted to include only cited sections]
4	Newton's Telecom Dictionary 27th ed. (2013) (OURA1478_00002999)
5	Dictionary of Computer Vision and Image Processing (2005) (OURA1478_00002989)
6	Declaration of Darrin Young, Ph.D.
7	Curriculum Vitae of Darrin J. Young, Ph.D.

I. INTRODUCTION

The Samsung Complainants hereby submit their Opening Claim Construction Brief addressing the construction of three disputed claim terms, which are in two of the four asserted patents¹: U.S. Patent Nos. 10,231,675 (“’675 Patent,” Ex. 1) and 10,642,359 (“’359 Patent,” Ex. 2). For all three terms, Oura’s proposals ignore the specification’s clear guidance and the fundamental purpose of the claimed inventions. Instead, Oura’s proposals either rely on alleged ambiguities in the claim language, which are clarified and resolved by the specification, or dictionary definitions, which are contradicted by the meanings dictated by the intrinsic record.

For the ’675 Patent, both terms—(1) “. . . a second period of time shortly before and/or immediately after a first period of time . . .” and (2) “acquiring additional information on the motion of the electronic device . . . for a third period of time . . .” — relate to the function of, and timing relationship, between three recited “periods of time” (i.e., “first,” “second,” and “third” “periods of time”). Oura proposes indefiniteness for both terms. But its proposals ignore the objective boundaries and triggering events that the specification describes, using multiple figures and embodiments, for when each successive “period” begins relative to the prior “period.”

For the ’359 Patent, the disputed term “the *biometric* signal” in Claim 14 is plainly intended to refer to a similar antecedent phrase earlier in the claim, “a *biological* signal.” Samsung thus proposes the term be read as “the biological signal” and construed consistently with the specification and other claims, all of which use the term “biological signal” and relate to signals for measuring “biological information.” Apparently seeking to manufacture a non-infringement defense where none exists, Oura attempts to completely alter the nature of the patented invention by construing the term as a signal for “identification” using “fingerprints, finger lengths, facial

¹ Samsung is also asserting U.S. Patent Nos. 10,978,789 and 10,945,677 in this investigation but no terms from those patents have been proposed for construction.

features, retinal patterns, voice recognition, and weight.” But Oura’s proposal is premised entirely on extrinsic dictionaries, completely divorced from and, in fact, contrary to the intrinsic record.

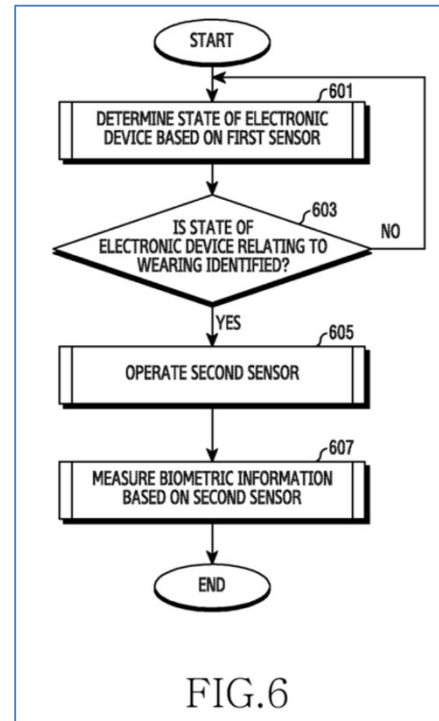
Respondents’ litigation-driven arguments should be rejected and Complainants’ constructions adopted.

II. PATENT BACKGROUND

Both the ’675 and ’359 Patents are directed to wearable electronic devices with two sensors, one to sense motion and a second to sense biological information. In both patents, the motion sensor measures movement and, depending on the amount or degree of movement, certain actions may be taken, including activating or deactivating the sensor for biological information (e.g., ’675 Patent at Claim 13), or deciding whether a biological signal measured by the second sensor is valid (e.g., ’359 Patent at Claim 14).

The ’675 Patent describes a wearable electronic device for measuring biological signals, such as heart rate, wherein a first sensor for measuring movement is used to selectively turn on or off the device’s second sensor that measures the user’s biological signals. ’675 Patent at 1:32-2:36. This selective activation of the second biological sensor reduces power consumption to conserve the battery and ensures that the device only measures biological signals when the user is wearing the device or when such signals can be accurately measured (e.g., measuring blood oxygen or resting heart rate when the user is resting/asleep). *Id.* at 1:45-56.

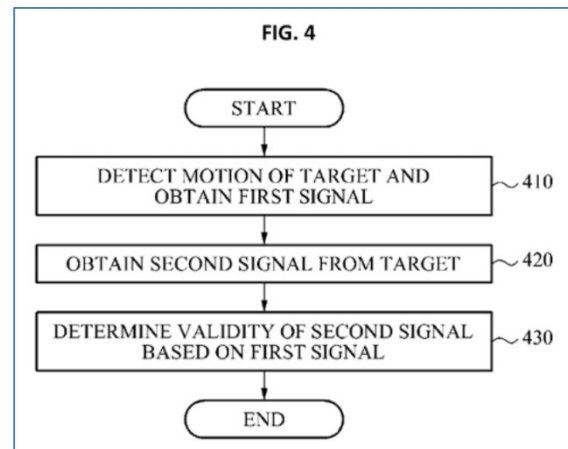
In the embodiment of Figure 6 (right), the device transitions between different states, depending on the motion of the device. In steps 601 and 603, the device uses the movement measured by the first sensor to determine if the device is being worn by the user. *Id.* at 23:20-52 (“Referring to FIG. 6, in operation 601 the electronic device ... may determine the state (e.g., wearing state, [or non-wearing state]) ... of the electronic device based on a first sensor.”). If the device is worn, the device transitions to steps 605 and 607, where the second sensor is activated to measure biological



signals, which could be “blood pressure, a heart rate, an ECG, skin moisture, saturation of peripheral oxygen, a body temperature, blood glucose, an EEG, an EMG, and the like.” *Id.* at 23:53-67.

The '675 Patent's claims recite different variations of this concept of using a first motion sensor to measure the device's movement and, based on the level or degree of movement, deciding whether to activate or deactivate a second sensor that measures the biological signals. For example, Claim 13 recites a method wherein: (1) a “first sensor” initially determines if a user is wearing a device based on its motion in a “first period of time”; (2) the “first sensor” also measures motion in a “second period of time” to determine whether to “activat[e] at least one second sensor”; (3) the “second sensor” then detects “biometric information”; and (4) if the “first sensor” measures a change in the motion level that is “more than a threshold motion,” the device “inactivat[es]” the “second sensor,” thereby ending the measurement of biometric information.

The '359 Patent is not in the same family as the '675 Patent, but the '359 Patent describes similar concepts of using a motion sensor to measure movement, and deciding whether biological signals measured by a second, biological sensor are valid based on the measured movement level. This process is depicted, for example, in the flowchart of Figure 4 below. In step 410, a wearable device measures a user's motion using a motion sensor; in step 420, the device measures the user's biological information, such as "muscle contraction and muscle relaxation," using a biosensor; and in step 430, the validity of the biological signal is decided based on the measured motion. '359 Patent at 10:7-22. In the last step, as an example, the biological signal may only be considered reliable and valid if the user is at rest and not in motion when the biological signal is measured. *Id.* at 1:28-45, 13:9-12.



The '359 Patent's asserted claims reflect these steps. Claim 14, for example, recites a wearable device where a (1) "motion sensor" obtains a "movement signal"; (2) a "biological sensor" obtains a "biological signal"; (3) the device determines a "select period" of time when the "movement signal" meets a "specified condition"; and (4) at least a portion of the "biological signal" (mistakenly recited as the "biometric signal") is determined to be a "valid signal" during the "select period."

III. PERSON OF ORDINARY SKILL IN THE ART ("POSITA")

The '359 Patent's earliest claimed filing date is May 8, 2014, and it further claims priority to a Korean application filed on August 20, 2013. The '675 Patent's earliest claimed filing date is June 3, 2016, and it further claims priority to a Korean application filed on August 12, 2015.

Samsung contends that a POSITA at the time of the priority dates of the '359 and '675 Patents would have had at least a four-year degree in electrical engineering, mechanical engineering, biomedical engineering, optical engineering, or related field of study, or equivalent experience, and at least two years' experience in academia or industry studying or developing sensor systems.

IV. DISPUTED CLAIM CONSTRUCTIONS

A. U.S. Patent No. 10,231,675 (“’675 Patent”)

1. **“acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user's body” (’675 Patent, Claim 13)**

Samsung	Oura
Within this phrase, “second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user's body” means: “second period of time which begins: (i) before the first period of time has ended when it is determined that the electronic device is worn, or (ii) immediately after the first period of time has ended when it is determined that the electronic device is worn”	Indefinite

Oura contends that the term “. . . second period of time shortly before and/or immediately after a first period of time . . .” is indefinite because the phrases “shortly before” and “immediately after” lack clear boundaries, making the timing relationship between the “first” and “second” periods of time unclear. *See* Ex. 3 (Oura’s Invalidity Contentions) at 166. Oura is mistaken. The specification provides multiple figures and embodiments to explain that the bounds of the start of the “second” period are (i) after the “first” period has started but before it has ended when it is determined that a user’s device is worn, or (ii) “immediately after” the end of that “first” period when it is determined that the electronic device is worn. *See generally* Young Decl. (Ex. 6) ¶¶ 30-35. A POSITA viewing the claims and the specification would understand these bounds with

“reasonable certainty,” as required for a claim term to be definite. *Nautilus, Inc. v. Biosig Instruments, Inc.*, 572 U.S. 898, 901 (2014).

A “claim is not indefinite” “[i]f a person of ordinary skill in the art, with the aid of the specification, would understand what is claimed.” *Biosig Instruments, Inc. v. Nautilus, Inc.*, 783 F.3d 1374, 1381 (Fed. Cir. 2015). This extends to words of degree, such as “shortly before,” which are not “inherently indefinite”; instead, “the court must determine whether the patent provides ‘some standard for measuring that degree.’” *Id.* at 1378 (internal quotations omitted). In particular, for claims with terms of degree, the intrinsic evidence “must provide objective boundaries for those of skill in the art” in the context of the invention. *Guangdong Alison Hi-Tech Co. v. Int’l Trade Comm’n*, 936 F.3d 1353, 1360 (Fed. Cir. 2019). Here, the claims and the specification provide the objective boundaries for “first” and “second” periods, including what it means for the “second” period to come “shortly before and/or immediately after” the “first” period.

Starting with the claims, the “first period” is when the device uses the “first sensor” to detect whether it is being worn on the user’s body, while the “second period” is when the same “first sensor” is used to acquire motion information for the electronic device, which in turn helps determine whether to activate a “second sensor.” *See also* Young Decl. (Ex. 6) ¶¶ 31-32. This can be seen in Claim 13, which recites (1) “determining whether an electronic device is worn” during the “first period” in the first step and parts of the second step (identified in red), and (2) “acquiring first information” about motion during a “second period” and then deciding whether to “activat[e] at least one second sensor” based on that motion in the second and third steps (identified in blue):

13. A method comprising:

determining whether an electronic device is worn on a part of a user's body using
at least one first sensor of the electronic device;

acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user's body;

activating at least one second sensor based on at least a part of the first information on the motion of the electronic device;

...

The claims thus permit a triggering event for the start of the “second period” to be either: (1) the end of the “first period” (i.e., “immediately after”) or (2) a wear detection event that occurs after the first period has started but before it has ended, in which case the two periods overlap (i.e., “shortly before”).

Echoing Claim 13, the specification describes “various embodiments” of an “electronic device” that determines whether the electronic device is worn during the “first period” (identified in red), and acquires information regarding motion of the electronic device during a “second period,” which motion information is then used in activating the “second sensor” (identified in blue):

[D]etermine whether the electronic device is worn on (or close to) a part of the user's body using the at least one first sensor; acquire information on the motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time selected (or specified) passes from the time when the electronic device is worn on (or close to) a part of the user's body; and activate (operate or trigger) the at least one second sensor based on at least a part of the information on the motion.

'675 Patent at 20:64-21:8. Thus, the specification confirms that the start of the “second” period is bound by two possibilities: (1) the end of the “first period” (i.e., “immediately after”) or (2) a wear detection event after the first period has started but before it has ended (“shortly before”).

The specification further provides clear, precise guidance for both types of triggering events. First, in the simplest case, the “second” period begins “immediately after” the “first” period ends. *See* Young Decl. (Ex. 6) ¶ 33. Figure 5B below depicts this scenario using sequential time periods 522 (first period) and 524 (second period).

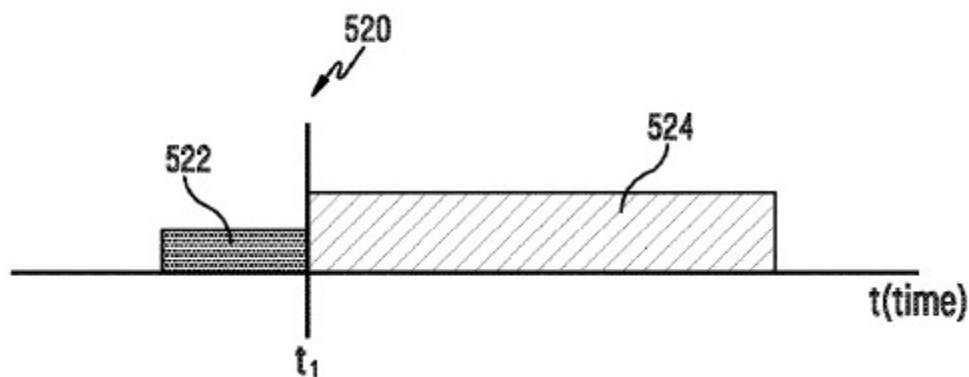
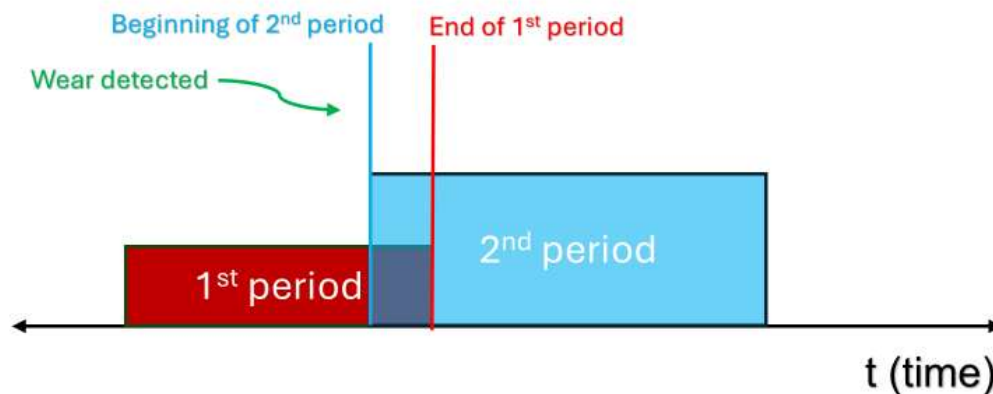


FIG. 5B

As the specification explains, in this embodiment, the claimed device will “measure the motion of the electronic device in at least one time interval 522.” ’675 Patent at 19:47-48. The device will likewise “additionally measure the motion of the electronic device for the time 524 during which the *second* timer operates.” *Id.* at 20:13-16. In Figure 5B, the boundary between the first and second time periods, 520, is when “the operation of the first timer ends (520).” *Id.* at 19:48-49. In other words, a first period ends and, immediately after, the second period begins. *See* Young Decl. (Ex. 6) ¶ 33.

Second, the specification describes an alternative timing, where the “second period” for obtaining motion information begins “shortly before” the end of the first period used for wear detection. *Id.* at ¶ 34. The specification explains that such overlap is possible, stating that “the second period of time may be set to *include the time point when the first period of time passes.*” ’675 Patent at 21:55-57 (emphasis added), 42:21-23 (same); *see also* 21:52-54, 42:18-20. Similarly, another passage explains that the device can acquire motion information “shortly before and/or immediately after a first period of time selected (or specified) passes from the time when the electronic device is worn.” *Id.* at 21:3-5. In these overlap embodiments, the triggering event for starting the second period is “when the electronic device is worn,” which can occur “shortly before” the first period of time ends. *Id.* As a result, the device can immediately enter the “second”

period of acquiring motion information (used in determining whether to activate a “second sensor”), without waiting for the “first” period to end, and the “second” period continues after the “first” period ends. *See* Young Decl. (Ex. 6) ¶ 34. Though not illustrated in the figures of the patent, this scenario may be shown in a modified version of Figure 5B, as follows:



See Young Decl. (Ex. 6) ¶ 34.

The '675 Patent thus fully defines the bounds of the second period. To the extent Oura demands more, such as defining “shortly before” or “immediately after” as a specific duration (e.g., seconds or minutes), the Federal Circuit has rejected such demands. For example, in *Enzo Biochem, Inc. v. Applera Corp.*, the Federal Circuit held that, where the intrinsic evidence provides “a general guideline and examples sufficient to enable a person of ordinary skill in the art to determine [the scope of the claims],” a term of degree can be definite even when defined “without reference to a precise numerical measurement,” notwithstanding the description of exemplary numeric values in the specification. 599 F.3d 1325, 1335 (Fed. Cir. 2010). As in *Enzo*, the specification provides numeric examples that would enable a POSITA to determine the scope of the claims. For example, with reference to Figure 5B, the specification explains that the electronic device “may attempt to measure biometric information according to a period (e.g., about 1

minute).” ’675 Patent at 19:35-37; *see also* 18:19-22 (same). As in *Enzo*, these numeric examples provide further support for the definiteness of the term, even if the proper construction “defines the term without reference to a precise numerical measurement.” 599 F.3d at 1335.

Accordingly, in view of the specification’s clear guidance, a POSITA would understand with reasonable certainty the bounds of the “first” and “second” periods. *See generally* Young Decl. (Ex. 6) ¶¶ 30-35. At the earlier end, in the “shortly before” case, the “second” period begins before the “first” period has ended, when it is determined that the electronic device is worn. *See* Young Decl. (Ex. 6) ¶ 34. On the later end, in the “immediately after” scenario, the “second” period begins immediately after the “first” period ends. *Id.* at ¶ 33. Samsung’s construction follows directly from this specification guidance, by explaining that the “second period of time which begins: (i) before the first period of time has ended when it is determined that the electronic device is worn, or (ii) immediately after the first period of time has ended when it is determined that the electronic device is worn.” *See generally id.* ¶¶ 30-35.

2. **“acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes; and activating the at least one second sensor based on at least a part of the additional information” (’675 Patent, Claim 15)**

Samsung	Oura
Within this phrase, “third period of time” means: “period of time that immediately follows the second period of time and during which additional information regarding the motion of the electronic device is collected by at least a part of the at least one first sensor and used to determine whether to activate the at least one second sensor”	Indefinite

Oura contends that the claimed “third period of time” in dependent Claim 15 is indefinite because a POSITA would not understand with reasonable certainty what the “third” period is or how it is distinct from the “second” period, because the same “first sensor” is making the same

measurement of “motion” in both periods. Ex. 3 (Oura’s Invalidity Contentions) at 167. But Oura fundamentally misunderstands the purpose and function of the recited “periods.” See Young Decl. (Ex. 6) ¶¶ 37-38. In all recited “periods,” the same “first sensor” is measuring motion. *Id.* at ¶¶ 39-43. The “periods” are distinguished from each other based on the purpose of the motion measurement in each “period,” including what the device does with information concerning the measured motion. In Claim 13’s “second period,” the device uses the measured motion to decide whether to activate a “second sensor.” *Id.* at ¶¶ 39, 41. Claim 13 then ends with the inactivation of the same “second sensor.” *Id.* In Claim 15’s “third” period, the device uses further information about the measured motion to decide whether to reactivate the “second sensor,” after it was inactivated in Claim 13’s last step. *Id.* at ¶¶ 39, 42-43.

The plain language of Claims 13 and 15 confirms the relationship and respective purposes of the “second” and “third” periods. Claim 13 recites that the “second sensor” is initially activated based on motion, and then inactivated when that motion exceeds a threshold level:

13. A method comprising:

...

activating at least one second sensor based on at least a part of the first information on the motion of the electronic device;

detecting a biometric information of the user using the activated at least one second sensor;

during detecting the biometric information, acquiring second information on a motion of the electronic device using at least the part of the at least one first sensor; and

in response to identifying that the motion of the electronic device included in the second information is more than a threshold motion, inactivating the at least one second sensor.

Claim 15, quoted below, begins after Claim 13 ends, as it recites a subsequent “third” period that comes “immediately after the second period” of Claim 13. Therefore, ahead of the step in Claim 15, the sensor has already been inactivated. Claim 15 recites that the purpose of the

“third” period is to decide whether to “activat[e]” the “second sensor” again based on “additional information on the motion”:

15. The method of claim 13, further comprising
acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor ***for a third period of time***, which is shorter than the first period of time, immediately after the second period of time passes; and
activating the at least one second sensor based on at least a part of the additional information.

The relationship between the “second” and “third” periods, as recited in Claims 13 and 15, is confirmed by the specification. In the context of Figure 5B (excerpted in Section IV.A.1 for the prior term), the specification explains that in ***both*** back-to-back time periods 522 and 524, the device measures motion and “may operate at least one sensor (e.g., biometric sensor) for measuring biometric information when the measured motion of the electronic device satisfies a selected (or specified) reference.” *See* ’675 Patent at 19:46-64, 20:14-29. The specification similarly describes the “third” period of Claim 15, where a device will:

[A]cquire additional information on the motion of the electronic device using at least a part of the at least one first sensor ***for a third period of time***, which is shorter than the first period of time, ***immediately after the second period of time passes; and activate (operate or trigger) the at least one second sensor based on at least a part of the additional information***, without activating (operating or triggering) the second sensor when the information on the motion does not satisfy a selected (or specified) condition.

’675 Patent at 21:42-51; *see id.* at 42:8-17 (similar). Thus, the specification plainly describes two consecutive periods (the “second” and “third” periods), where a first (motion) sensor measures motion to determine whether to activate or inactivate a second (biometric) sensor.

A POSITA, therefore, would understand “third period of time” as Samsung has proposed: “period of time that immediately follows the second period of time and during which additional information regarding the motion of the electronic device is collected by at least a part of the at

least one first sensor and used to determine whether to activate the at least one second sensor.”

See generally Young Decl. (Ex. 6) ¶¶ 36-44.

B. U.S. Patent No. 10,642,359 (“359 Patent”):

1. “[the] biometric signal” (’359 Patent, Claim 14)

Samsung	Oura
Correcting “the biometric signal” to be “the biological signal” and applying the plain and ordinary meaning (i.e., a signal that is associated with biological information of the user).	Correcting “the biometric signal” to be “a biometric signal” and applying the plain and ordinary meaning (i.e., a signal relating to the Identification of an Individual based on physical characteristics, such as fingerprints, finger lengths, facial features, retinal patterns, voice recognition, and weight).

Reading Claim 14 itself, the specification, and other claims, it is clear that the recitation of “the *biometric* signal” in Claim 14 is an obvious, minor clerical or typographical error, and should instead read “the *biological* signal,” consistent with the antecedent recitation of “a *biological* signal” obtained by “the *biological* sensor” earlier in the claim. Consistent with the surrounding claim language and the specification, Samsung’s construction makes this correction and applies the plain and ordinary meaning of the term. See generally Young Decl. (Ex. 6) ¶¶ 45-55. Oura agrees that there is an obvious typographical error that requires correction. But, rather than propose a correction based on the intrinsic evidence, in a transparent effort to manufacture a non-infringement defense, Oura proposes a correction and interpretation wholly untethered to the intrinsic record. Such a strained and ungrounded approach to claim construction should be rejected out of hand.

a. Samsung’s Construction Is Firmly Supported By All Of The Intrinsic Evidence

Considering the intrinsic evidence, all strongly supports Samsung’s proposed construction and leaves no doubt as to the appropriate correction. See Young Decl. (Ex. 6) ¶¶ 45-52.

First, it is obvious from Claim 14 itself that the antecedent basis of “*the* biometric signal” is “a biological signal,” recited earlier in the claim:

obtain a movement signal of an object external to the wearable device using the motion sensor;

obtain *a biological signal* of the object using *the biological sensor*;

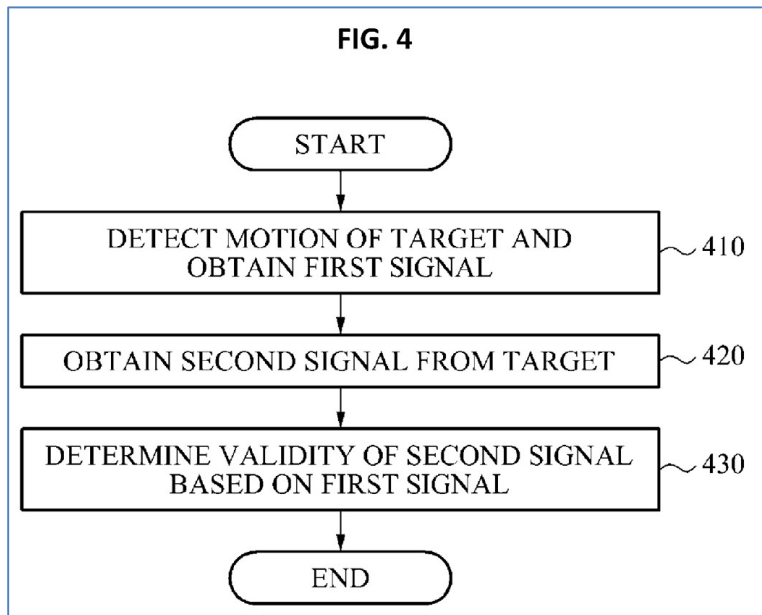
determine a select period, of the movement signal, that meets a specified condition; determine, as a valid signal, at least a portion of *the biometric signal* that is determined to correspond to the determined select period; and

perform a specified function using the valid signal.

Only two signals, “a movement signal” and “a biological signal,” are recited earlier in Claim 14, before “*the* biometric signal.” See Young Decl. (Ex. 6) ¶ 48. Because the “biometric signal” is introduced using the article “the,” it is evident that “biometric signal” refers back to one of these prior signals. *Tuna Processors, Inc. v. Haw. Int’l Seafood, Inc.*, 327 F. App’x 204, 210 (Fed. Cir. 2009). And because the phrase “biometric signal” is etymologically closer to “biological signal” than to “movement signal,” the self-evident choice is “biological signal.”

The correction to “biological signal” over “movement signal” is also logically supported by the function of the claimed signals. The “movement signal” is used to “determine a select period” “that meets a specified condition.” See Young Decl. (Ex. 6) ¶ 48. After this “select period” is determined, “at least a portion of the biometric signal that is determined to correspond to the determined select period” is identified as the “valid signal.” *Id.* Thus, the “movement signal” serves the purpose of identifying a “select period” of time, during which a “portion” of the “biometric signal” is considered a “valid signal.” *Id.* The “movement signal” cannot, therefore, also *be* the “biometric signal” as the “movement signal” is instead used to *identify* a valid portion of the “biometric signal.” *Id.* Rather, the only earlier-recited phrase that provides plausible antecedent basis for “the biometric signal” is the “biological signal.” *Id.*

Second, the specification describes these exact steps of Claim 14, which confirms that the movement signal is used to determine a valid portion of a second, biological signal. *See* Young Decl. (Ex. 6) ¶ 49. Figure 4 (reproduced below), discloses obtaining a first (movement) signal at step 410, obtaining a second (biological) signal at step 420, and determining the validity of the second (biological) signal based on the first (movement) signal at step 430. As the specification explains, in step 410, the wearable device “may monitor, using a motion sensor” “the motion of the target” and “receive the first signal to perceive a movement involved with the motion.” ’359 Patent at 10:9-22. In step 420, the wearable device may “obtain the second signal generated by at least one of



[the] muscle contraction and muscle relaxation” using a second sensor. *Id.* at 10:63-11:2. Thus, the “second signal” corresponds to a “biological signal” as it measures biological information, such as a user’s “muscle contraction and muscle relaxation.” Finally, at step 430, the wearable device may “determine a validity of the second signal based on the first signal” “in the event that the magnitude of the first signal satisfies a first threshold.” *Id.* at 11:45-54. Figure 4’s close correspondence to Claim 14 further confirms that the recited “biometric signal”—which is validated and filtered using a “first signal” measuring movement—corresponds to the earlier-recited “biological signal.” *See* Young Decl. (Ex. 6) ¶ 49.

More generally, beyond Figure 4, the specification describes “biological signal[s] or physiological signal[s]” obtained by a “biosignal sensor” that relate to biological information, such as “muscle contraction or muscle relaxation of a body part, such as a wrist or a heart.” *See, e.g., id.* at 3:52-64; Young Decl. (Ex. 6) ¶ 52. And consistent with Samsung’s proposed construction, the specification explains that a biological signal is “*associated with biological information*” such as, for example, muscle contraction and muscle relaxation,” consistent with Samsung’s construction. ’359 Patent at 5:14-17.

Third, other claims of the ’359 Patent also confirm that “biometric signal” in Claim 14 should be “biological signal.” *See* Young Decl. (Ex. 6) ¶¶ 50-51. Claims 1 and 7 generally parallel Claim 14 in relevant respects but, like the description in the specification, recite “biological signal” rather than “biometric signal.” *Id.* As quoted below, Claim 7 in particular tracks the same steps as Claim 14, including: (1) measuring both movement and biological signals; (2) identifying a time period in which the movement signal satisfies some condition; and (3) performing a specified function based on a portion of the biological signal corresponding to the identified time period.

Claim 7	Claim 14 (disputed)
“obtain a movement signal of an object external to the wearable device using the motion sensor;	“obtain a movement signal of an object external to the wearable device using the motion sensor;
obtain a biological signal of the object using the biological sensor ;	obtain a biological signal of the object using the biological sensor;
“identify a period, of the movement signal, the movement signal is determined to be less than a specified condition; and”	“determine a select period, of the movement signal, that meets a specified condition;”

“perform a specified function using a select portion of the <i>biological signal</i> corresponding to the period.”	“determine, as a valid signal, at least a portion of the <i>biometric signal</i> that is determined to correspond to the determined select period; and perform a specified function using the valid signal.”
--	---

As quoted above, Claim 7 identifies a “select portion” of the “biological signal,” which is a time segment of the “biological signal” that corresponds to a “select period” when the “movement signal” meets a “specified condition.” *See* Young Decl. (Ex. 6) ¶ 51. The device then performs a “specified function” using this “select portion” of the “biological signal.” Claim 14 recites similar steps. *Id.* The only difference is that Claim 14 refers to the “select portion” of the “biometric/biological” signal as the “valid signal.” But this “valid signal” is the same as the “select portion” because it is a “portion of the biometric signal” corresponding to the “select period” identified based on the “movement signal.” *Id.*

b. Oura’s Construction Ignores The Intrinsic Evidence And Would Recite An Invention Not Described Anywhere In The Patent

In sharp contrast to Samsung’s well-supported construction, Oura proposes a construction wholly unrelated to, and unsupported by, the intrinsic record. Oura proposes correcting “*the* biometric signal” to “*a* biometric signal” and construing “biometric signal” as “(i.e., a signal relating to the identification of an individual based on physical characteristics, such as fingerprints, finger lengths, facial features, retinal patterns, voice recognition, and weight).” But Oura’s proposal is based entirely on dictionary definitions,² and is untethered from the specification and

² In particular, Oura’s interpretation appears cobbled together from five different dictionary definitions of “biometric,” “biometrics,” or “biometric device.” Exs. 4, 5. For example, one dictionary defines biometrics as “the study of unique and measurable physical (or biological) characteristics such as fingerprints, retinal pattern, or handwriting, biometric authentication devices capture, encrypt and use these unique and measurable characteristics as the basis for confirming identity and determining whether to grant or deny access to physical or logical assets.”

the fundamental purpose of the claimed invention. *See* Young Decl. (Ex. 6) ¶ 53. The '359 Patent has nothing to do with identifying users for security purposes, but everything to do with measuring a user's biological information. Indeed, other than the one obvious typographical error in Claim 14, the '359 Patent does not anywhere use the term "biometric" or "biometric signal," or reference any signals that correspond to fingerprints, facial features, voices, and the like, for identifying individuals. *Id.* Instead, as discussed above, the '359 Patent describes only "biological signal[s] or physiological signal[s]" for measuring "biological information" like muscle patterns. *See, e.g.,* '359 Patent at 3:52-64, 5:14-17; Young Decl. (Ex. 6) ¶ 52. Thus, Oura's dictionary-based, litigation-inspired interpretation should be rejected as contrary to the specification. *See Dow Chem. Co. v. Sumitomo Chem. Co.*, 257 F.3d 1364, 1373 (Fed. Cir. 2001) (holding that "any definition found in or ascertained by a reading of the intrinsic evidence may not be contradicted by any meaning found in dictionaries or technical treatises.").

Even if Oura's reliance on cherry-picked extrinsic evidence over intrinsic evidence were appropriate, its construction should still be rejected because there are other, contrary extrinsic definitions of "biometric signal" that are consistent with the '359 Patent and Samsung's construction. *See* Young Decl. (Ex. 6) ¶ 54. Indeed, the '675 Patent discussed in this brief describes "biometric information" as biological information "includ[ing] blood pressure, a heart rate, an electrocardiogram, skin moisture, saturation of peripheral oxygen, and the like." '675 Patent at 1:29-35.

* * * * *

Ex. 4 (Newton's Telecom Dictionary 27th ed. (2013) (OURA1478_00002999 at 3001)). Another defines "biometrics" as "[t]he science of discriminating individuals from accurate measurements of their physical features. Example biometric measurements are retinal lines, finger lengths, fingerprints, voice characteristics and facial features." Ex. 5, Dictionary of Computer Vision and Image Processing (2005) (OURA1478_00002989 at 2991).

Based on the specification and context of the claim language, Oura's proposal should be rejected and "biometric signal" should be construed as "biological signal," which is "a signal associated with biological information."

V. CONCLUSION

For the reasons discussed above, Samsung respectfully requests that its constructions be adopted for the disputed terms in the '675 and '359 Patents.

Dated: May 1, 2026

Respectfully submitted,

/s/ D. Sean Trainor

D. Sean Trainor

Renia Zervos

Khanh Leon

Jacob Boesch

O'MELVENY & MYERS

1625 Eye Street

Washington D.C. 20006

Telephone: +1 202 383-5300

Ryan K. Yagura

Ben Haber

Nancy Schroeder

Gregory Sefian

Grace McFee

Ohona Chowdhury

O'MELVENY & MYERS LLP

400 South Hope Street

Los Angeles, CA 90071

Telephone: (213) 430-6000

Mark Liang

Bill Trac

Rui Li

Sorin Zaharia

Eric T. Ong

O'MELVENY & MYERS LLP

Two Embarcadero Center, 28th Floor

San Francisco, CA 94111

Telephone: (415) 984-8700

Marc Pensabene

Laura Gore

Caitlin P. Hogan

Nate Legum

Joanne Bae

O'MELVENY & MYERS LLP

1301 Avenue of the Americas, 17th Floor

New York, NY 10019

Telephone: (212) 326-2000

John Kappos

Cason Cole

Frances Mackay

O'MELVENY & MYERS LLP
2801 North Harwood Street, Suite 1600
Dallas, TX 75201
Telephone: (972) 360-1900

Email: OMM-Samsung-Oura@omm.com

*Counsel for Complainants Samsung
Electronics Co., Ltd. and Samsung
Electronics America, Inc.*

CERTIFICATE OF WORD COUNT

The undersigned hereby certifies that the foregoing Opening Claim Construction Brief complies with the word limitations of Ground Rule 7.2. Excluding the items listed in Ground Rule 1.7 as not included in the word count, and using the word count of the word-processing system used to prepare the paper (Microsoft Word), Complainants' Opening Claim Construction Brief contains 5,470 words, which is under the 10,000 word limit.

/s/ D. Sean Trainor

D. Sean Trainor

EXHIBIT 1



US010231675B2

(12) **United States Patent**
Chung et al.

(10) **Patent No.:** **US 10,231,675 B2**

(45) **Date of Patent:** **Mar. 19, 2019**

(54) **METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME**

(58) **Field of Classification Search**

USPC 600/300-301
See application file for complete search history.

(71) Applicant: **Samsung Electronics Co., Ltd.**,
Suwon-si, Gyeonggi-do (KR)

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Primary Examiner — Nicole F Johnson

(74) *Attorney, Agent, or Firm* — Jefferson IP Law, LLP

(72) Inventors: **Gihsung Chung**, Suwon-si (KR);
Jooman Han, Seongnam-si (KR); **So Hyun Chung**, Seoul (KR); **Dae-Yong Lee**, Suwon-si (KR); **Hongsig Kim**, Seongnam-si (KR); **Hyunwoo Jang**, Suwon-si (KR); **Cheolho Cheong**, Seoul (KR); **Jae-Woong Chun**, Suwon-si (KR); **Won Suk Choi**, Seoul (KR)

(73) Assignee: **Samsung Electronics Co., Ltd.**,
Suwon-si (KR)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 217 days.

(21) Appl. No.: **15/173,099**

(22) Filed: **Jun. 3, 2016**

(65) **Prior Publication Data**

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(30) **Foreign Application Priority Data**

Aug. 12, 2015 (KR) 10-2015-0113972

(51) **Int. Cl.**

A61B 5/04 (2006.01)

A61B 5/00 (2006.01)

(Continued)

(52) **U.S. Cl.**

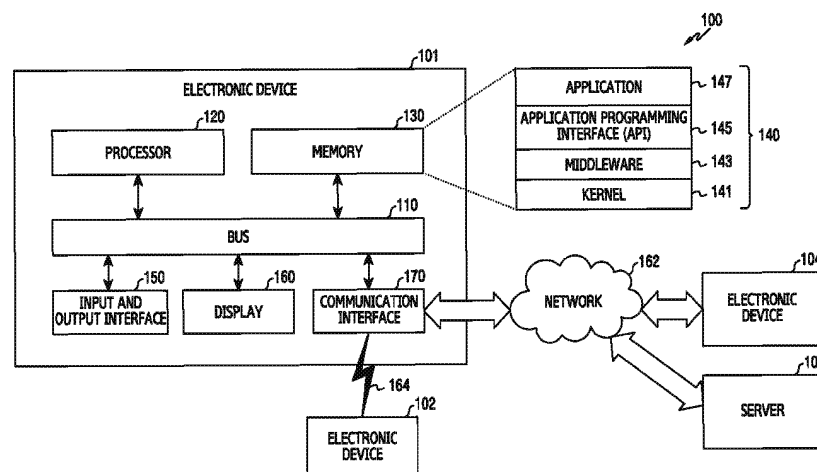
CPC **A61B 5/7285** (2013.01); **A61B 5/0205** (2013.01); **A61B 5/0402** (2013.01);

(Continued)

ABSTRACT

A device for detecting biometric information is provided. The device includes a housing, a fastener coupled to a part of the housing and detachably attached to a part of a user's body, at least one first sensor configured to detect the motion of the electronic device, at least one second sensor configured to detect a change in the user's body, a memory, and a processor electrically connected to the at least one first sensor, the at least one second sensor, and the memory. The memory may store instructions that configure the processor, when being executed, to determine whether the electronic device is worn on a part of the user's body using the at least one first sensor, acquire information on the motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time selected passes from the time when the electronic device is worn on a part of the user's body, and activate the at least one second sensor based on at least a part of the information on the motion.

20 Claims, 23 Drawing Sheets



- (51) **Int. Cl.**
A61B 5/0205 (2006.01)
A61B 5/0402 (2006.01)
A61B 5/11 (2006.01)
G06F 1/3231 (2019.01)
G06F 1/16 (2006.01)
G06F 1/3215 (2019.01)
G06F 1/3234 (2019.01)
G04G 21/02 (2010.01)
A61B 5/021 (2006.01)
A61B 5/024 (2006.01)
A61B 5/145 (2006.01)
- (52) **U.S. Cl.**
 CPC *A61B 5/11* (2013.01); *A61B 5/1118* (2013.01); *A61B 5/443* (2013.01); *A61B 5/6801* (2013.01); *A61B 5/681* (2013.01); *A61B 5/6843* (2013.01); *G04G 21/025* (2013.01); *G06F 1/163* (2013.01); *G06F 1/1684* (2013.01); *G06F 1/1694* (2013.01); *G06F 1/325* (2013.01); *G06F 1/3215* (2013.01); *G06F 1/3231* (2013.01); *A61B 5/021* (2013.01); *A61B 5/02438* (2013.01);
- A61B 5/14542* (2013.01); *A61B 2560/0209* (2013.01); *A61B 2562/0219* (2013.01); *A61B 2562/0223* (2013.01); *A61B 2562/0257* (2013.01); *Y02D 10/173* (2018.01)
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- * cited by examiner

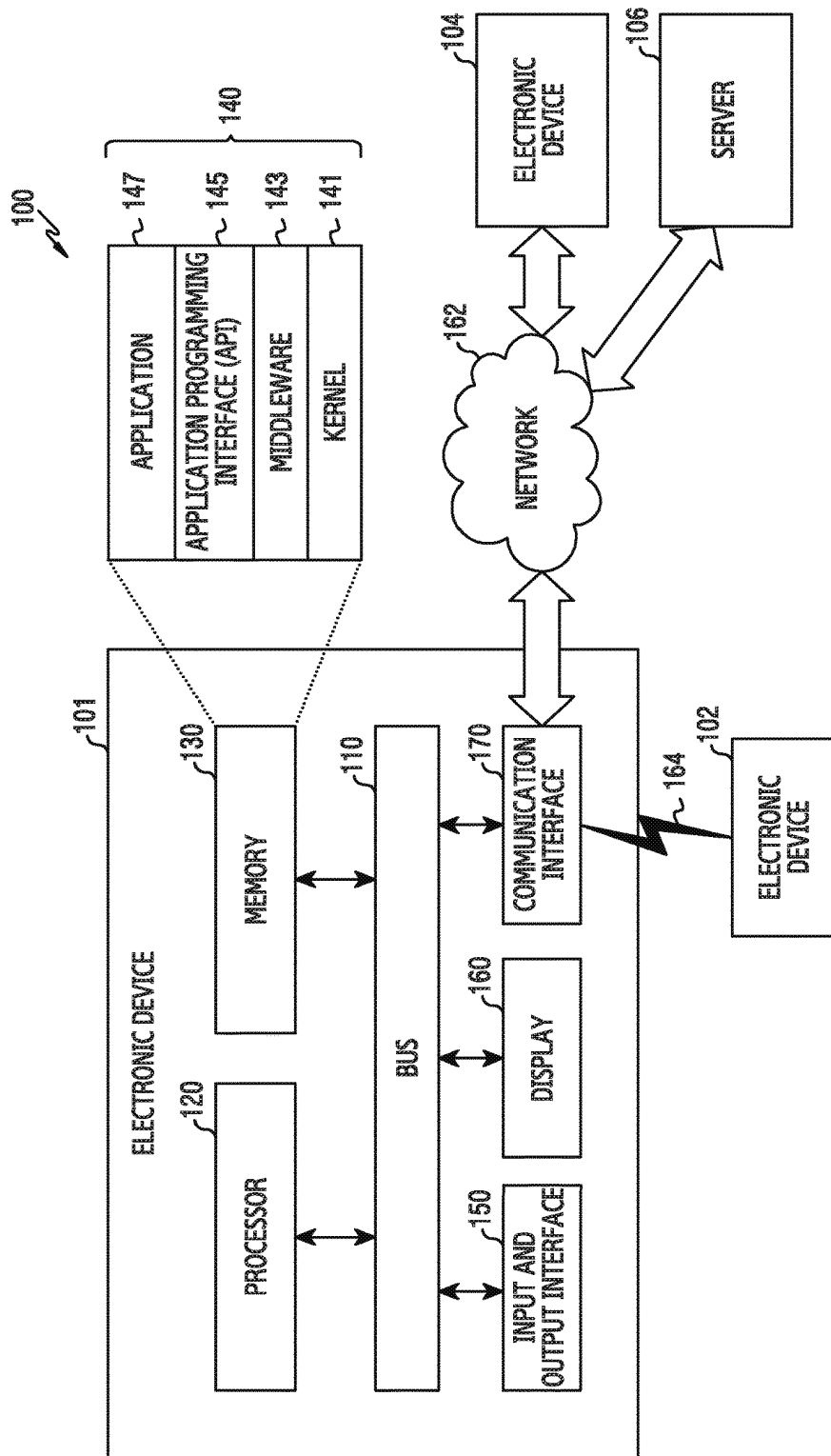


FIG.1

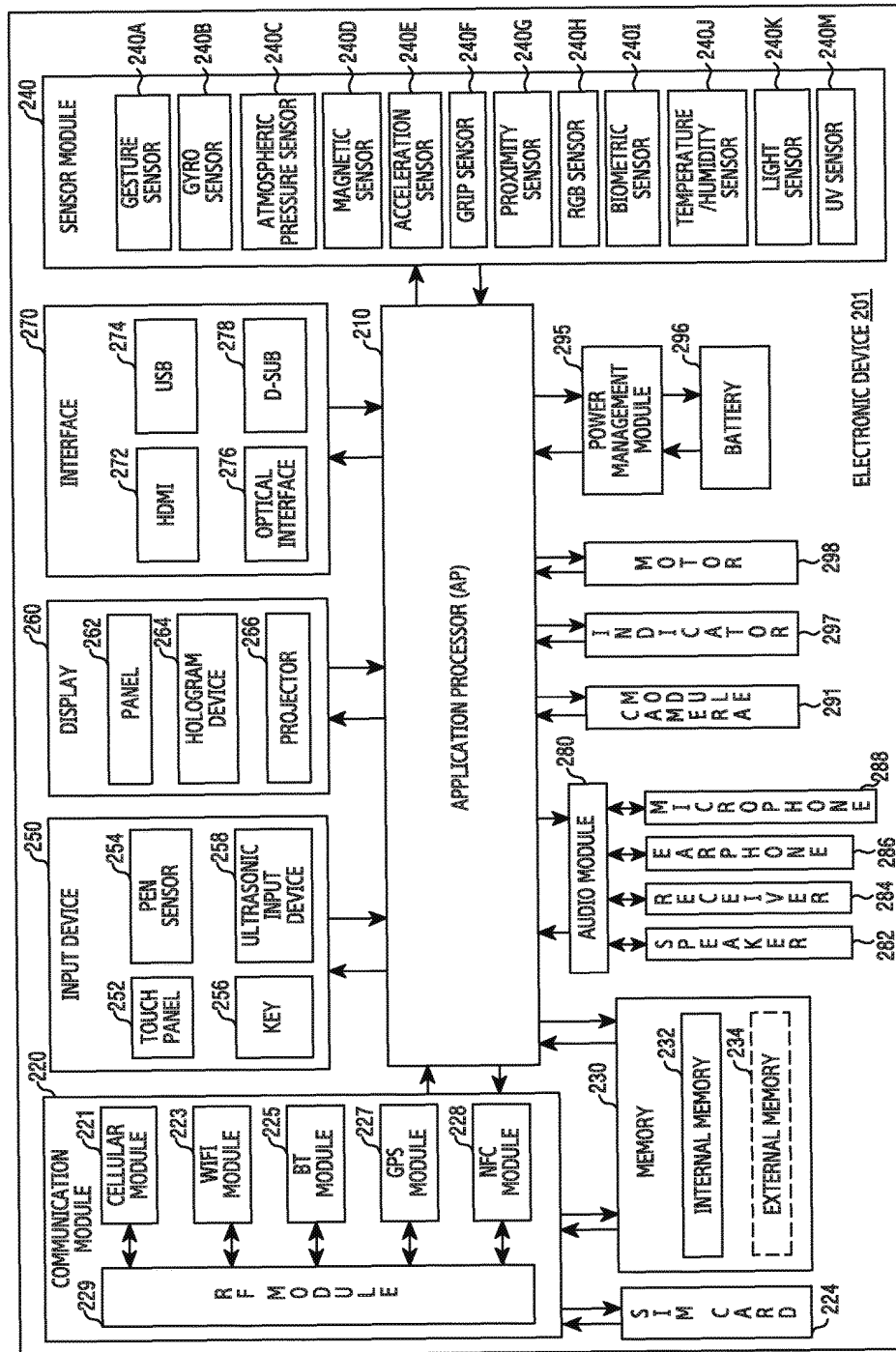


FIG. 2

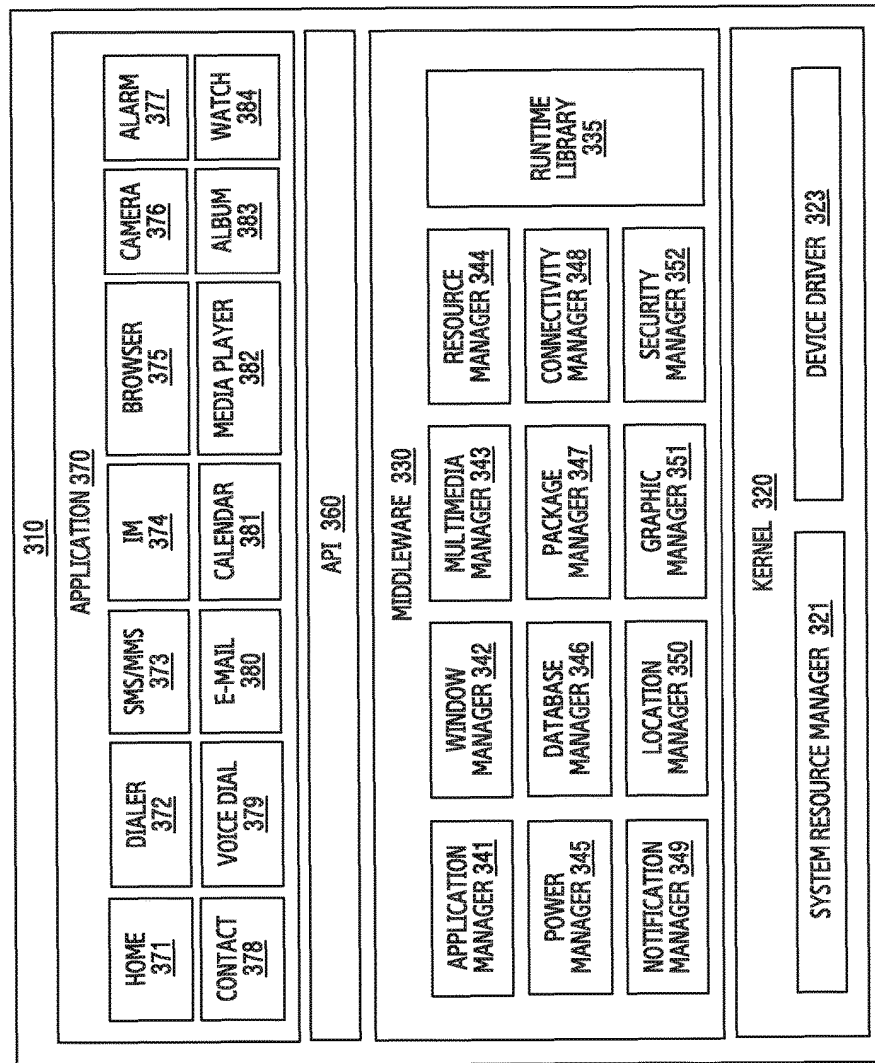


FIG.3

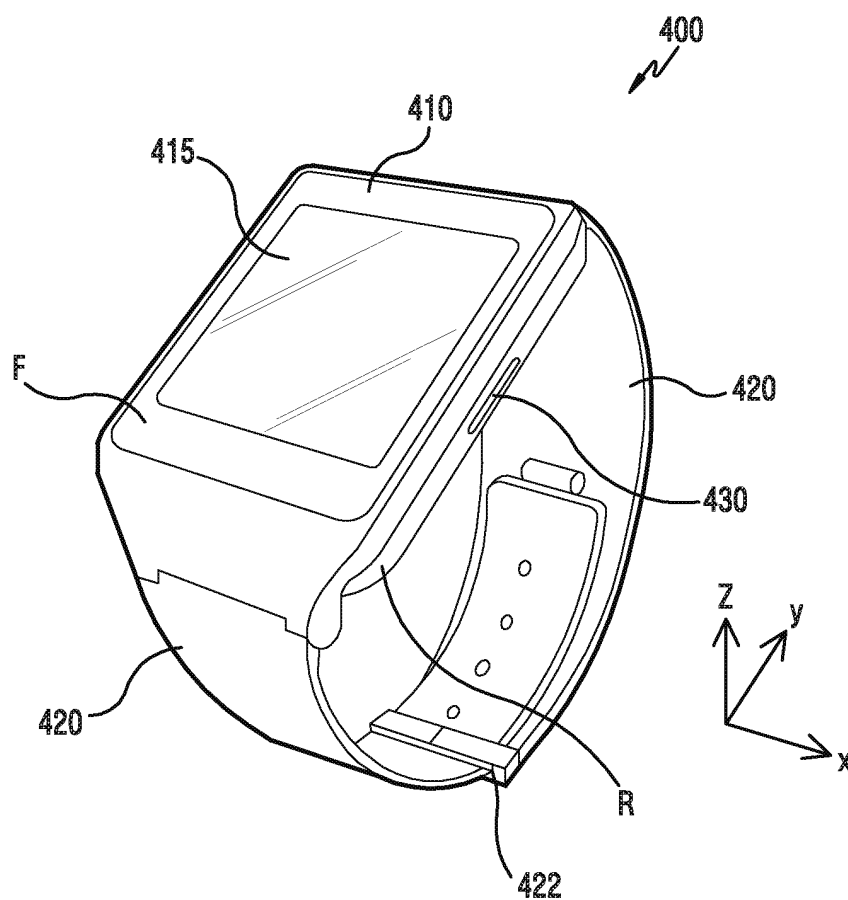


FIG. 4A

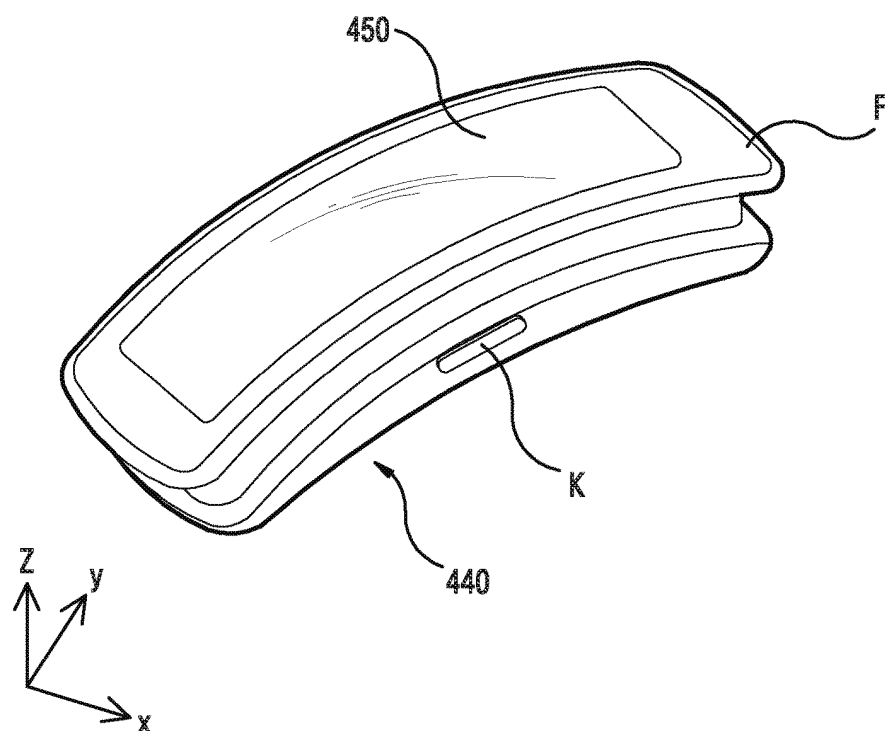


FIG. 4B

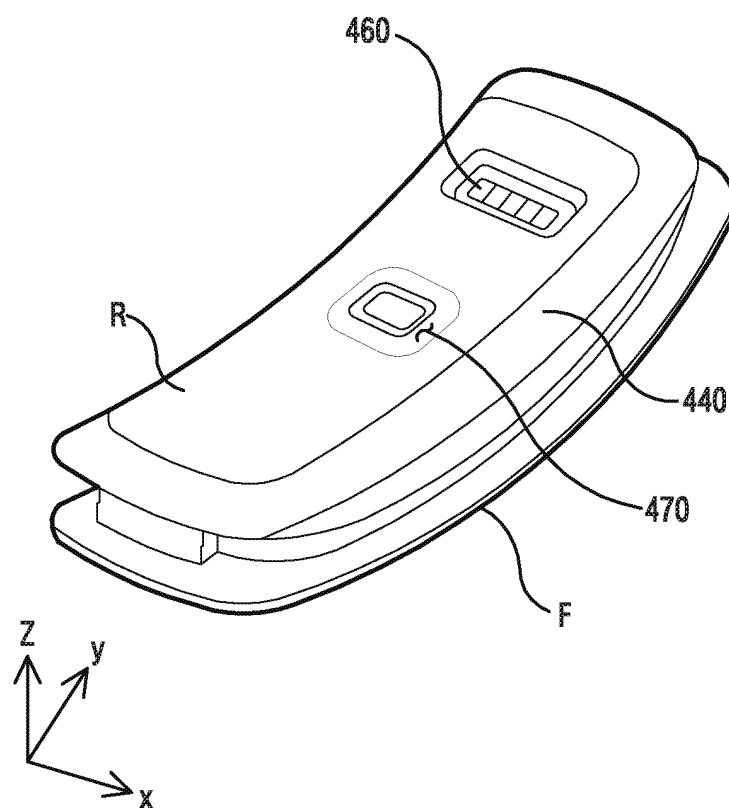


FIG. 4C

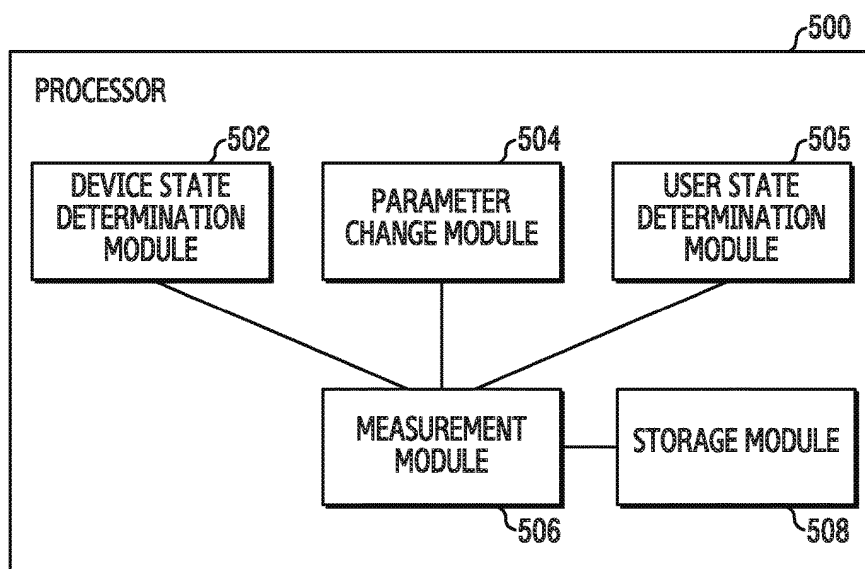


FIG.5A

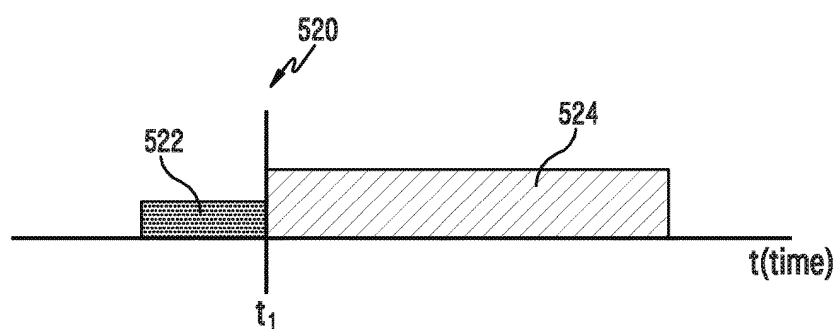


FIG.5B

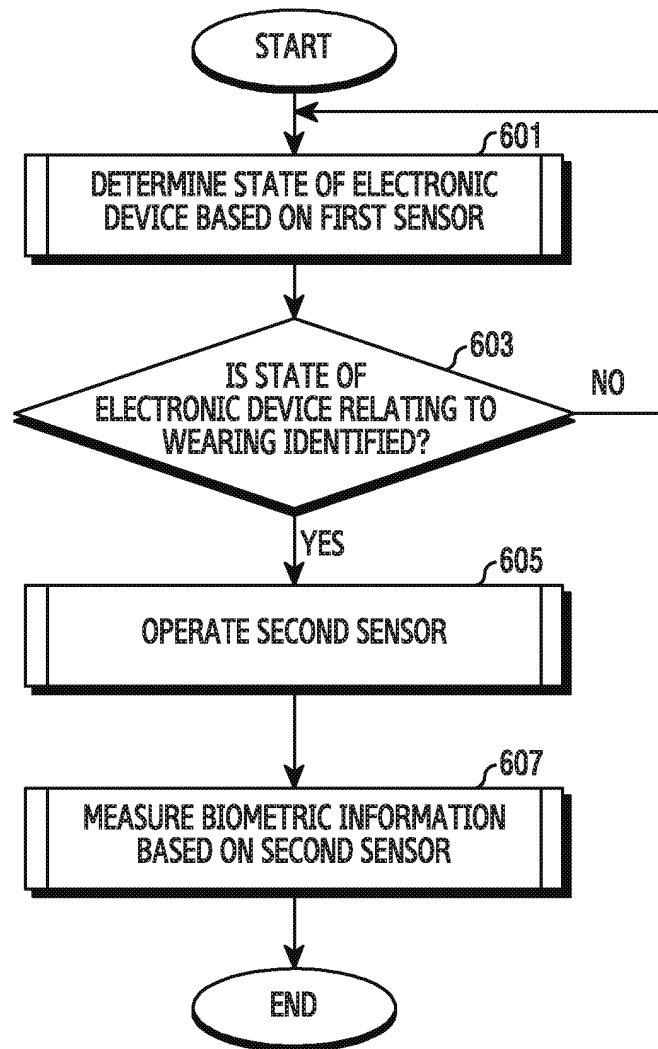


FIG. 6

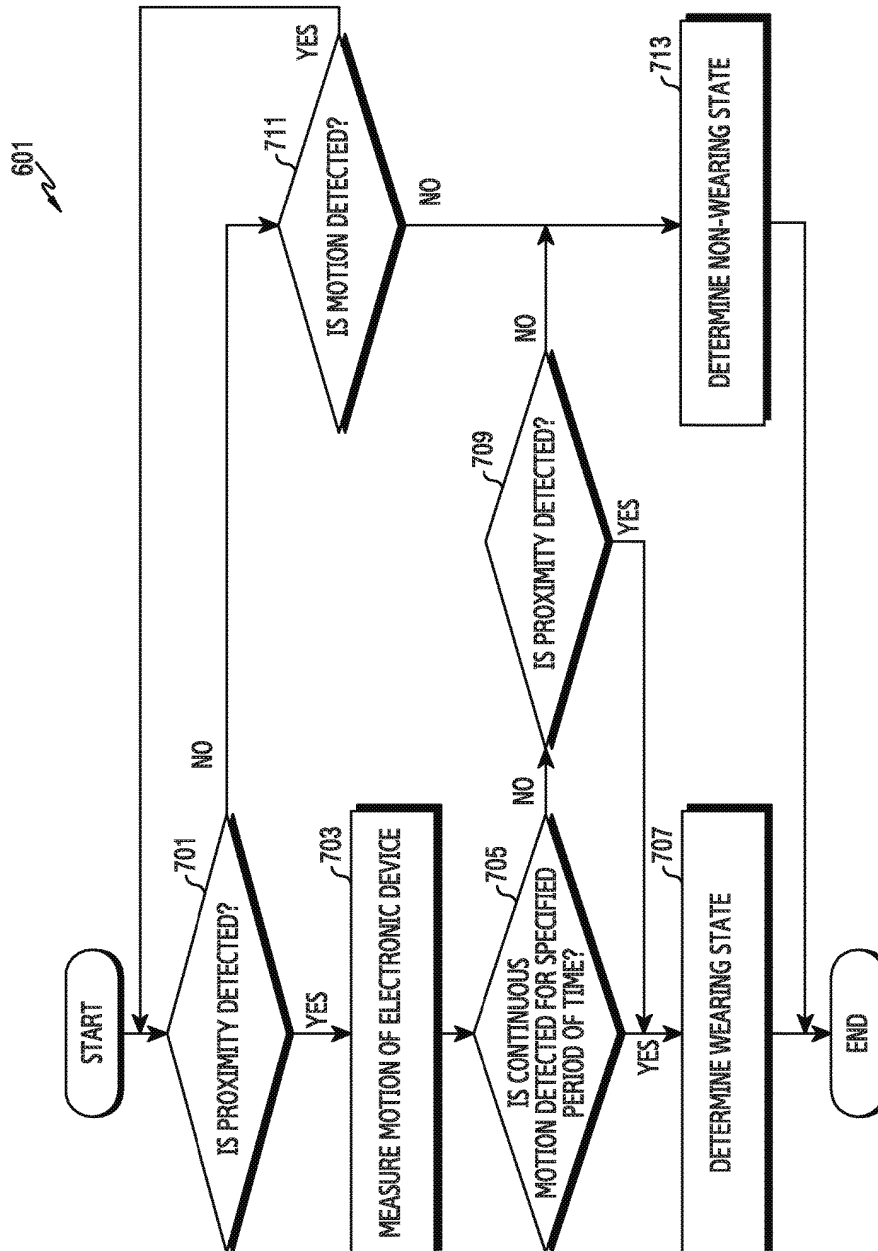


FIG.7

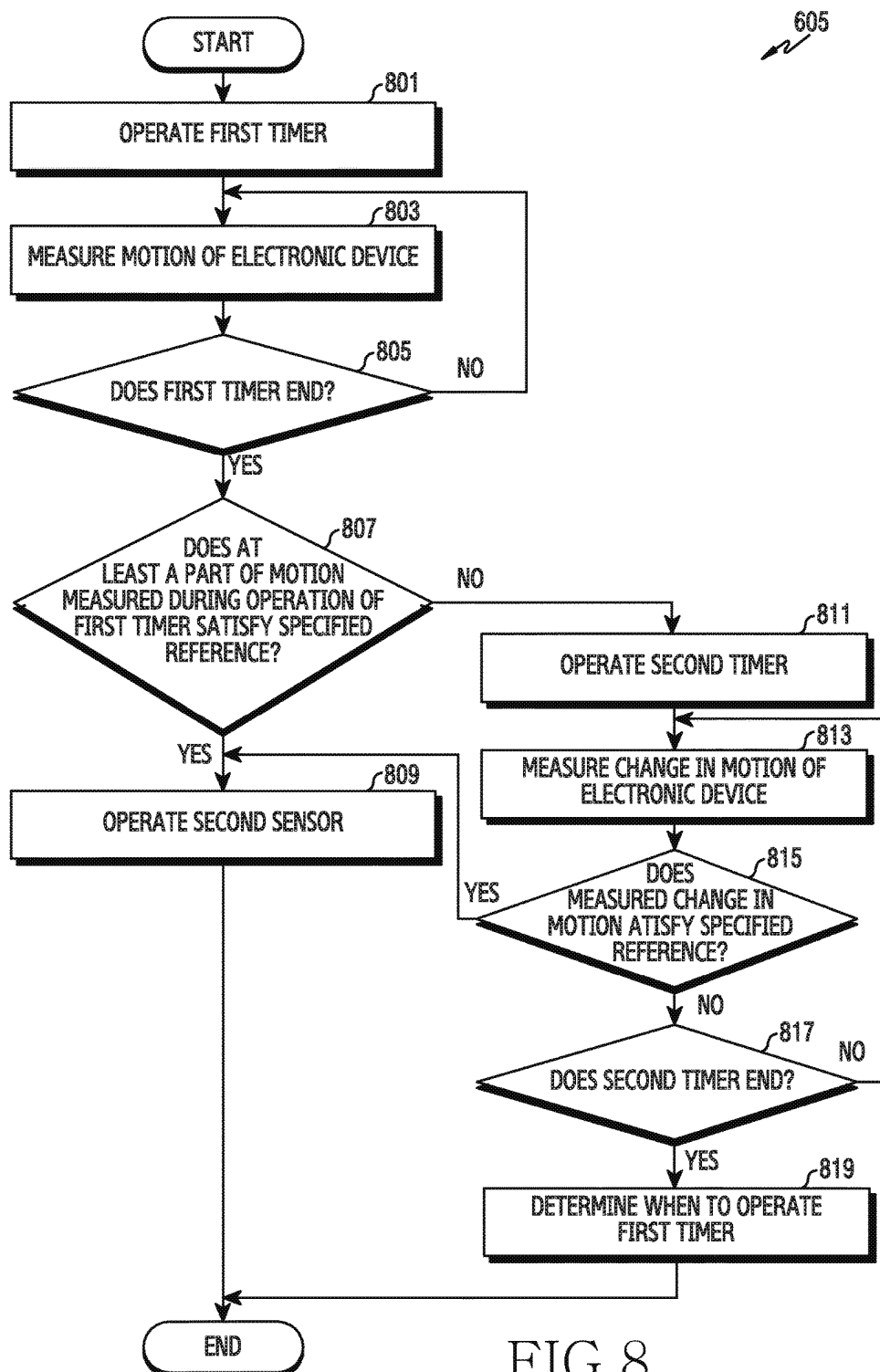


FIG.8

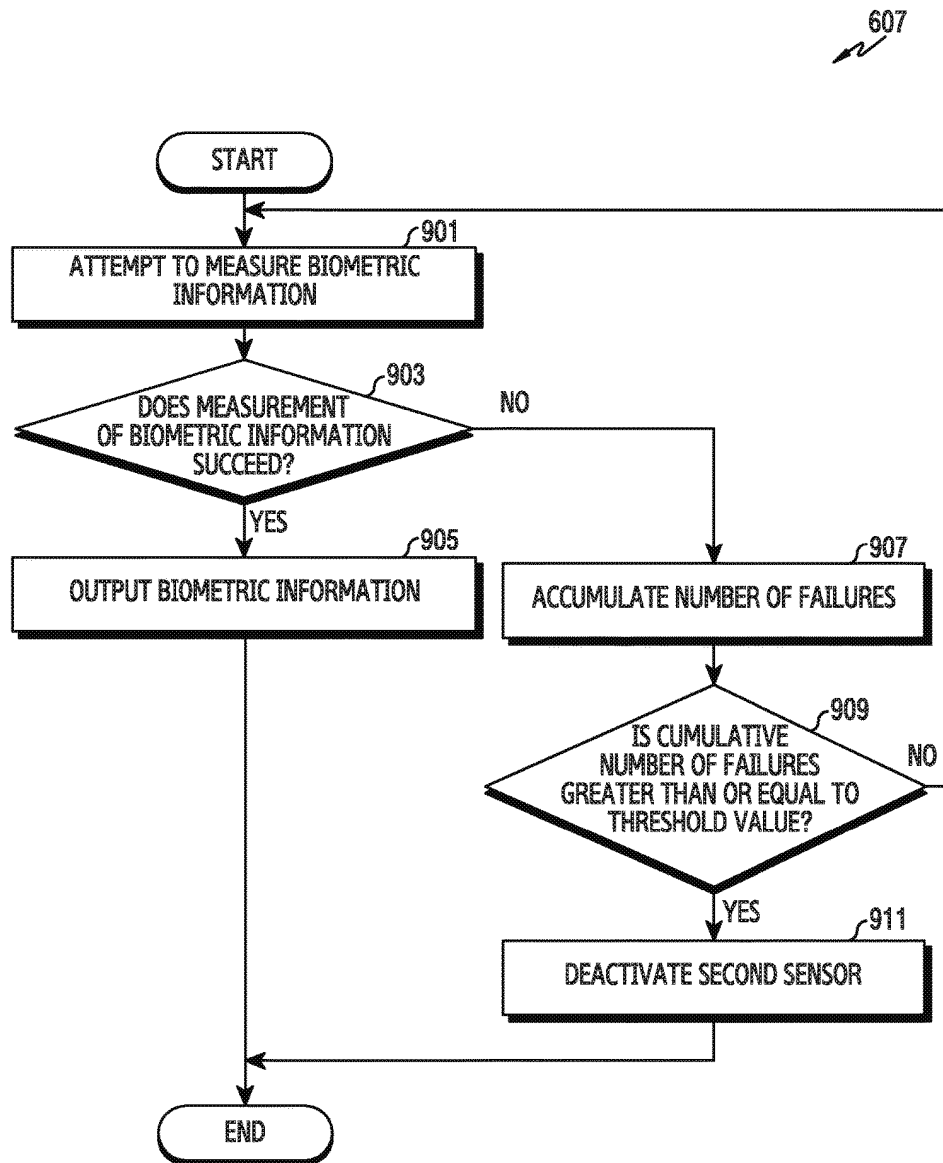


FIG.9

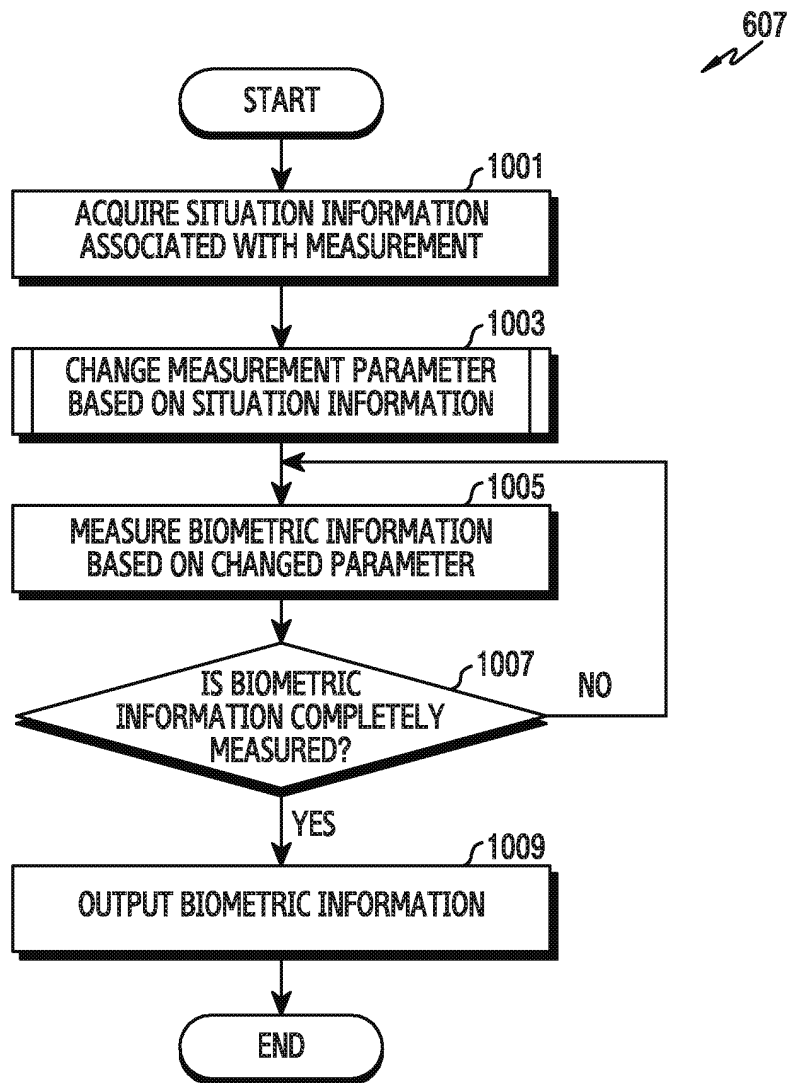


FIG.10

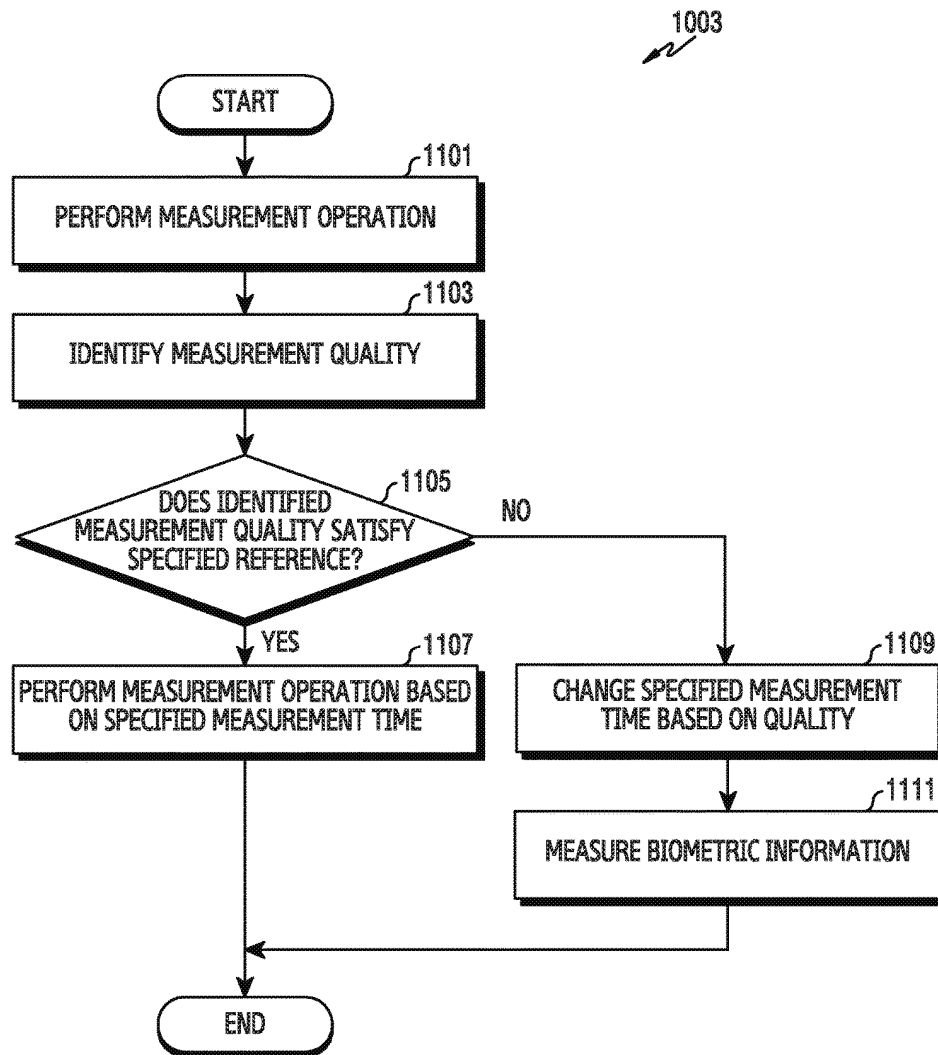


FIG.11

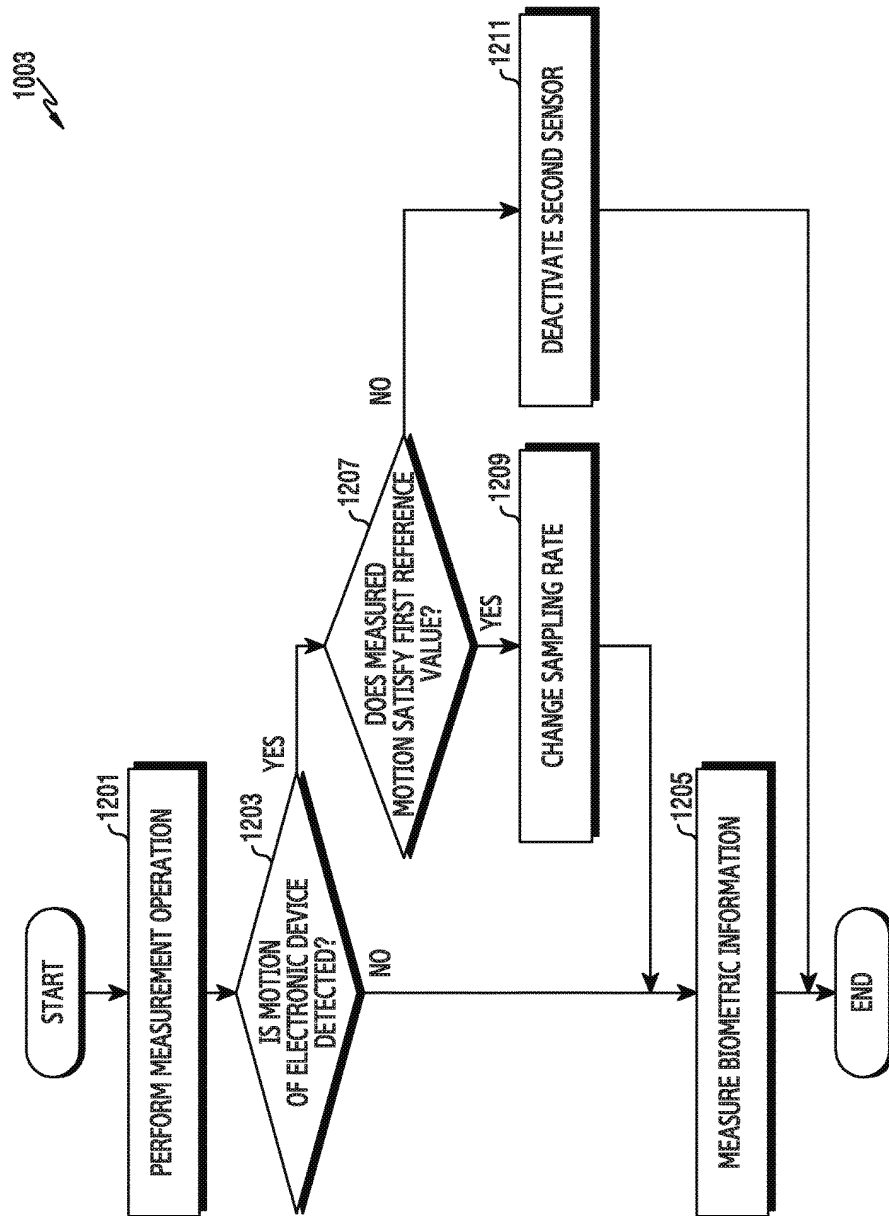


FIG.12

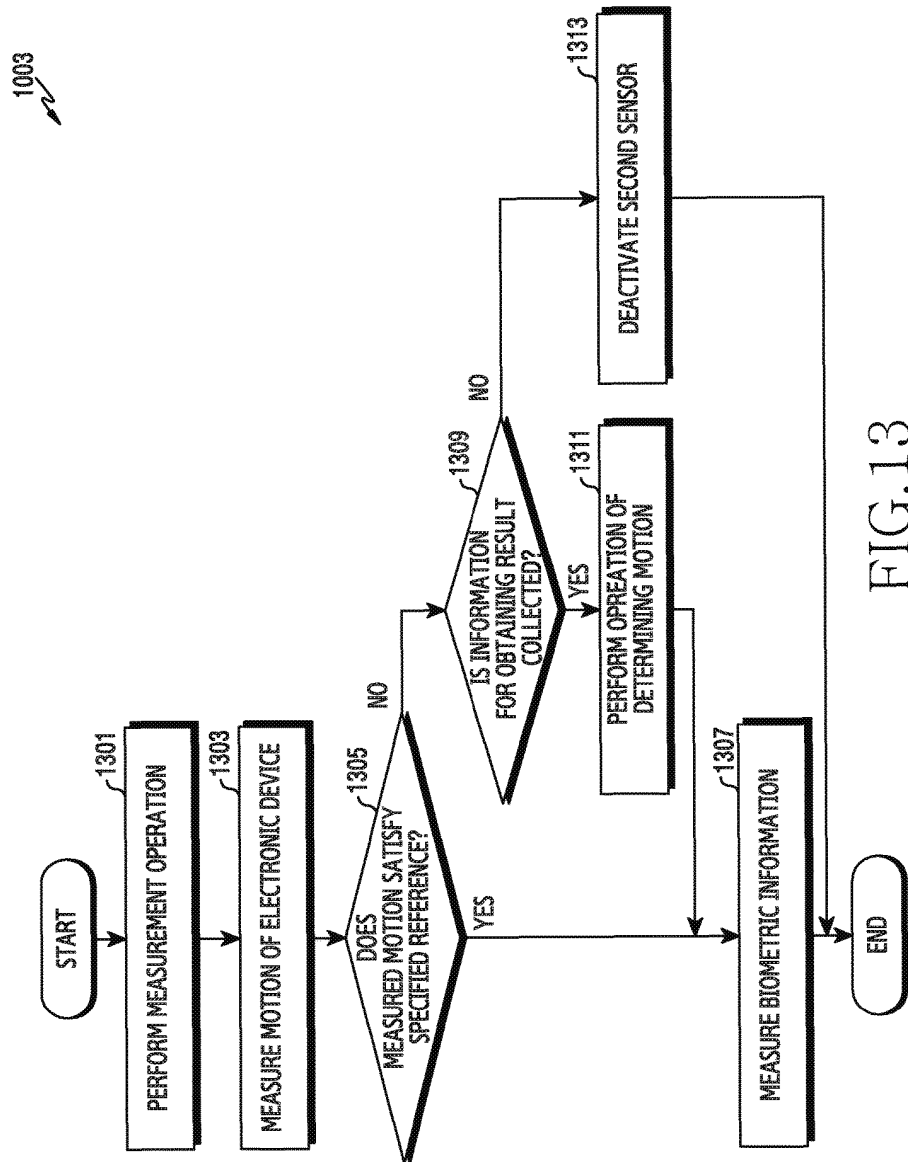


FIG.13

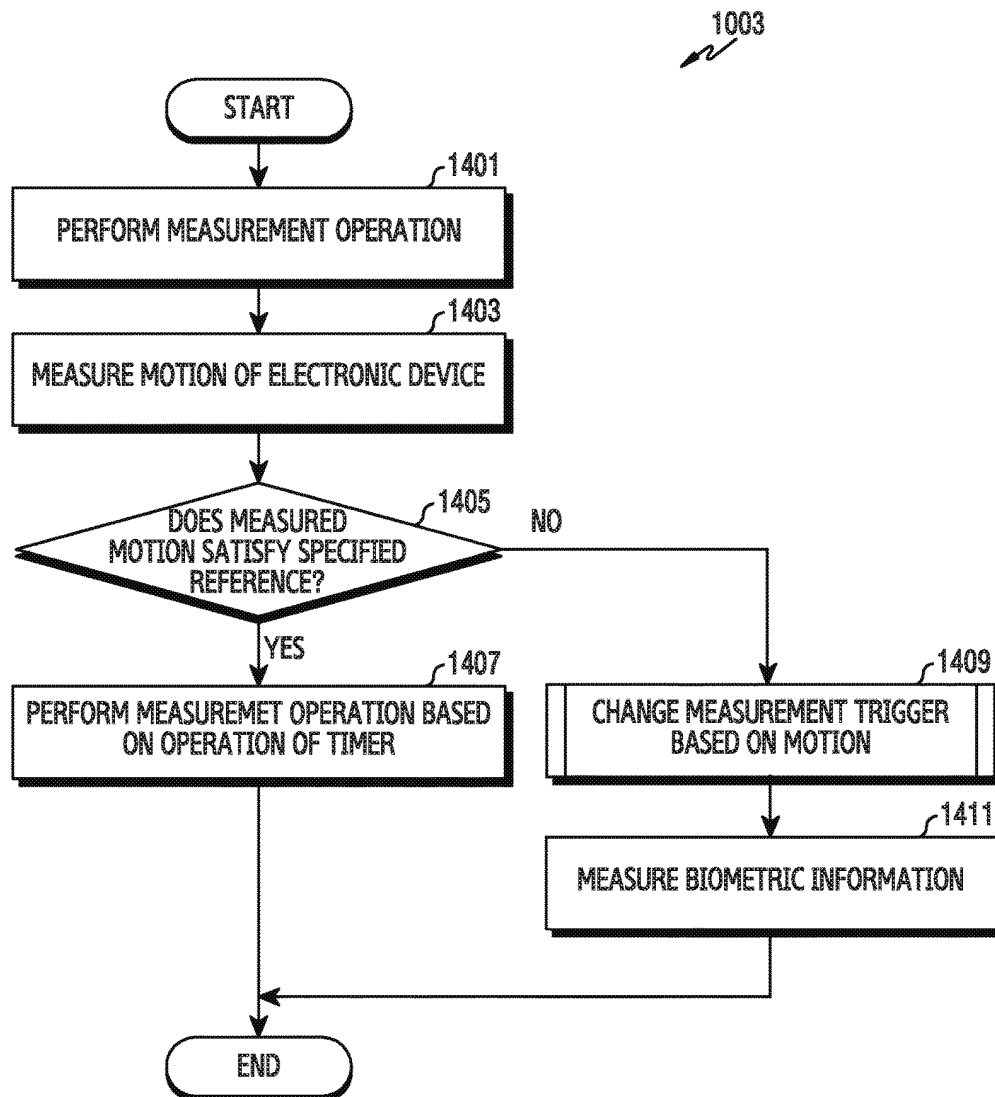


FIG. 14

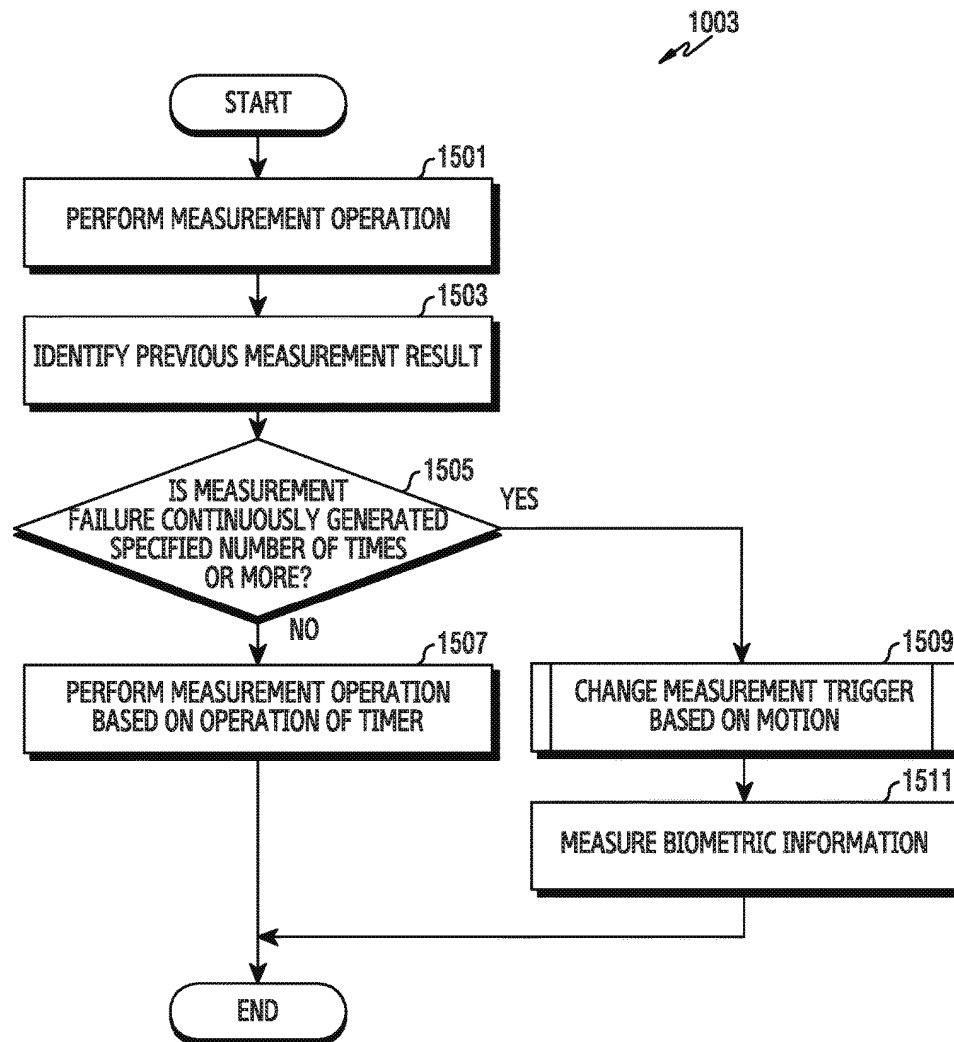


FIG.15

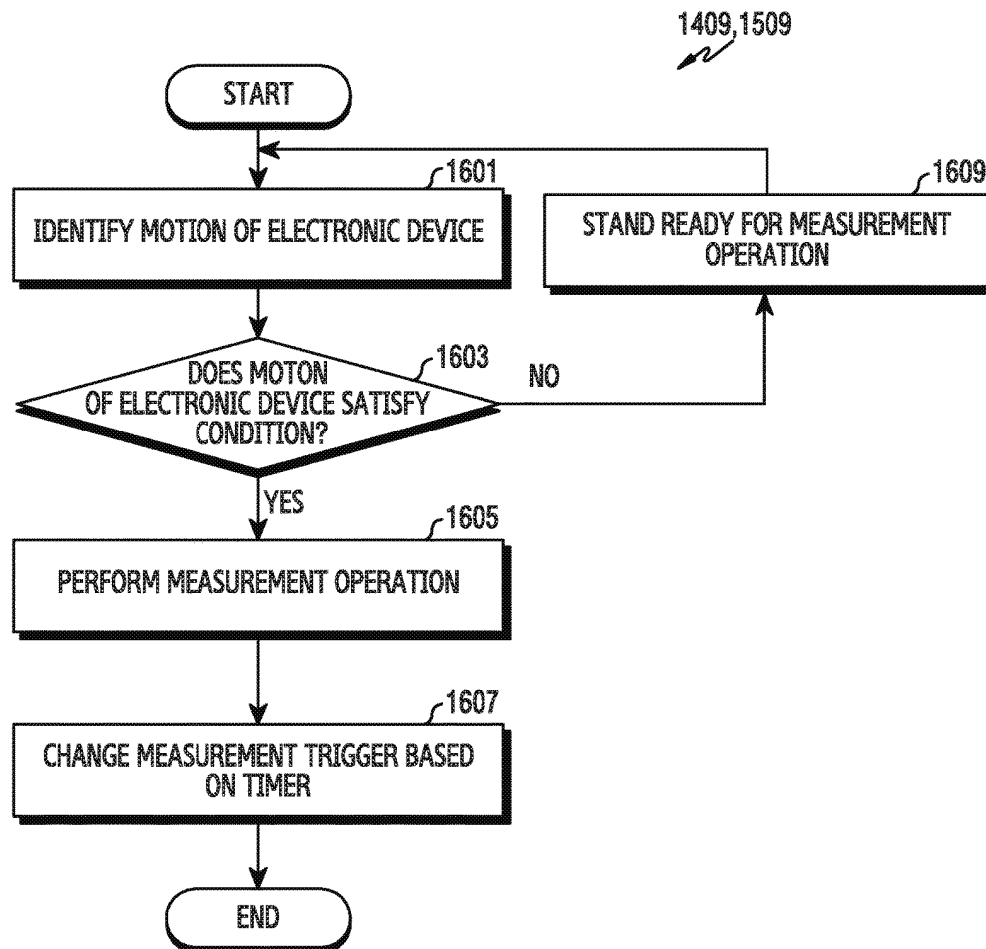


FIG.16

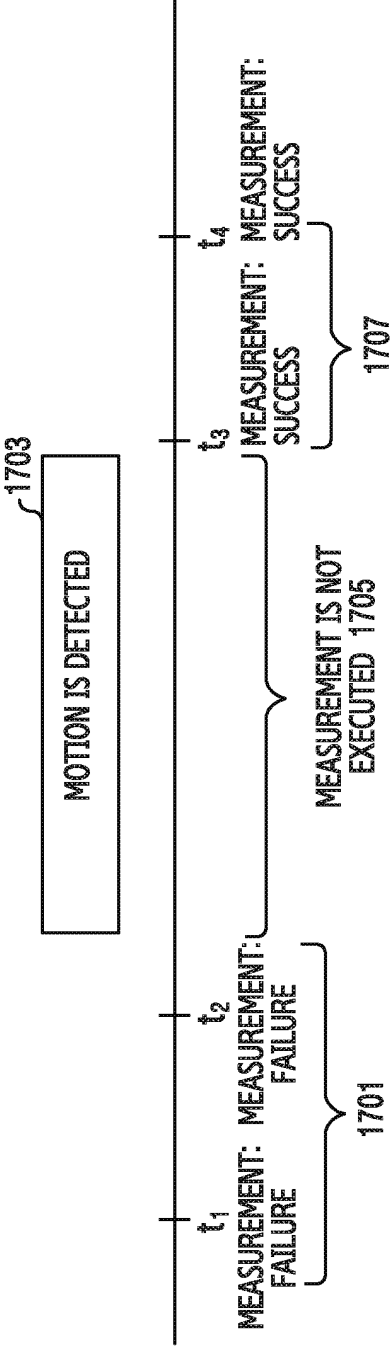


FIG.17

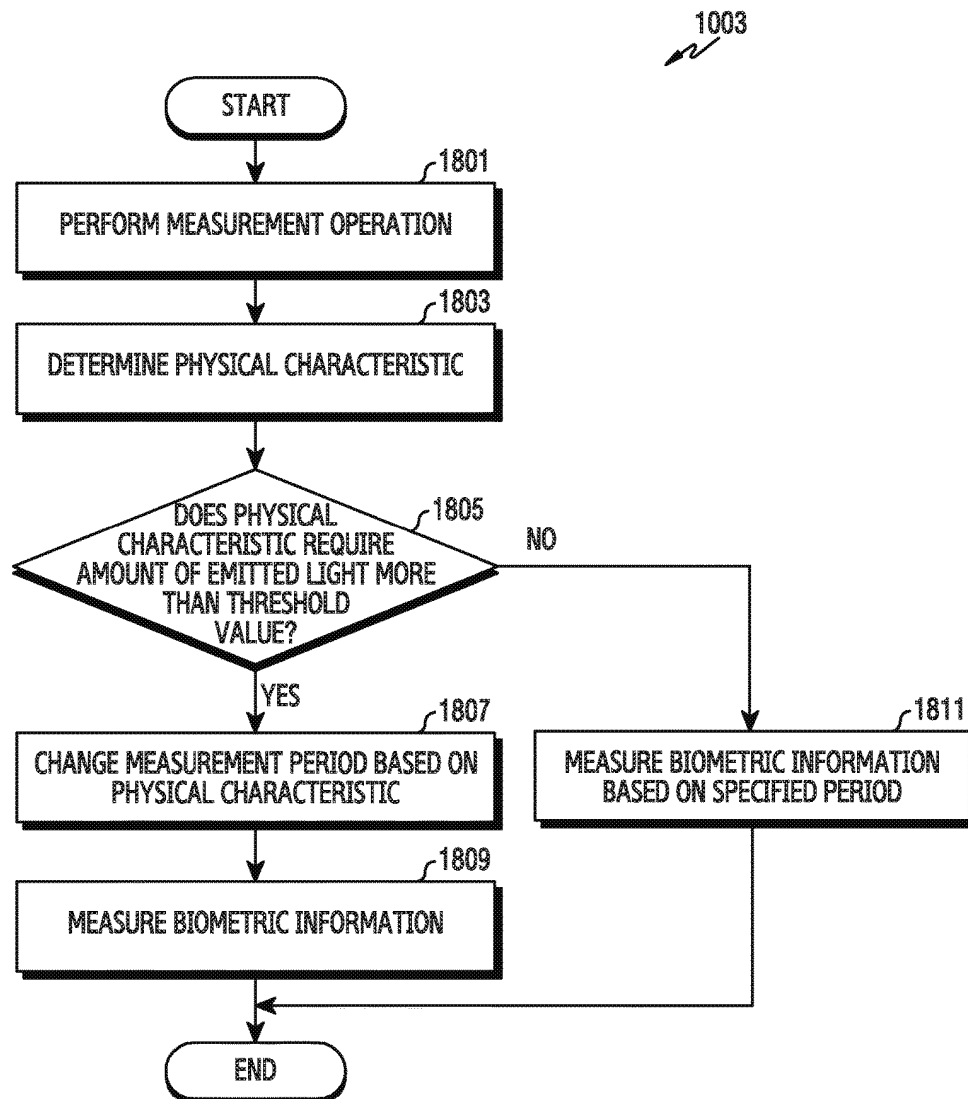


FIG.18

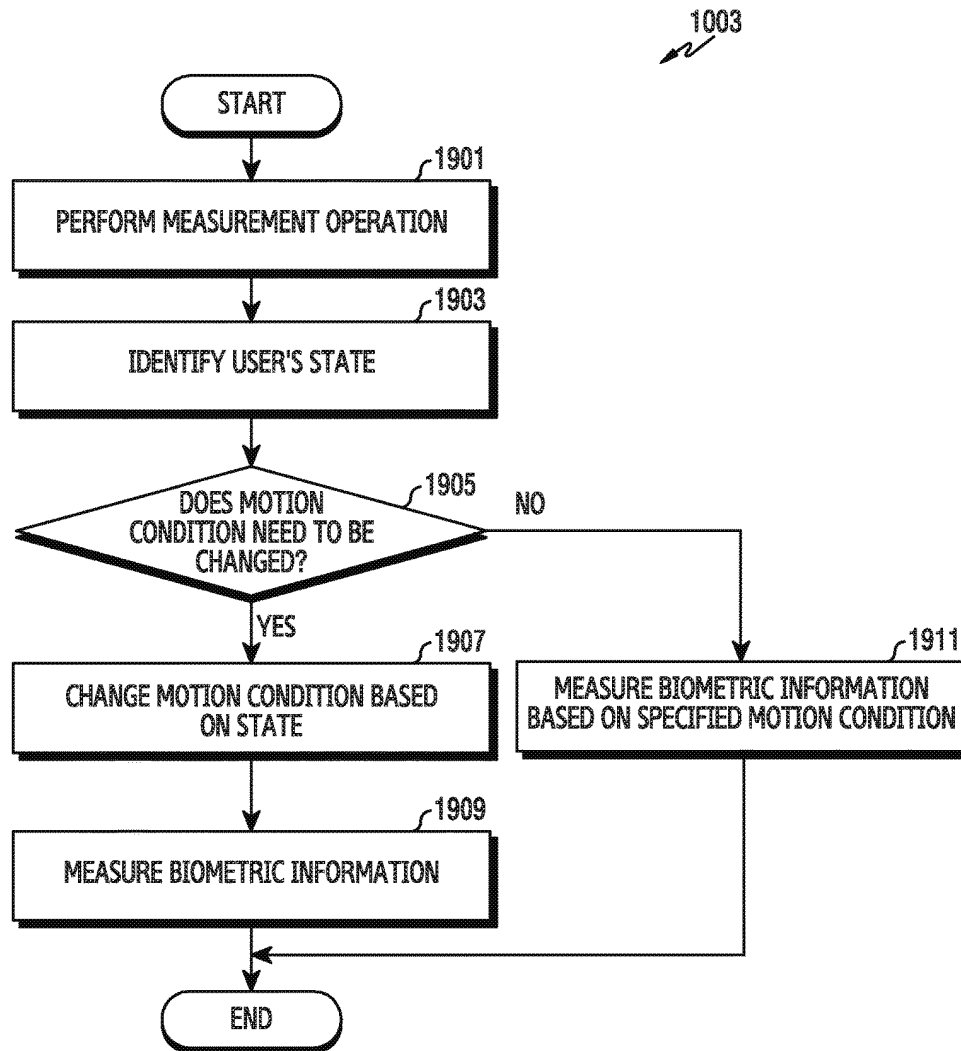


FIG.19

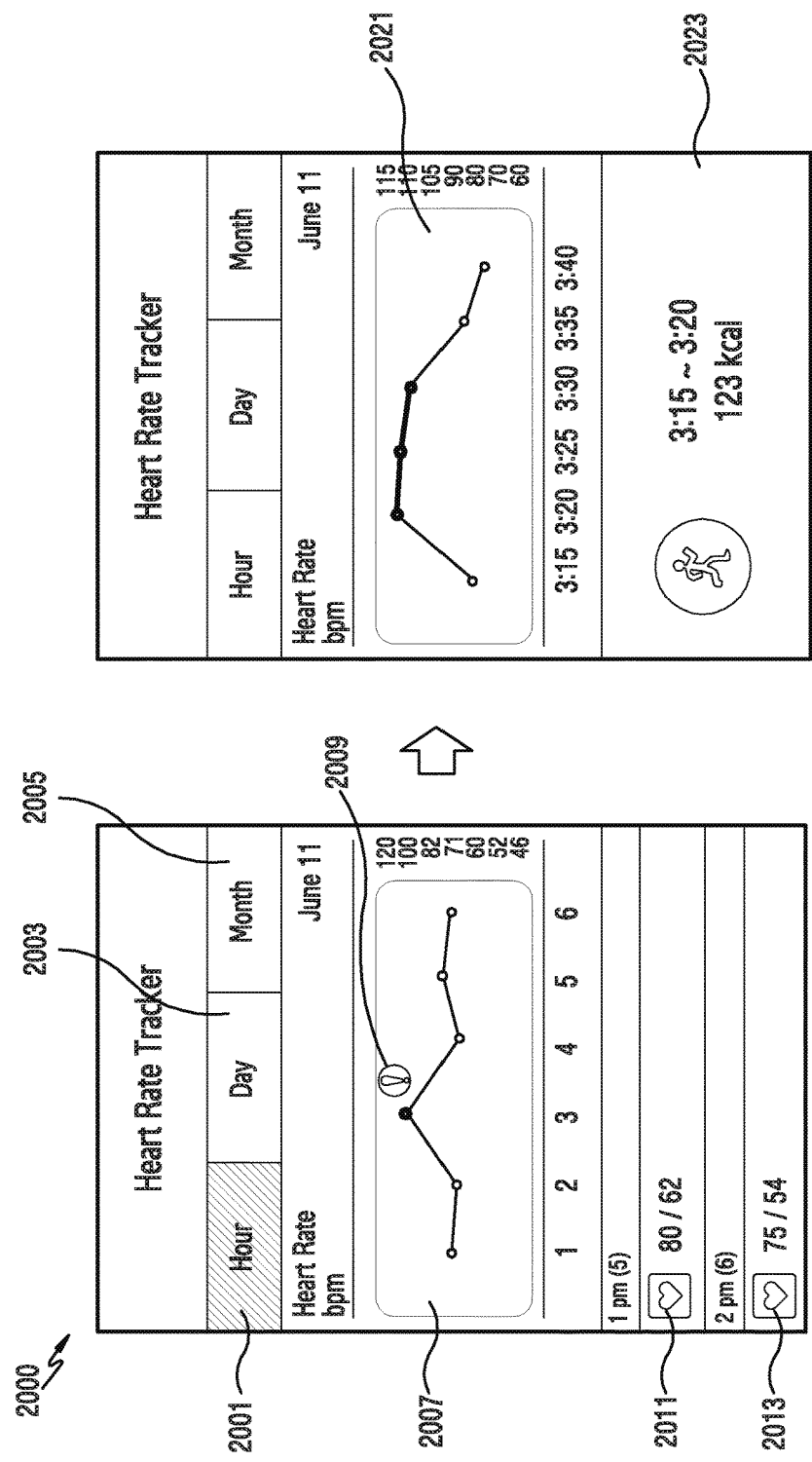


FIG.20

1

METHOD FOR DETECTING BIOMETRIC INFORMATION AND ELECTRONIC DEVICE USING SAME

CROSS-REFERENCE TO RELATED APPLICATION(S)

This application claims the benefit under 35 U.S.C. § 119(a) of a Korean patent application filed on Aug. 12, 2015 in the Korean Intellectual Property Office and assigned Serial number 10-2015-0113972, the entire disclosure of which is hereby incorporated by reference.

TECHNICAL FIELD

The present disclosure relates to an electronic device. More particularly, the present disclosure relates to a device and method for detecting biometric information.

BACKGROUND

Electronic devices have become smaller and slimmer, and have evolved to be easily carried while performing the same or more diverse functions. Although compact electronic devices are generally carried while being received in a user's pocket, they may be worn on the wrist, the head, or the arm of a human body.

In addition, electronic devices may be equipped with biometric sensors (e.g., health care sensors) in order to measure biometric information (e.g., health data) that individuals manage. For example, the biometric information may include blood pressure, a heart rate, an electrocardiogram, skin moisture, saturation of peripheral oxygen, and the like.

The above information is presented as background information only to assist with an understanding of the present disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the present disclosure.

SUMMARY

Electronic devices may operate biometric sensors (e.g., heart rate sensors) in order to measure biometric information (e.g., a heart rate). The operation of the biometric sensors may consume the battery power of the electronic devices. Accordingly, the electronic devices may have difficulty in measuring biometric information for a long period of time (e.g., while users wear the electronic devices).

In another example, when electronic devices measure users' biometric information, the measurement of biometric information is impossible, or the accuracy of the measured biometric information may be deteriorated according to the motion of the electronic devices.

Aspects of the present disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the present disclosure is to provide an electronic device and a method that can selectively operate a biometric sensor based on a selected (or specified) condition.

Another aspect of the present disclosure is to provide an electronic device and a method that can continually (e.g., periodically (e.g., in 1-minute periods)) operate a biometric sensor according to a selected (or specified) period.

2

Another aspect of the present disclosure is to provide an electronic device and a method that can measure biometric information based on the motion of the electronic device.

In accordance with an aspect of the present disclosure, an electronic device is provided. The electronic device includes a housing, a fastener coupled to a part of the housing and detachably attached to a part of a user's body, at least one first sensor configured to detect a motion of the electronic device, at least one second sensor configured to detect a change in the user's body, a memory, and a processor electrically connected to the first sensor, the second sensor, and the memory, wherein the memory may store instructions that configure the processor, when being executed, to determine whether the electronic device is worn on a part of the user's body using the at least one first sensor, acquire information on the motion of the electronic device using at least a part of the at least one first sensor for a second period of time, which is shorter than the first period of time, shortly before and/or immediately after a first period of time, which is selected (or specified), passes from a time when the electronic device is worn on a part of the user's body, and activate (operate or trigger) the at least one second sensor based on at least a part of the information on the motion.

In accordance with another aspect of the present disclosure, a method of measuring biometric information by an electronic device is provided. The method includes determining whether the electronic device is worn on a part of a user's body using at least one first sensor, activating (operating or triggering) at least one second sensor based on at least a part of the a motion of the electronic device from a time when the electronic device is worn on a part of the user's body to the time when a first period of time selected (or specified) passes, and measuring biometric information based on the at least one second sensor.

Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the present disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and other aspects, features, and advantages of certain embodiments of the present disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

FIG. 1 is a diagram illustrating an electronic device in a network environment according to various embodiments of the present disclosure;

FIG. 2 is a block diagram of an electronic device according to various embodiments of the present disclosure;

FIG. 3 is a block diagram of a program module according to various embodiments of the present disclosure;

FIGS. 4A, 4B, and 4C are views illustrating an electronic device according to various embodiments of the present disclosure;

FIG. 5A is a diagram illustrating the configuration of a processor according to various embodiments of the present disclosure;

FIG. 5B is a view illustrating an operation of measuring biometric information by an electronic device according to various embodiments of the present disclosure;

FIG. 6 is a flowchart illustrating a biometric information measurement operation of an electronic device according to various embodiments of the present disclosure;

3

FIG. 7 is a flowchart illustrating a performance procedure of a method of determining the state of an electronic device according to various embodiments of the present disclosure;

FIG. 8 is a flowchart illustrating the performance procedure of a method of operating a sensor by an electronic device according to various embodiments of the present disclosure;

FIG. 9 is a flowchart illustrating the performance procedure of a method of measuring biometric information by an electronic device according to various embodiments of the present disclosure;

FIG. 10 is a flowchart illustrating the performance procedure of a method of measuring biometric information by an electronic device according to various embodiments of the present disclosure;

FIG. 11 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device according to various embodiments of the present disclosure;

FIG. 12 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device according to various embodiments of the present disclosure;

FIG. 13 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device according to various embodiments of the present disclosure;

FIG. 14 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device according to various embodiments of the present disclosure;

FIG. 15 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device according to various embodiments of the present disclosure;

FIG. 16 is a flowchart illustrating the performance procedure of a method of changing a measurement trigger according to various embodiments of the present disclosure;

FIG. 17 is a view illustrating an operation of changing a measurement trigger by an electronic device according to various embodiments of the present disclosure;

FIG. 18 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device according to various embodiments of the present disclosure;

FIG. 19 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device according to various embodiments of the present disclosure; and

FIG. 20 is a view illustrating a screen configuration of an electronic device according to various embodiments of the present disclosure.

Throughout the drawings, it should be noted that like reference numbers are used to depict the same or similar elements, features, and structures.

DETAILED DESCRIPTION

The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the present disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without depart-

4

ing from the scope and spirit of the present disclosure. In addition, descriptions of well-known functions and constructions are omitted for clarity and conciseness.

The terms and words used in the following description and claims are not limited to the bibliographical meanings, but, are merely used by the inventor to enable a clear and consistent understanding of the present disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the present disclosure is provided for illustration purpose only and not for the purpose of limiting the present disclosure as defined by the appended claims and their equivalents.

It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a component surface” includes reference to one or more of such surfaces.

The terms “have”, “may have”, “include”, or “may include” used in the various embodiments of the present disclosure indicate the presence of disclosed corresponding functions, operations, elements, and the like, and do not limit additional one or more functions, operations, elements, and the like. In addition, it should be understood that the terms “include” or “have” used in the various embodiments of the present disclosure are to indicate the presence of features, numbers, operations, elements, parts, or a combination thereof described in the specifications, and do not preclude the presence or addition of one or more other features, numbers, operations, elements, parts, or a combination thereof.

The terms “A or B”, “at least one of A or/and B” or “one or more of A or/and B” used in the various embodiments of the present disclosure include any and all combinations of words enumerated with it. For example, “A or B”, “at least one of A and B” or “at least one of A or B” means (1) including at least one A, (2) including at least one B, or (3) including both at least one A and at least one B.

Although the term such as “first” and “second” used in various embodiments of the present disclosure may modify various elements of various embodiments, these terms do not limit the corresponding elements. For example, these terms do not limit an order and/or importance of the corresponding elements. These terms may be used for the purpose of distinguishing one element from another element. For example, a first user device and a second user device all indicate user devices and may indicate different user devices. For example, a first element may be named a second element without departing from the scope of right of various embodiments of the present disclosure, and similarly, a second element may be named a first element.

It will be understood that when an element (e.g., first element) is “connected to” or “(operatively or communicatively) coupled with/to” to another element (e.g., second element), the element may be directly connected or coupled to another element, and there may be an intervening element (e.g., third element) between the element and another element. To the contrary, it will be understood that when an element (e.g., first element) is “directly connected” or “directly coupled” to another element (e.g., second element), there is no intervening element (e.g., third element) between the element and another element.

The expression “configured to (or set to)” used in various embodiments of the present disclosure may be replaced with “suitable for”, “having the capacity to”, “designed to”, “adapted to”, “made to”, or “capable of” according to a situation. The term “configured to (set to)” does not necessarily mean “specifically designed to” in a hardware level.

Instead, the expression “apparatus configured to . . .” may mean that the apparatus is “capable of . . .” along with other devices or parts in a certain situation. For example, “a processor configured to (set to) perform A, B, and C” may be a dedicated processor, e.g., an embedded processor, for performing a corresponding operation, or a generic-purpose processor, e.g., a central processing unit (CPU) or an application processor (AP), capable of performing a corresponding operation by executing one or more software programs stored in a memory device.

Further, all the terms used herein, including technical and scientific terms, should be interpreted to have the same meanings as commonly understood by those skilled in the art to which the present disclosure pertains, and should not be interpreted to have ideal or excessively formal meanings unless explicitly defined in various embodiments of the present disclosure.

An electronic device according to various embodiments of the present disclosure may be a device. For example, the electronic device according to various embodiments of the present disclosure may include at least one of: a smart phone; a tablet personal computer (PC); a mobile phone; a video phone; an e-book reader; a desktop PC; a laptop PC; a netbook computer; a workstation, a server, a personal digital assistant (PDA); a portable multimedia player (PMP); a Moving Picture Experts Group phase 1 or phase 2 (MPEG-1 or MPEG-2) audio layer 3 (MP3) player; a mobile medical device; a camera; or a wearable device (e.g., a head-mount-device (HMD), an electronic glasses, an electronic clothing, an electronic bracelet, an electronic necklace, an electronic appcessory, an electronic tattoo, a smart mirror, or a smart watch).

In other embodiments, an electronic device may be a smart home appliance. For example, of such appliances may include at least one of: a television (TV); a digital versatile disc (DVD) player; an audio component; a refrigerator; an air conditioner; a vacuum cleaner; an oven; a microwave oven; a washing machine; an air cleaner; a set-top box; a home automation control panel; a security control panel; a TV box (e.g., Samsung HomeSync™, Apple TV™, or Google TV™); a game console (e.g., Xbox™ Play Station™); an electronic dictionary; an electronic key; a camcorder; or an electronic frame.

In other embodiments, an electronic device may include at least one of: a medical equipment (e.g., a mobile medical device (e.g., a blood glucose monitoring device, a heart rate monitor, a blood pressure monitoring device or a temperature meter), a magnetic resonance angiography (MRA) machine, a magnetic resonance imaging (MRI) machine, a computed tomography (CT) scanner, or an ultrasound machine); a navigation device; a global positioning system (GPS) receiver; an event data recorder (EDR); a flight data recorder (FDR); an in-vehicle infotainment device; an electronic equipment for a ship (e.g., ship navigation equipment and/or a gyrocompass); an avionics equipment; a security equipment; a head unit for vehicle; an industrial or home robot; an automatic teller’s machine (ATM) of a financial institution, point of sale (POS) device at a retail store, or an internet of things device (e.g., a lightbulb, various sensors, an electronic meter, a gas meter, a sprinkler, a fire alarm, a thermostat, a streetlamp, a toaster, a sporting equipment, a hot-water tank, a heater, or a boiler and the like).

In certain embodiments, an electronic device may include at least one of: a piece of furniture or a building/structure; an electronic board; an electronic signature receiving device; a projector; and various measuring instruments (e.g., a water meter, an electricity meter, a gas meter, or a wave meter).

An electronic device according to various embodiments of the present disclosure may also include a combination of one or more of the above-mentioned devices. Further, it will be apparent to those skilled in the art that an electronic device according to various embodiments of the present disclosure is not limited to the above-mentioned devices.

FIG. 1 is a view illustrating a network environment 100 including an electronic device 101 according to various embodiments of the present disclosure.

Referring to FIG. 1, the electronic device 101 may include a bus 110, a processor 120, a memory 130, an input/output (I/O) interface 150, a display 160, and a communication interface 170.

The bus 110 may be a circuit for connecting the above-described elements (e.g., the processor 120, the memory 130, the I/O interface 150, the display 160 or the communication interface 170, etc.) with each other, and transferring communication (e.g., a control message) between the above-described elements.

The processor 120 may include a CPU, a communication processor (CP), a graphics processing unit (GPU).

The processor 120 may receive, for example, an instruction from the above-described other elements (e.g., the memory 130, the I/O interface 150, the display 160, or the communication interface 170, etc.) via the bus 110, decipher the received instruction, and execute an operation or a data process corresponding to the deciphered instruction.

The memory 130 may include any suitable type of volatile or non-volatile memory. The memory 130 may store an instruction or data received from the processor 120 or other elements (e.g., the I/O interface 150, the display 160, or the communication interface 170, etc.), or generated by the processor 120 or other elements. The memory 130 may include, for example, programming modules 140 such as a kernel 141, a middleware 143, an application programming interface (API) 145, or an application 147. The each of the programming modules may be configured using a software, a firmware, a hardware, or a combination of two or more of these.

The kernel 141 may control or manage system resources (e.g., the bus 110, the processor 120, or the memory 130, etc.) used for executing an operation or a function implemented in the rest of the programming modules, for example, the middleware 143, the API 145, or the application 147. Also, the kernel 141 may provide an interface for allowing the middleware 143, the API 145, or the application 147 to access an individual element of the electronic device 101 and control or manage the same.

The middleware 143 may perform a mediation role so that the API 145 or the application 147 may communicate with the kernel 141 to give and take data. Also, in connection with task requests received from the application 147, the middleware 143 may perform a control (e.g., scheduling or load balancing) for a task request using, for example, a method of assigning priority that may use a system resource (e.g., the bus 110, the processor 120, or the memory 130, etc.) of the electronic device 101 to at least one application 134.

The API 145 is an interface for allowing the application 147 to control a function provided by the kernel 141 or the middleware 143, and may include at least one interface or function (e.g., an instruction) for file control, window control, image processing, or character control, etc.

The I/O interface 150 may transfer an instruction or data input from a user via an I/O unit (e.g., a sensor, a keyboard, or a touchscreen) to the processor 120, the memory 130, or the communication interface 170 via the bus 110, for example. For example, the I/O interface 150 may provide

data regarding a user's touch input via the touchscreen to the processor 120. Also, the I/O interface 150 may, for example, output an instruction or data received via the bus 110 from the processor 120, the memory 130, or the communication interface 170 via the I/O unit (e.g., a speaker or a display). For example, the I/O interface 150 may output voice data processed by the processor 120 to a user via a speaker.

The display 160 may include, for example, a liquid crystal display (LCD), a light emitting diode (LED) display, an organic LED (OLED) display, a micro electro mechanical system (MEMS) display, or an electronic paper display. The display 160 may display various types of contents (for example, text, images, videos, icons, or symbols) for users. The display 160 may include a touch screen, and may receive, for example, a touch, gesture, proximity, or hovering input by using an electronic pen or a part of the user's body.

The communication interface 170 may connect communication (e.g., communication 164) between the electronic device 101 and an external device (for example, the electronic device 102, electronic device 104, or the server 106). For example, the communication interface 170 may be connected to a network 162 through wireless communication or wired communication, and may communicate with an external device.

The wireless communication may use at least one of, for example, long term evolution (LTE), LTE-advanced (LTE-A), code division multiple access (CDMA), wideband CDMA (WCDMA), universal mobile telecommunications system (UMTS), wireless broadband (WiBro), and global system for mobile communications (GSM) as a cellular communication protocol.

The wired communication may include, for example, at least one of universal serial bus (USB), high definition multimedia interface (HDMI), recommended standard 232 (RS-232), and plain old telephone service (POTS).

The network 162 may include at least one of communication networks such as a computer network (for example, a local area network (LAN) or a wide area network (WAN)), the internet, and a telephone network.

The electronic devices 102 and 104 may be devices of the same type as that the electronic device 101 or devices of different types from that of the electronic device 101. According to an embodiment, the server 106 may include a group of one or more servers. According to various embodiments, all or some of the operations executed in the electronic device 101 may be carried out in another electronic device or a plurality of electronic devices (for example, the electronic device 102 or 104 and the server 106). According to an embodiment, when the electronic device 101 should perform some functions or services automatically or by a request, the electronic device 101 may make a request for performing at least some functions related to the functions or services to another device (for example, the electronic device 102 or 104, or the server 106) instead of performing the functions or services by itself or additionally. The electronic device (for example, the electronic device 102 or 104, or the server 106) may carry out the functions requested by the electronic device 101 or additional functions and provide results thereof to the electronic device 101. The electronic device 101 may provide the requested functions or services to another electronic device based on the received results or after additionally processing the received results. To this end, for example, cloud computing, distributed computing, or client-server computing technology may be used.

FIG. 2 is a block diagram 200 illustrating an electronic device 201 according to various embodiments of the present disclosure. The electronic device 201 may configure, for example, all or a portion of the electronic device 101 illustrated in FIG. 1.

Referring to FIG. 2, the electronic device 201 may include one or more APs 210, a communication module 220, a subscriber identification module (SIM) card 224, a memory 230, a sensor module 240, an input unit 250, a display 260, an interface 270, an audio module 280, a camera module 291, a power management module 295, a battery 296, an indicator 297, or a motor 298.

The AP 210 may drive an OS or an application to control a plurality of hardware or software elements connected to the AP 210, and perform various data processes including multimedia data and operations. The AP 210 may be implemented, for example, as a system on chip (SoC). According to an embodiment, the AP 210 may further include at least one of a GPU or image signal processor (ISP). According to an embodiment, the AP 210 may be implemented to include at least a portion (e.g., the cellular module 221) of the above-described elements. Also, the AP 210 may store data received from at least one of other elements or generated by at least one of other elements in a non-volatile memory.

The communication module 220 (e.g., the communication interface 170) may perform data transmission/reception in communication between the electronic device 201 (e.g., the electronic device 101) and other electronic devices (e.g., the electronic device 104 or the server 106) connected via a network. According to an embodiment, the communication module 220 may include a cellular module 221, a Wi-Fi module 223, a Bluetooth (BT) module 225, a GPS module 227, a near field communication (NFC) module 228, and a radio frequency (RF) module 229.

The cellular module 221 may provide voice communication, image communication, a short message service, or an internet service, etc. via a communication network (e.g., LTE, LTE-A, CDMA, WCDMA, UMTS, WiBro, or GSM, etc.). Also, the cellular module 221 may perform discrimination and authentication of an electronic device within a communication network using, for example, a subscriber identify module (e.g., a SIM card 224). According to an embodiment, the cellular module 221 may perform at least a portion of functions that may be provided by the AP 210. According to an embodiment, the cellular module 221 may include a CP. Also, the cellular module 221 may be, for example, implemented as an SoC. Though elements such as the cellular module 221 (e.g., a CP), the memory 230, or the power management module 295, etc. are illustrated as elements separated from the AP 210 in FIG. 2, according to an embodiment, the AP 210 may be implemented to include at least a portion (e.g., the cellular module 221) of the above-described elements.

Each of the Wi-Fi module 223, the BT module 225, the GPS module 227, or the NFC module 228 may include, for example, a processor for processing data transmitted/received via a relevant module. Though the cellular module 221, the Wi-Fi module 223, the BT module 225, the GPS module 227, or the NFC module 228 are illustrated as separate blocks in FIG. 2, according to an embodiment, at least a portion (e.g., two or more elements) of the cellular module 221, the Wi-Fi module 223, the BT module 225, the GPS module 227, or the NFC module 228 may be included in one integrated circuit (IC) or an IC package. For example, at least a portion (e.g., a CP corresponding to the cellular module 221 and a Wi-Fi processor corresponding to the Wi-Fi module 223) of processors corresponding to each of

the cellular module 221, the Wi-Fi module 223, the BT module 225, the GPS module 227, or the NEC module 228 may be implemented as one SoC.

The RF module 229 may perform transmission/reception of data, for example, transmission/reception of an RF signal. The RF module 229 may include, for example, a transceiver, a power amp module (PAM), a frequency filter, or a low noise amplifier (LNA), etc., though not shown. Also, the RF module 229 may further include a part for transmitting/receiving an electromagnetic wave on a free space in wireless communication, for example, a conductor or a conducting line, etc. Though FIG. 2 illustrates the cellular module 221, the Wi-Fi module 223, the BT module 225, the GPS module 227, and the NEC module 228 share one RF module 229, according to an embodiment, at least one of the cellular module 221, the Wi-Fi module 223, the BT module 225, the GPS module 227, or the NEC module 228 may perform transmission/reception of an RF signal via a separate RF module.

The SIM card 224 may be a card including a subscriber identify module, and may be inserted into a slot formed in a specific position of the electronic device. The SIM card 224 may include unique identify information (e.g., integrated circuit card identifier (ICCID)) or subscriber information (e.g., international mobile subscriber identity (IMSI)).

The memory 230 (e.g., the memory 20) may include an internal memory 232 or an external memory 234. The internal 232 may include, for example, at least one of a volatile memory (e.g., dynamic random access memory (DRAM), static RAM (SRAM), synchronous dynamic RAM (SDRAM)) and a non-volatile memory (e.g., one time programmable read only memory (OTPROM), programmable ROM (PROM), erasable and programmable ROM (EPROM), electrically erasable and programmable ROM (EEPROM), mask ROM, flash ROM, NAND flash memory, NOR flash memory, etc.).

According to an embodiment, the internal memory 232 may be a solid state drive (SSD). The external memory 234 may further include a flash drive, for example, compact flash (CF), secure digital (SD), micro-SD, mini-SD, extreme digital (xD), or a memory stick. The external memory 234 may be functionally connected with the electronic device 201 via various interfaces. According to an embodiment, the electronic device 201 may further include a storage device (or a storage medium) such as a hard drive.

The sensor module 240 may measure a physical quantity or detect an operation state of the electronic device 201, and convert the measured or detected information to an electric signal. The sensor module 240 may include, for example, at least one of a gesture sensor 240A, a gyro sensor 240B, an atmospheric pressure sensor 240C, a magnetic sensor 240D, an acceleration sensor 240E, a grip sensor 240F, a proximity sensor 240G, a color sensor 240H (e.g., red, green, blue (RGB) sensor), a biometric sensor 240I, a temperature/humidity sensor 240J, an illuminance sensor 240K, or an ultra violet (UV) sensor 240M. Additionally or alternatively, the sensor module 240 may include, for example, an E-nose sensor (not shown), an electromyography (EMG) sensor (not shown), an electroencephalogram (EEG) sensor (not shown), an electrocardiogram (ECG) sensor (not shown), an infrared (IR) sensor (not shown), an iris sensor (not shown), or a fingerprint sensor (not shown), etc. The sensor module 240 may further include a control circuit for controlling at least one sensor belonging thereto.

The input unit 250 may include a touch panel 252, a (digital) pen sensor 254, a key 256, or an ultrasonic input

unit 258. The touch panel 252 may recognize a touch input using at least one of capacitive, resistive, infrared, or ultrasonic methods. Also, the touch panel 252 may further include a control circuit. A capacitive touch panel may perform detection by a physical contact or proximity recognition. The touch panel 252 may further include a tactile layer. In this case, the touch panel 252 may provide a tactile reaction to a user.

The (digital) pen sensor 254 may be implemented using, for example, a method which is the same as or similar to receiving a user's touch input, or using a separate sheet for detection. The key 256 may include, for example, a physical button, an optical key or keypad. The ultrasonic input unit 258 is a unit for recognizing data by detecting a sound wave using a microphone (e.g., a microphone 288) in the electronic device 201 via an input tool generating an ultrasonic signal, and enables wireless recognition. According to an embodiment, the electronic device 201 may receive a user input from an external device (e.g., a computer or a server) connected to the communication module 220 using the communication module 220.

The display 260 (e.g., the display 160) may include a panel 262, a hologram device 264, or a projector 266. The panel 262 may be, for example, an LCD, or an active-matrix OLED (AM-OLED), etc. The panel 262 may be implemented, for example, such that it is flexible, transparent, or wearable. The panel 262 may be configured as one module together with the touch panel 252. The hologram device 264 may show a three-dimensional image in the air using interferences of light. The projector 266 may project light onto a screen to display an image. The screen may be positioned, for example, inside or outside the electronic device 201. According to an embodiment, the display 260 may further include a control circuit for controlling the panel 262, the hologram device 264, or the projector 266.

The interface 270 may include, for example, a high-definition multimedia interface (HDMI) 272, a USB 274, an optical interface 276, or a D-subminiature (D-sub) 278. The interface 270 may be included, for example, in the communication interface 170 illustrated in FIG. 1. Additionally or alternatively, the interface 270 may include, for example, a mobile high-definition link (MHL) interface, an SD card/multi-media card (MMC) interface, or an infrared data association (IrDA) standard interface.

The audio module 280 may convert a sound and an electric signal in dual directions. At least a partial element of the audio module 280 may be included, for example, in the I/O interface 150 illustrated in FIG. 1. The audio module 280 may process sound information input or output via, for example, a speaker 282, a receiver 284, an earphone 286, or a microphone 288, etc.

The camera module 291 is a device that may shoot a still image and a moving picture. According to an embodiment, the camera module 291 may include one or more image sensors (e.g., a front sensor or a rear sensor), a lens (not shown), an ISP (not shown), or a flash (not shown) (e.g., an LED or xenon lamp).

The power management module 295 may manage power of the electronic device 201. Though not shown, the power management module 295 may include, for example, a power management IC (PMIC), a charger IC, or a battery or a battery or fuel gauge.

The PMIC may be mounted, for example, inside an IC or a SoC semiconductor. A charging method may be classified into a wired charging method and a wireless charging method. The charging IC may charge a battery and prevent introduction of an overvoltage or an overcurrent from a

charger. According to an embodiment, the charging IC may include a charging IC for at least one of the wired charging method and the wireless charging method. The wireless charging method may be, for example, a magnetic resonance method, a magnetic induction method, or an electromagnetic wave method, etc., and may additionally include an additional circuit for wireless charging, for example, a circuit such as a coil loop, a resonance circuit, or a rectifier, etc.

The battery gauge may measure, for example, a remnant of the battery 296, a voltage, a current, or a temperature while charging. The battery 296 may store or generate electricity, and supply power to the electronic device 201 using the stored or generated electricity. The battery 296 may include, for example, a rechargeable battery or a solar battery.

The indicator 297 may display a specific state of the electronic device 201 or a portion thereof (e.g., the AP 210), for example, a booting state, a message state, or a charging state, etc. The motor 298 may convert an electric signal to mechanical vibration. Though not shown, the electronic device 201 may include a processor (e.g., a GPU) for supporting a mobile TV. The processor for supporting the mobile TV may process media data corresponding to standards, for example, such as digital multimedia broadcasting (DMB), digital video broadcasting (DVB), or a media flow, etc.

The aforementioned elements of the electronic device according to various embodiments of the present disclosure may be constituted by one or more components, and the name of the corresponding element may vary with a type of electronic device. The electronic device according to various embodiments of the present disclosure may include at least one of the aforementioned elements. Some elements may be omitted or other additional elements may be further included in the electronic device. Further, some of the components of the electronic device according to the various embodiments of the present disclosure may be combined to form a single entity, and thus, may equivalently execute functions of the corresponding elements prior to the combination.

FIG. 3 is a block diagram 300 of a program module 310 according to various embodiments of the present disclosure.

Referring to FIG. 3, the program module 310 (for example, the programming modules 140) may include an operating system (OS) for controlling resources related to the electronic device (for example, the electronic device 101) and/or various applications (for example, the application 147) executed in the operating system. The OS may be, for example, Android, iOS, Windows, Symbian, Tizen, Bada, or the like.

The program module 310 may include a kernel 320, middleware 330, an API 360, and/or applications 370. At least some of the program module 310 may be preloaded in the electronic device or downloaded from the server.

The kernel 320 (for example, the kernel 141 of FIG. 1) may include, for example, a system resource manager 321 or a device driver 323. The system resource manager 321 may control, allocate, or collect the system resources. According to an embodiment, the system resource manager 321 may include a process management unit, a memory management unit, or a file system management unit. The device driver 323 may include, for example, a display driver, a camera driver, a BT driver, a shared-memory driver, a USB driver, a keypad driver, a Wi-Fi driver, an audio driver, or an inter-process communication (IPC) driver.

The middleware 330 may provide a function used by the applications 370 in common or provide various functions to the applications 370 through the API 360 so that the appli-

cations 370 can efficiently use limited system resources within the electronic device. According to an embodiment, the middleware 330 (for example, the middleware 143) may include, for example, at least one of a runtime library 335, an application manager 341, a window manager 342, a multimedia manager 343, a resource manager 344, a power manager 345, a database manager 346, a package manager 347, a connectivity manager 348, a notification manager 349, a location manager 350, a graphic manager 351, and a security manager 352.

The runtime library 335 may include, for example, a library module that a compiler uses to add new functions through a programming language while the application 370 is executed. The runtime library 335 may perform I/O management, memory management, or a function for an arithmetic function.

The application manager 341 may manage, for example, a life cycle of at least one of the applications 370. The window manager 342 may manage graphical user interface (GUI) resources used by a screen. The multimedia manager 343 may grasp formats used for the reproduction of various media files, and may perform an encoding or decoding of the media file by using a codec suitable for the corresponding format. The resource manager 344 may manage resources such as a source code, a memory, and a storage space of at least one of the applications 370.

The power manager 345 may operate together with a basic input/output system (BIOS) to manage a battery or power and may provide power information used for the operation of the electronic device. The database manager 346 may generate, search for, or change a database to be used by at least one of the applications 370. The package manager 347 may manage the installation or the updating of applications distributed in the form of package file.

The connectivity manager 348 may manage wireless connection of, for example, Wi-Fi or BT. The notification manager 349 can display or notify of an event such as an arrival message, promise, proximity notification, and the like in such a way that does not disturb a user. The location manager 350 may manage location information of the electronic device. The graphic manager 351 may manage graphic effects to be provided to a user and user interfaces related to the graphic effects. The security manager 352 may provide one or more security functions used for system security or user authentication. According to an embodiment, when the electronic device (for example, electronic device 101) has a call function, the middleware 330 may further include a telephony manager for managing a voice call function or a video call function of the electronic device.

The middleware 330 may include a middleware module for forming a combination of various functions of the aforementioned components. The middleware 330 may provide modules specialized according to types of operating systems in order to provide differentiated functions. Further, the middleware 330 may dynamically remove some of the existing components or add new components.

The API 360 (for example, the API 145) is, for example, a set of API programming functions, and a different configuration thereof may be provided according to an operating system. For example, Android or iOS may provide one API set per platform, and Tizen may provide two or more API sets per platform.

The applications 370 (for example, the application 147) may include, for example, one or more applications which can provide functions such as home 371, dialer 372, short message service (SMS)/multimedia message service (MMS) 373, instant message (IM) 374, browser 375, camera 376,

alarm 377, contacts 378, voice dialer 379, email 380, calendar 381, media player 382, album 383, clock 384, health care (for example, measure exercise quantity or blood sugar), or environment information (for example, atmospheric pressure, humidity, or temperature information).

According to an embodiment, the applications 370 may include an application (hereinafter, referred to as an “information exchange application” for convenience of the description) supporting information exchange between the electronic device (for example, the electronic device 101) and an external electronic device. The information exchange application may include, for example, a notification relay application for transferring predetermined information to an external electronic device or a device management application for managing an external electronic device.

For example, the notification relay application may include a function of transferring, to the external electronic device, notification information generated from other applications of the electronic device 101 (for example, an SMS/MMS application, an e-mail application, a health management application, or an environmental information application). Further, the notification relay application may receive notification information from, for example, a control device and provide the received notification information to the user. The device management application may manage (for example, install, delete, or update), for example, a function for at least a part of the external electronic device communicating with the electronic device (for example, turning on/off the external electronic device itself (or some elements thereof) or adjusting brightness (or resolution) of a display), applications executed in the external electronic device, or services provided from the external electronic device (for example, a telephone call service or a message service).

According to an embodiment, the applications 370 may include an application (for example, health management application) designated according to attributes of the external electronic device (for example, attributes of the electronic device such as the type of electronic device which corresponds to a mobile medical device). According to an embodiment, the applications 370 may include an application received from the external electronic devices (for example, the server or the electronic device). According to an embodiment, the applications 370 may include a pre-loaded application or a third party application which can be downloaded from the server. The names of the components of the program module 310 according to the embodiment illustrated in FIG. 3 may vary according to the type of operating system.

According to various embodiments, at least some of the program module 310 may be implemented by software, firmware, hardware, or a combination of two or more thereof. At least some of the program module 310 may be implemented (for example, executed) by, for example, the processor (for example, the application program). At least some of the program module 310 may include, for example, a module, program, routine, sets of instructions, or process for performing one or more functions.

FIGS. 4A, 4B, and 4C are views illustrating an electronic device according to various embodiments of the present disclosure.

A three-dimensional X-Y-Z Cartesian coordinate system is illustrated in FIGS. 4A, 4B, and 4C, where the ‘Z-axis’ may denote the vertical direction (thickness direction) of the main body of the electronic device; the ‘X-axis’ may denote the lateral direction of the main body; and the ‘Y-axis’ may denote the longitudinal direction of the main body.

FIG. 4A is a perspective view of the electronic device according to various embodiments of the present disclosure.

Referring to FIG. 4A, the electronic device 400 (e.g., the electronic device 101 of FIG. 1) may be a wrist-mounted wearable electronic device that can be worn on a user’s wrist. However, the various embodiments of the present disclosure are not limited thereto, and the electronic device may also be one of a bracelet, a strip, a band, an attachment type (band-aid type) band, a belt, ear-mounted earphones, headphones, a clothing type electronic device, a shoe type electronic device, an HMD, a hat type electronic device, a glove type electronic device, a finger-tip type electronic device, a clip type electronic device, an arm band type electronic device, a contact lens device, digital clothing, and a remote controller.

According to various embodiments, the electronic device 400 may include the main body 410 and connection parts 420 (e.g., straps) that are equipped to the main body 410.

According to various embodiments, the electronic device 400 may have a battery (e.g., a rechargeable battery, etc.) therein as a power supply means. For example, the electronic device 400 may be implemented to be selectively mounted on a portable charging cradle in order to recharge the battery.

According to various embodiments, the main body 410 may include a display 415, at least one key button 430 (e.g., a side key K, a touch sensor, or a pressure sensor), a sensor module (e.g., a biometric sensor, etc.), and/or the like. According to various embodiments, the display 415 may include a touch screen to receive a touch input. According to an embodiment, the main body 410 may include the front surface F and the rear surface R which makes contact with the user’s body while the main body 410 is worn on the user’s body. For example, the display 415 may be disposed on the front surface F of the main body 410, and the sensor module may be disposed on the rear surface R of the main body 410. However, the various embodiments of the present disclosure are not limited thereto, and the display 415 or the sensor module may be disposed in various regions of the electronic device 400. For example, the sensor module may also be disposed on the connection parts 420.

According to various embodiments, the main body 410 may have a bar shape and may at least partially have a curvature that corresponds to the user’s body. For example, the main body 410 may have a curved rectangular shape extending in the longitudinal direction (Y-axis direction). According to an embodiment, the main body 410 may have fastening recesses formed on opposite sides thereof with which the connection parts 420 are engaged.

According to various embodiments, the connection parts 420 may be configured to be coupled to, or separated from, the main body 410. According to various embodiments, the connection parts 420 may be formed of a resilient material. Accordingly, it is possible to bring the main body 410 close to the skin of the user’s body. According to an embodiment, the connection parts 420 may have a changeable structure and may be implemented in various designs or colors. Accordingly, the user may change the connection parts 420 according to his/her preference.

According to various embodiments, the connection parts 420 may be coupled to the main body 410 and may include a fastener 422 that enables the connection parts 420 to be detachably attached to a part of the user’s body. According to an embodiment, the electronic device 400 may be worn on the wrist by a method of tying and fastening the connection parts 420 together while the electronic device 400 is placed on the wrist. For example, one of the connection parts 420

15

may have a plurality of openings for wrist adjustment that are formed therein at a constant interval, and the other connection part 420 may include the fastener 422 that is coupled through one of the openings for wrist adjustment.

According to various embodiments, the main body 410 may have a rotary body (not illustrated) (e.g., a rotary input module) mounted thereon for adjusting various user interface environments that are displayed on the display 415. According to various embodiments, the rotary body may have a protrusion shape similar to a crown and may be rotatably disposed on the side surface of the main body 410. According to various embodiments, in a case where the main body 410 has a circular shape, the rotary body may be rotatably mounted on the top, bottom, or side surface of the main body 410. For example, the rotary body may be disposed on a bezel that surrounds the edge of the display 415 that is disposed on the top of the main body 410.

FIG. 4B is a perspective view of the main body of the electronic device 400 according to various embodiments of the present disclosure, and FIG. 4C is a perspective view of the main body of the electronic device 400 according to various embodiments of the present disclosure, when viewed in a different direction.

Referring to FIGS. 4B and 4C, the main body may include a main body housing 440 and a display 450 mounted on the main body housing 440. According to various embodiments, the main body housing 440 may have the front surface F, the rear surface R, and the side surface that connects the front surface F and the rear surface R. However, the various embodiments of the present disclosure are not limited thereto, and the main body housing 440 may be constituted by the front surface F and the rear surface R. Furthermore, the front surface F, the rear surface R, and the side surface of the main body housing 440 may be integrally formed with each other. According to an embodiment, the front surface F and the rear surface R may be configured to have different curvatures.

According to various embodiments, the main body housing 440 may further include a key button (e.g., the key button 430, the side key K, a touch sensor, or a pressure sensor) for inputting various types of information. According to an embodiment, the front surface F may be a surface on which the display 450 is mounted, and the rear surface R may be a wearing surface that makes contact with the user's body. However, the various embodiments of the present disclosure are not limited thereto, and the display 415 or the sensor module may be disposed in various regions of the electronic device 400.

According to various embodiments, the display 450 may be disposed on the front surface F of the main body housing 440. Accordingly, it may be easy for the user to view a screen displayed through the electronic device. The display 450 is illustrated in a shape that reflects the curved surface of the user's body, but may be constituted by a flat panel display (e.g., an LCD, an OLED, etc.), a curved display, or a flexible display. For example, in the specific embodiments of the present disclosure, the main body 410 is illustrated as including a curved display, but the main body 410 may include a flat panel display or a flexible display.

Referring to FIG. 4C, a sensor module 470 (e.g., a biometric sensor) may be disposed on the rear surface R of the main body housing 440 according to an embodiment. For example, the sensor module may have a shape that is brought close to the user's wrist. Accordingly, it is possible to enhance the comfort that the user experiences when wearing the electronic device.

16

The sensor module 470 (e.g., the sensor module 240 or the biometric sensor) provided in the main body housing 440 may include at least one of an acceleration sensor, a gyro sensor, a terrestrial magnetism sensor, a heart rate sensor, a proximity sensor, an optical sensor, a galvanic skin response (GSR) sensor, an ECG sensor, an EMG sensor, a blood glucose sensor, a blood pressure sensor, a pressure sensor, and a temperature sensor. In addition, the sensor module may also include other sensors that detect a sensing value for determining whether the electronic device 400 is worn. According to various embodiments, the sensor module 470 may be provided on the rear surface R of the main body housing 440 in a single module shape that includes: a sensor (e.g., an acceleration sensor, a terrestrial magnetism sensor, or a gyro sensor) for measuring the motion of the electronic device 400; a sensor (e.g., a proximity sensor or a pressure sensor) that is used to determine whether the electronic device 400 is worn on a body; and a biometric sensor (e.g., a heart rate sensor, a blood pressure sensor, or a body temperature sensor) for measuring a biometric signal.

According to various embodiments, connection members 460 (e.g., a charging terminal, the interface 270, etc.) may be disposed on the rear surface R of the main body housing 440.

FIG. 5A is a diagram illustrating the configuration of a processor 500 according to various embodiments of the present disclosure. According to various embodiments, the processor may be a processor (e.g., the processor 120) of an electronic device (e.g., the electronic device 101 or 400).

Referring to FIG. 5A, the processor 500 may include a device state determination module 502, a parameter change module 504, a user state determination module 505, a measurement module 506, and a storage module 508.

According to various embodiments, the device state determination module 502 may determine the state of the electronic device (e.g., a wearing or non-wearing state, a proximity or non-proximity state, or an in-use (holding) or not-in-use (non-holding) state). According to various embodiments, the device state determination module 502 may determine the state of the electronic device based on at least one sensor. According to an embodiment, the device state determination module 502 may determine the state of the electronic device based on information acquired through at least one of an acceleration sensor, a terrestrial magnetism sensor, a gyro sensor, a proximity sensor, an atmospheric pressure sensor (or a pressure sensor), and a temperature sensor.

For example, when the motion of the electronic device is detected through, for example, an acceleration sensor, a terrestrial magnetism sensor, or a gyro sensor, the device state determination module 502 may determine that the electronic device has been worn on a user's body. Furthermore, when the motion of the electronic device is not detected through, for example, an acceleration sensor, a terrestrial magnetism sensor, or a gyro sensor, the device state determination module 502 may determine that the electronic device has not been worn on the user's body.

For example, when an object (e.g., a user) is detected within a selected (or specified) distance (e.g., about 10 cm) from the electronic device through, for example, a proximity sensor, the device state determination module 502 may determine that the electronic device is close to the object (e.g., the electronic device is used or held by the user). Further, when an object (e.g., a user) is not detected within a selected (or specified) distance from the electronic device through, for example, a proximity sensor, the device state determination module 502 may determine that the electronic

device is not close to the object (e.g., the electronic device is not used or held by the user).

For example, when a selected (or specified) pressure or a selected (or specified) temperature (e.g., about 37 degrees Celsius) is detected through, for example, a pressure sensor or a temperature sensor, the device state determination module 502 may determine that the electronic device is used or held by a user. Further, when a selected pressure or a selected temperature is not detected through a pressure sensor or a temperature sensor, the device state determination module 502 may determine that the electronic device is not used or held by the user.

According to various embodiments, the device state determination module 502 may determine the state of the electronic device based on whether the electronic device is electrically or mechanically connected or not. For example, in a case where the electronic device is a pair of smart eyeglasses, a switch circuit included in the smart eyeglass temples may be connected when the smart eyeglasses are worn on a user's head. Accordingly, the device state determination module 502 may determine the state of the electronic device to be a wearing state. Further, in a case where the smart eyeglasses are not worn on the user's head, the switch circuit included in the smart eyeglass temples may not be connected. Accordingly, the device state determination module 502 may determine the state of the electronic device to be a non-wearing state.

For example, in a case where the electronic device is a smart watch, the device state determination module 502 may determine the state of the electronic device to be a wearing state when the buckle of the smart watch is connected. Further, the device state determination module 502 may determine the state of the electronic device to be a non-wearing state when the buckle of the smart watch is not connected.

According to various embodiments, the user state determination module 505 may determine a user's state (e.g., a sleep state). According to an embodiment, the user state determination module 505 may determine the user's state based on the user's pattern. For example, the user's pattern may include the user's motion pattern, a life pattern relating to a specified schedule, etc.

According to various embodiments, the user state determination module 505 may determine, for example, the user's motion, the strength of the motion, etc. when biometric information is measured. For example, the user state determination module 505 may determine the user's state to be a sleep state when the user's motion strength is lower than a selected (or specified) strength. Further, the user state determination module 505 may determine the user's state to be an active state (e.g., a state in which the user does not sleep) when the user's motion strength is higher than, or equal to, the selected (or specified) strength. According to an embodiment, the user state determination module 505 may provide information on the user's motion to the measurement module 506 or the parameter change module 504 in order to change a biometric information measurement method.

According to various embodiments, the user state determination module 505 may determine the user's state based on schedule information stored in the electronic device. For example, a first time (e.g., from about 11:00 P.M. to about 8:00 A.M.) may be specified as a 'sleeping time,' and a second time may be specified as an 'active time' (e.g., from about 8:00 A.M. to about 11:00 P.M.) in the schedule stored in the electronic device. In this case, the user state determination module 505 may determine the user's state at the first

time to be a 'sleep state' based on the schedule. Further, the user state determination module 505 may determine the user's state at the second time to be an 'active state' based on the schedule.

According to various embodiments, the measurement module 506 may measure a user's biometric information. According to various embodiments, the measurement module 506 may measure biometric information based on at least one sensor. According to an embodiment, the measurement module 506 may measure at least one of a blood pressure, a heart rate, an ECG, skin moisture, saturation of peripheral oxygen, a body temperature, blood glucose, an EEG, an EMG, and the like based on a sensor that is brought into contact with at least a part of a body. According to various embodiments, the measurement module 506 may be activated when the state of the electronic device is determined to be a selected (or specified) state (e.g., a wearing state) through the device state determination module 502.

According to various embodiments, the measurement module 506 may perform a measurement operation when a biometric information measurement period (e.g., about 1 minute) arrives. According to various embodiments, the measurement module 506 may perform the measurement operation based on the motion of the electronic device or a user when the biometric information measurement period arrives. According to an embodiment, when the measurement period arrives, the measurement module 506 may identify the motion of the user or the electronic device that was measured before the period arrives, and when the motion satisfies a selected (or specified) condition (e.g., when the strength of the motion is lower than a selected (or specified) strength), the measurement module 506 may measure biometric information. According to another embodiment, when the motion of the electronic device or the user does not satisfy the selected (or specified) condition (e.g., when the strength of the motion is higher than the selected (or specified) strength), the measurement module 506 may monitor a change in the motion for a predetermined period of time (e.g., about 10 seconds) with respect to the time when the period arrives. According to an embodiment, when the change in the motion for the selected (or specified) period of time satisfies the selected (or specified) condition (e.g., when the average of motion data is smaller than a selected (or specified) value), the measurement module 506 may measure biometric information.

According to various embodiments, the parameter change module 504 may change a measurement parameter used for measuring biometric information. According to an embodiment, the parameter change module 504 may change the light-emitting time, the biometric information measurement period, the sampling rate, the biometric information measurement trigger, the motion determination condition, etc. of at least one sensor. Accordingly, it is possible to reduce power consumption that is generated when the electronic device measures biometric information, or to enhance the accuracy of measurement of biometric information. Further, according to an embodiment, when biometric information is measured, the parameter change module 504 may change a measurement parameter based on situation information that is associated with the motion strength of the electronic device or a body, the physical characteristic or state of an object to be measured, a measurement quality, a measurement result, or the like while the biometric information is being measured.

According to various embodiments, the storage module 508 may store at least one program for the operation of the electronic device. The storage module 508 may store infor-

mation for the operation of at least one of the device state determination module 502, the parameter change module 504, the user state determination module 505, and the measurement module 506. According to various embodiments, the device state determination module 502, the parameter change module 504, the user state determination module 505, the measurement module 506, or the storage module 508 may be at least one software element.

According to various embodiments, the processor 500 may include at least one of the device state determination module 502, the parameter change module 504, the user state determination module 505, the measurement module 506, and the storage module 508 in order to perform the above-described measurement operations.

Although not illustrated, according to various embodiments, the device state determination module 502, the parameter change module 504, the user state determination module 505, the measurement module 506, and the storage module 508 may be included in a plurality of processors (or chips). For example, the device state determination module 502 and the user state determination module 505 may be included in a first processor (or a first chip). The parameter change module 504 may be included in a second processor (or a second chip). Further, the measurement module 506 may be included in a third processor (or a third chip).

According to various embodiments, the electronic device may perform the above-described operation of measuring biometric information through a separate module that is different from the processor 500.

FIG. 5B is a view illustrating an operation of measuring biometric information by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the electronic device 400, or the processor 500), according to various embodiments of the present disclosure.

Referring to FIG. 5B, the electronic device (e.g., the processor 500) may attempt to measure biometric information according to a period (e.g., about 1 minute) having a selected (or specified) period of time. According to an embodiment, the electronic device (e.g., the processor 500) may operate a first timer that represents a selected (or specified) period of time in order to determine when to attempt to measure biometric information. For example, the electronic device (e.g., the processor 500) may attempt to measure biometric information at every 1 minute by operating the first timer.

According to various embodiments, the electronic device (e.g., the processor 500) may measure the motion of the electronic device in at least one time interval 522 before the operation of the first timer ends (520) (e.g., for about 10 seconds before the operation of the first timer ends), and when the operation of the first timer ends, the electronic device (e.g., the processor 500) may determine whether at least a part of the measured motion satisfies a selected (or specified) reference. For example, the electronic device (e.g., the processor 120 or 500) may measure the average of the motion that is acquired in at least one time interval 522 before the first timer ends, and may determine whether the measured average of the motion is less than, or equal to, a selected (or specified) threshold value.

According to various embodiments, the electronic device (e.g., the processor 500) may operate at least one sensor (e.g., biometric sensor) for measuring biometric information when the measured motion of the electronic device satisfies a selected (or specified) reference.

According to various embodiments, when the measured motion of the electronic device does not satisfy the selected (or specified) reference, the electronic device (e.g., the

processor 500) may determine whether the motion of the electronic device is changed to satisfy the selected (or specified) reference for the time 524 (e.g., about 30 seconds) during which a second timer operates. According to an embodiment, the electronic device (e.g., the processor 500) may not measure biometric information when the previously measured motion (e.g., the average of the motion) is maintained at a threshold value or more. According to another embodiment, the electronic device (e.g., the processor 500) may operate at least one sensor (e.g., biometric sensor) for measuring biometric information when the previously measured motion (e.g., the average of the motion) is changed to the threshold value or less. According to yet another embodiment, the electronic device (e.g., the processor 500) may additionally measure the motion of the electronic device for the time 524 during which the second timer operates. According to an embodiment, the electronic device (e.g., the processor 500) may additionally measure the motion of a selected (or specified) time interval (e.g., a time interval that has a magnitude that is the same as, or similar to, that of the time interval 522 in which the average of the motion has been measured) for the time 524 during which the second timer operates, and may determine whether the additionally measured motion satisfies a selected (or specified) reference. For example, the electronic device (e.g., the processor 500) may operate at least one sensor (e.g., biometric sensor) for measuring biometric information when the additionally measured motion satisfies the selected (or specified) reference.

According to various embodiments, the electronic device (e.g., the processor 500) may change the time 524 during which the second timer operates. According to an embodiment, the electronic device (e.g., the processor 500) may increase the time 524 during which the average of the previously measured motion is identified, and may determine whether the measurement of biometric information is possible for the increased time. According to an embodiment, based on the motion (e.g., the strength of the motion) of the electronic device, the electronic device (e.g., the processor 500) may change the time 524 during which the second timer operates. For example, the electronic device (e.g., the processor 500) may increase or decrease the operating time 524 of the second timer based on the previously measure motion (e.g., the average of the motion). For example, when the motion (e.g., the average of the motion) of the electronic device decreases or closely approaches a threshold value, the electronic device (e.g., the processor 500) may increase the operating time 524 of the second timer. For example, when the motion (e.g., the average of the motion) of the electronic device increases or does not closely approach the threshold value, the electronic device (e.g., the processor 500) may decrease the operating time 524 of the second timer.

An electronic device, according to various embodiments, may include: a housing; a fastener coupled to a part of the housing and detachably attached to a part of a user's body; at least one first sensor (e.g., an acceleration sensor, a terrestrial magnetism sensor, a gyro sensor, a proximity sensor, or a combination thereof) configured to detect the motion of the electronic device; at least one second sensor (e.g., a heart rate sensor) configured to detect a change in the user's body; a memory; and a processor electrically connected to the at least one first sensor, the at least one second sensor, and the memory. According to an embodiment, the memory may store instructions that configure the processor to: determine whether the electronic device is worn on (or close to) a part of the user's body using the at least one first

sensor; acquire information on the motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time selected (or specified) passes from the time when the electronic device is worn on (or close to) a part of the user's body; and activate (operate or trigger) the at least one second sensor based on at least a part of the information on the motion.

According to various embodiments, the determining of whether the electronic device is worn on the user's body part may include determining liveness of the body. According to an embodiment, the determining of the liveness of the body may be performed using at least one of a heart rate sensor, a blood pressure sensor, a body temperature sensor, a skin moisture sensor, an oxygen saturation sensor, a blood glucose sensor, an ECG sensor, an EEG sensor, and an EMG sensor. According to an embodiment, even though the determining of whether the electronic device is worn on (or close to) the user's body part is performed using the at least one first sensor, when it is determined that a signal received from a heart rate sensor is not a signal relating to the body, the electronic device may be determined to not be worn on the user's body part. According to an embodiment, even though the determining of whether the electronic device is worn on (or close to) the user's body part is performed using the at least one first sensor, when it is determined that a signal received from a body temperature sensor is not within a range suitable for the body, the electronic device may be determined to not be worn on the user's body part.

According to various embodiments, the second period of time may be shorter than the first period of time.

According to various embodiments, the at least one first sensor may include a proximity sensor, a gyro sensor, a terrestrial magnetism sensor, or an acceleration sensor.

According to various embodiments, the at least one second sensor may include a biometric sensor. According to an embodiment, the biometric sensor may include at least one of a heart rate sensor, a blood pressure sensor, an ECG sensor, a skin moisture sensor, and an oxygen saturation sensor.

According to various embodiments, the instructions may be set to configure the processor to: acquire additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes; and activate (operate or trigger) the at least one second sensor based on at least a part of the additional information, without activating (operating or triggering) the second sensor when the information on the motion does not satisfy a selected (or specified) condition.

According to various embodiments, at least one of the second period of time and the third period of time may be set to include the time point when the first period of time passes.

According to various embodiments, the second period of time may be set to include the time point when the first period of time passes.

According to various embodiments, when data is not received from the activated (operated or triggered) second sensor within a selected (or specified) time, the instructions may be set to configure the processor to activate (operate or trigger) the second sensor again after the selected (or specified) time passes.

According to various embodiments, the instructions may be set to configure the processor to: acquire additional information on the motion of the electronic device using at least a part of the at least one first sensor for a period of time

that is substantially the same as the second period of time before the selected (or specified) time passes; and activate (operate or trigger) the at least one second sensor again based on at least a part of the additional information.

According to various embodiments, the instructions may be set to configure the processor to: acquire data from at least one of the at least one first sensor and the at least one second sensor; and change the first period of time and/or the second period of time and/or the third period of time based on at least a part of the acquired data.

According to various embodiments, the instructions may be set to configure the processor to: acquire data from at least one of the at least one first sensor and the at least one second sensor; and change at least one condition for activating (operating or triggering) the at least one second sensor based on at least a part of the data.

According to various embodiments, the instructions may be set to configure the processor to: acquire data from at least one of the at least one first sensor and the at least one second sensor; and change at least one of a period of time and a sampling rate for activating (operating or triggering) the at least one second sensor based on at least a part of the data.

According to various embodiments, an electronic device may include a memory and a processor electrically connected to the memory. According to an embodiment, the memory may store instructions that configure the processor to: determine whether the electronic device is worn on (or close to) a part of a user's body using at least one first sensor; activate (operate or trigger) at least one second sensor based on at least a part of the motion of the electronic device within a selected (specified) time range at the time when a selected (or specified) period of time passes in a case where the electronic device is worn on (or close to) a part of the user's body; and measure the user's biometric information through the at least one second sensor.

According to various embodiments, the at least one first sensor may include at least one of a proximity sensor, a gyro sensor, a terrestrial magnetism sensor, and an acceleration sensor.

According to various embodiments, the instructions may be set to configure the processor to: acquire additional information on the motion of the electronic device using at least a part of the at least one first sensor for a selected (or specified) time from the time when the period of time passes; and activate (operate or trigger) the at least one second sensor based on at least a part of the additional information, without activating (operating or triggering) the second sensor when at least a part of the motion does not satisfy a selected (or specified) condition.

According to various embodiments, the instructions may be set to configure the processor to change the selected (or specified) time for acquiring the additional information based on data acquired from at least one of the at least one first sensor and the at least one second sensor.

According to various embodiments, the instructions may be set to configure the processor to: accumulate the number of failures when the measurement of the biometric information fails; and deactivate the second sensor when the cumulative number of failures satisfies a selected (or specified) condition.

According to various embodiments, the instructions may be set to configure the processor to acquire data from at least one of the at least one first sensor and the at least one second sensor and activate (operate or trigger) the at least one second sensor based on at least a part of the data.

According to various embodiments, the instructions may be set to configure the processor to acquire data from at least one of the at least one first sensor and the at least one second sensor and change at least one of a period of time or a sampling rate for activating (operating or triggering) the at least one second sensor based on at least a part of the data.

According to various embodiments, an electronic device may include a memory and a processor electrically connected to the memory. According to an embodiment, the memory may be set to configure the processor to: determine whether the electronic device is close to a user's body; acquire information on the motion of the electronic device using at least one first sensor within a selected (or specified) time range from the time when a selected (or specified) period passes in a case where the electronic device is close to the user's body part; and acquire the user's biometric information through at least one second sensor based on at least a part of the information on the motion.

FIG. 6 is a flowchart illustrating a biometric information measurement operation of an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

Referring to FIG. 6, in operation 601 the electronic device (e.g., the processor 120 or 500 or the device state determination module 502) may determine the state (e.g., wearing state, proximity state, in-use state, or holding state) of the electronic device based on a first sensor. According to various embodiments, the electronic device (e.g., the processor 120 or 500) may identify the state of the electronic device relating to wearing (or proximity) based on at least one first sensor. According to an embodiment, the state of the electronic device relating to wearing may be a state in which the electronic device makes contact with at least a part of a body, a state in which the electronic device accesses at least a part of a body, etc. According to various embodiments, the first sensor may be constituted by at least one of an acceleration sensor, a terrestrial magnetism sensor, a gyro sensor, a proximity sensor, an atmospheric pressure sensor, and a temperature sensor.

In operation 603, the electronic device (e.g., the processor 120 or 500) may identify the state (e.g., wearing state or non-wearing state) of the electronic device relating to wearing through a state determination operation.

When it is identified in operation 603 that the state of the electronic device relating to wearing does not correspond to a selected (or specified) state (e.g., when the state of the electronic device corresponds to a non-wearing state), the electronic device (e.g., the processor 120 or 500) may perform the operation of determining the state of the electronic device in operation 601.

When it is identified in operation 603 that the state of the electronic device relating to wearing corresponds to the selected (or specified) state (e.g., when the state of the electronic device corresponds to a wearing state), the electronic device (e.g., the processor 120 or 500) may operate at least one second sensor (e.g., the sensor module 470 or the biometric sensor 240I) in operation 605. According to various embodiments, at least one second sensor may be a sensor for measuring biometric information, and may measure at least one of a blood pressure, a heart rate, an ECG, skin moisture, saturation of peripheral oxygen, a body temperature, blood glucose, an EEG, an EMG, and the like. For example, the second sensor may be disposed on one surface of the electronic device so as to be brought into contact with a body.

According to various embodiments, the electronic device may operate at least one second sensor based on at least one of when a selected (or specified) period of time has elapsed, when the motion of the electronic device that is acquired using at least a part of at least one first sensor satisfies a selected (or specified) reference, and when a selected (or specified) time arrives. According to an embodiment, when the state of the electronic device is determined to be a wearing state, the electronic device may operate a first timer that represents a selected (or specified) period of time and may operate at least one second sensor when the first timer ends. According to an embodiment, when the state of the electronic device is determined to be a wearing state, the electronic device may identify the motion of the electronic device and may operate at least one second sensor when the motion satisfies a selected (or specified) reference. According to an embodiment, when the state of the electronic device is determined to be a wearing state, the electronic device may identify whether a selected (or specified) time (e.g., about 6:11 in a case where a wearing determination time is about 6:10) arrives and may operate at least one second sensor.

In operation 607, the electronic device (e.g., the processor 120 or 500 or the measurement module 506) may measure biometric information based on the second sensor. According to various embodiments, the electronic device (e.g., the processor 120 or 500) may output the biometric information measured based on the second sensor on a screen thereof. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may provide a health care function (e.g., providing a heart rate or a work rate) based on the measured biometric information.

FIG. 7 is a flowchart illustrating a performance procedure of a method of determining the state of an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

Referring to FIG. 7, the performance procedure of the state determination method may be a detailed operation for the operation 601 illustrated in FIG. 6.

In operation 701, the electronic device (e.g., the processor 120 or 500) may identify whether the proximity of an external object (e.g., a user) to the electronic device is detected based on at least one first sensor (e.g., proximity sensor). According to various embodiments, the proximity may be associated with a state in which the electronic device is close to a body, a state in which the electronic device makes contact with a body, etc. For example, the electronic device (e.g., the processor 120 or 500) may determine whether the electronic device is close to a body based on information acquired through the proximity sensor, which is the first sensor.

When the proximity of the user (e.g., body) to the electronic device is detected in operation 701, the electronic device (e.g., the processor 120 or 500) may measure the motion of the electronic device based on at least one first sensor (e.g., an acceleration sensor, a terrestrial magnetism sensor, or a gyro sensor) in operation 703. According to various embodiments, the motion may be associated with the motion of the electronic device or user. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may measure the motion based on information acquired through at least one of an acceleration sensor, a terrestrial magnetism sensor, and a gyro sensor in response to the detection of the proximity.

In operation 705, the electronic device (e.g., the processor 120 or 500) may identify whether a motion that satisfies a selected (or specified) condition is detected. According to various embodiments, the electronic device (e.g., the processor 120 or 500) may identify whether a continuous motion is detected for a selected (or specified) period of time while the electronic device is close to a body. For example, when the continuous motion is detected for the selected (or specified) period of time, the electronic device (e.g., the processor 120 or 500) may determine that the motion that satisfies the condition is detected.

When the motion that satisfies the selected (or specified) condition is detected in operation 705, the electronic device (e.g., the processor 120 or 500) may determine the state of the electronic device to be a state relating to wearing in operation 707.

When the motion that satisfies the selected (or specified) condition is not detected in operation 705, the electronic device (e.g., the processor 120 or 500) may identify whether the proximity of a body to the electronic device is detected based on at least one first sensor (e.g., a proximity sensor) in operation 709. For example, the electronic device may determine whether the electronic device is close to the body based on information acquired through the proximity sensor, which is the first sensor.

When the proximity of the body to the electronic device is detected in operation 709, the electronic device (e.g., the processor 120 or 500) may determine the state of the electronic device to be a state relating to wearing (e.g., wearing state) in operation 707. In one embodiment, when the electronic device makes contact with, or is close to, a body in a motionless state (e.g., sleep state), the electronic device (e.g., the processor 120 or 500) may determine the state of the electronic device to be a state relating to wearing.

When the proximity of a body to the electronic device is not detected in operation 709, the electronic device (e.g., the processor 120 or 500) may determine the state of the electronic device to be a state not relating to wearing (e.g., non-wearing state) in operation 713.

When the proximity of an external object to the electronic device is not detected in operation 701, the electronic device (e.g., the processor 120 or 500) may identify whether the motion of the electronic device is detected in operation 711. According to various embodiments, in the state in which no proximity is detected, the electronic device (e.g., the processor 120 or 500) may determine whether to perform the operation of determining the detection of proximity again based on the motion of the electronic device. For example, although no proximity is detected, the electronic device (e.g., the processor 120 or 500) may determine that proximity is likely to be detected when the motion of the electronic device is detected.

When the motion of the electronic device is detected in operation 711, the electronic device (e.g., the processor 120 or 500) may determine whether the electronic device is close to a body based on at least one first sensor in operation 701.

When the motion of the electronic device is not detected in operation 711, the electronic device (e.g., the processor 120 or 500) may determine the state of the electronic device to be a state not relating to wearing (e.g., non-wearing state) in operation 713. For example, the electronic device (e.g., the processor 120 or 500) may determine that the electronic device is unlikely to access a body and may determine the state of the electronic device to be a state not relating to wearing.

According to various embodiments, at least one of the operations 701 to 713 may be omitted. For example, in

operation 701, the electronic device (e.g., the processor 120 or 500) may identify whether the proximity of a user to the electronic device is detected. When the proximity of the user to the electronic device is detected, the electronic device (e.g., the processor 120 or 500) may determine the state of the electronic device to be a state relating to wearing (e.g., wearing state) in operation 707. Furthermore, when the proximity of the user to the electronic device is not detected, the electronic device (e.g., the processor 120 or 500) may determine the state of the electronic device to be a state not relating to wearing (e.g., non-wearing state) in operation 713.

According to various embodiments, in operation 703, the electronic device (e.g., the processor 120 or 500) may measure the motion of the electronic device. In operation 705, the electronic device (e.g., the processor 120 or 500) may identify whether the motion (e.g., continuous motion) is detected for a specified (or selected) period of time. When the motion (continuous motion) is detected for the selected (or specified) period of time, the electronic device (e.g., the processor 120 or 500) may determine the state of the electronic device to be a state (e.g., wearing state) relating to wearing (or a state in which the user holds or uses the electronic device) in operation 707. When the continuous motion is not detected for the selected (or specified) period of time, the electronic device (e.g., the processor 120 or 500) may determine the state of the electronic device to be a state (non-wearing state) not relating to wearing (or a state in which the user does not hold or use the electronic device) in operation 713.

FIG. 8 is a flowchart illustrating the performance procedure of a method of operating a second sensor by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

Referring to FIG. 8, the performance procedure of the method of operating the second sensor may be a detailed operation for the operation 605 illustrated in FIG. 6.

In operation 801, the electronic device (e.g., the processor 120 or 500) may operate a first timer that operates for a selected (or specified) period of time. According to various embodiments, the electronic device (e.g., the processor 120 or 500) may attempt to measure biometric information based on the operation of the first timer. According to an embodiment, the first timer may be time on the basis of which when to attempt to measure biometric information is determined. For example, the first timer may operate in about 1-minute periods.

In operation 803, the electronic device (e.g., the processor 120 or 500) may measure the motion of the electronic device in response to the operation of the timer. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may detect the motion of the electronic device based on at least one first sensor (e.g., an acceleration sensor or a gyro sensor). According to various embodiments, based on the operation of the first timer, the electronic device (e.g., the processor 120 or 500) may measure the motion of the electronic device until the first timer ends.

In operation 805, the electronic device (e.g., the processor 120 or 500) may identify whether the first timer ends. The time when the first timer ends may be, for example, the time when whether to perform an operation of measuring biometric information is determined. The operation of measuring the biometric information may be, for example, an operation of driving at least one second sensor for measuring biometric information.

When the first timer does not end in operation **805**, the electronic device (e.g., the processor **120** or **500**) may measure the motion of the electronic device in operation **803**. For example, the electronic device (e.g., the processor **120** or **500**) may measure the motion of the electronic device until the first timer ends.

When the first timer ends in operation **805**, the electronic device (e.g., the processor **120** or **500**) may determine whether the motion of the electronic device, which is measured during the operation of the first timer, satisfies a selected (or specified) reference in operation **807**. According to an embodiment, the electronic device (e.g., the processor **120** or **500**) may determine whether a motion by which biometric information can be accurately measured is generated. According to various embodiments, the electronic device (e.g., the processor **120** or **500**) may determine whether at least a part of the motion measured during the operation of the first timer satisfies a selected (or specified) reference. According to an embodiment, the electronic device (e.g., the processor **120** or **500**) may measure the average of the motion measured during the operation of the first timer and may determine whether the measured average of the motion satisfies a selected (or specified) reference. According to another embodiment, the electronic device (e.g., the processor **120** or **500**) may measure the average of the motion acquired in at least one time interval before the first timer ends and may determine whether the measured average of the motion satisfies a selected (or specified) reference. According to various embodiments, the situation in which the selected (or specified) condition is satisfied may be a situation in which the motion of the electronic device that is less than, or equal to, a selected (or specified) threshold value is measured. For example, the situation in which the motion of the electronic device that is less than, or equal to, the selected (or specified) threshold value is measured may be a state in which the motion of the electronic device is not generated, or a state in which the motion of the electronic device is so small that it is recognized as not being generated.

When the motion of the electronic device, which is measured during the operation of the first timer, satisfies the selected (or specified) reference in operation **807**, the electronic device (e.g., the processor **120** or **500**) may operate at least one second sensor for measuring biometric information in operation **809**.

When the motion of the electronic device, which is measured during the operation of the first timer, does not satisfy the selected (or specified) reference in operation **807**, the electronic device (e.g., the processor **120** or **500**) may operate a second timer that operates for a selected (or specified) period of time in operation **811**. According to an embodiment, the electronic device (e.g., the processor **120** or **500**) may operate the second timer in order to determine whether the measured motion of the electronic device that does not satisfy the reference is changed to satisfy the selected (or specified) reference. The operating time of the second timer may be shorter than that of the first timer. For example, the electronic device (e.g., the processor **120** or **500**) may identify whether the motion of the electronic device changes until the next period for the measurement of biometric information arrives.

In operation **813**, the electronic device (e.g., the processor **120** or **500**) may measure a change in the motion of the electronic device.

In operation **815**, the electronic device (e.g., the processor **120** or **500**) may identify whether the measured change in the motion satisfies a selected (or specified) reference.

According to an embodiment, the electronic device (e.g., the processor **120** or **500**) may determine whether the measured motion (e.g., the motion of the electronic device that does not satisfy the reference) is changed into a motion by which biometric information can be accurately measured. For example, the electronic device (e.g., the processor **120** or **500**) may identify whether the average value of the motion that is measured when the first timer ends is changed to a selected (or specified) threshold value or less.

When the change in the motion satisfies the selected (or specified) reference in operation **815**, the electronic device (e.g., the processor **120** or **500**) may measure biometric information. For example, the electronic device (e.g., the processor **120** or **500**) may operate at least one second sensor for measuring biometric information in operation **809**.

When the change in the motion does not satisfy the selected (or specified) reference in operation **815**, the electronic device (e.g., the processor **120** or **500**) may identify whether the second time ends in operation **817**.

When the second timer does not end in operation **817**, the electronic device (e.g., the processor **120** or **500**) may identify whether the change in the motion of the electronic device (e.g., the processor **120** or **500**) satisfies the selected (or specified) reference again while the second timer operates. For example, the electronic device (e.g., the processor **120** or **500**) may measure a change in the motion of the electronic device in operation **813**.

When the second timer ends in operation **817**, the electronic device (e.g., the processor **120** or **500**) may determine when to operate the first timer in operation **819**. For example, the electronic device (e.g., the processor **120** or **500**) may determine the state of the electronic device relating to wearing. In another example, as the electronic device (e.g., the processor **120** or **500**) determines the state (e.g., wearing state) of the electronic device relating to wearing, the electronic device may omit the operation of determining the state of the electronic device and may operate the first timer.

According to various embodiments, at least one of the operations **801** to **819** may be omitted. When the motion of the electronic device **101**, which is measured during the operation of the first timer, satisfies the selected (or specified) reference in operation **807**, the electronic device **101** (e.g., the processor **120** or **500**) may operate at least one second sensor for measuring biometric information in operation **809**. When the motion of the electronic device **101**, which is measured during the operation of the first timer, does not satisfy the selected (or specified) reference in operation **807**, the electronic device (e.g., the processor **120** or **500**) may determine when to operate the first timer in operation **819**. For example, the electronic device (e.g., the processor **120** or **500**) may determine the state of the electronic device relating to wearing. In another example, as the electronic device (e.g., the processor **120** or **500**) determines the state (e.g., wearing state) of the electronic device relating to wearing, the electronic device may omit the operation of determining the state of the electronic device and may operate the first timer.

FIG. **9** is a flowchart illustrating the performance procedure of a method of measuring biometric information by an electronic device (e.g., the electronic device **101**, the processor **120** of the electronic device **101**, the processor **500**, or the electronic device **400**) according to various embodiments of the present disclosure.

Referring to FIG. **9**, the performance procedure of the method of measuring biometric information may be a detailed operation for the operation **607** illustrated in FIG. **6**.

In operation 901, the electronic device (e.g., the processor 120 or 500) may attempt to measure biometric information. According to various embodiments, the measurement of biometric information may be performed by operating at least one second sensor. According to various embodiments, the second sensor may include a light emitting part that emits light and a light receiving part that receives the light of the light emitting part, and the electronic device (e.g., the processor 120 or 500) may measure biometric information based on a signal output from the second sensor. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may measure biometric information (e.g., a heart rate) based on the amount of light output through at least one second sensor that makes contact with at least a part of a body. For example, slightly dark light may be received for a systolic period of the heart cycle due to a large amount of blood in a body part, and bright light may be received for a diastolic period of the heart cycle due to a relatively small amount of blood in the body part. The electronic device (e.g., the processor 120 or 500) may measure heart rate information based on the amount of light.

In operation 903, the electronic device (e.g., the processor 120 or 500) may identify whether the biometric information is successfully measured. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may identify whether a normal measurement result is obtained. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may identify whether a normal measurement result is obtained based on the time devoted to measuring the biometric information. For example, when a measurement result is not obtained within a selected (or specified) time after the measurement of the biometric information starts, the electronic device (e.g., the processor 120 or 500) may determine that the measurement of the biometric information has failed. According to another embodiment, the electronic device (e.g., the processor 120 or 500) may identify whether a normal measurement result is obtained based on the motion of the electronic device. For example, when a motion that does not satisfy a selected (or specified) condition is detected after the measurement of the biometric information starts, the electronic device (e.g., the processor 120 or 500) may determine that the measurement of the biometric information has failed.

When the biometric information is successfully measured in operation 903, the electronic device (e.g., the processor 120 or 500) may output the measured biometric information in operation 905. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may display the measurement result on a display thereof (e.g., a screen) based on the second sensor. According to another embodiment, the electronic device (e.g., the processor 120 or 500) may provide a health care function (e.g., providing whether a heart rate is normal or not) based on the measurement result.

When no biometric information is measured in operation 903, the electronic device (e.g., the processor 120 or 500) may accumulate the number of failures in operation 907. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may accumulate the number of failures every time a normal result is not obtained after the electronic device attempts to measure biometric information. For example, when a normal measurement result on biometric information is not obtained once, the electronic device (e.g., the processor 120 or 500) may determine the number of failures to be one in operation 907. Furthermore, when a normal measurement result on biometric information

is not obtained three times, the electronic device (e.g., the processor 120 or 500) may determine the number of failures to be three in operation 907.

In operation 909, the electronic device (e.g., the processor 120 or 500) may identify the cumulative number of failures. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may identify whether the cumulative number of failures is greater than, or equal to, a threshold value (e.g., about three times). According to various embodiments, when the cumulative number of failures is greater than, or equal to, the threshold value, the electronic device may determine that the measurement has continually failed so that it is impossible to measure biometric information.

When it is identified in operation 909 that the cumulative number of failures is smaller than the threshold value, the electronic device (e.g., the processor 120 or 500) may attempt to measure biometric information again. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may perform an operation associated with operation 901 again. According to another embodiment, the electronic device (e.g., the processor 120 or 500) may measure biometric information based on the motion of the electronic device. For example, the electronic device (e.g., the processor 120 or 500) may perform at least one of the operations 801 to 819.

When it is identified in operation 909 that the cumulative number of failures is greater than, or equal to, the threshold value, the electronic device (e.g., the processor 120 or 500) may deactivate at least one second sensor. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may determine that it is impossible to measure biometric information, and may stop the operation of the second sensor accordingly in operation 911.

According to various embodiments, at least one of the operations 901 to 911 may be omitted. For example, the operations 907 to 911 may be omitted. Accordingly, in a case where the measurement of biometric information fails in operation 903, the electronic device (e.g., the processor 120 or 500) may perform operation 901 again.

FIG. 10 is a flowchart illustrating the performance procedure of a method of measuring biometric information by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

Referring to FIG. 10, the performance procedure of the method of measuring biometric information may be a detailed operation for the operation 607 illustrated in FIG. 6.

In operation 1001, the electronic device (e.g., the processor 120 or 500) may acquire situation information associated with a measurement. According to various embodiments, the electronic device (e.g., the processor 120 or 500) may acquire the situation information based on at least one first sensor and/or at least one second sensor. According to an embodiment, the situation information may include the motion of the electronic device or a body while biometric information is being measured, the physical characteristic of an object to be measured, the state of the object to be measured, a measurement quality, a measurement result, or the like.

In operation 1003, the electronic device (e.g., the processor 120 or 500) may change a measurement parameter based on the situation information. For example, the measurement parameter that is changed based on the situation information may be defined as listed in Table 1 below. However, various embodiments of the present disclosure are not limited to Table 1. For example, in a case where the situation infor-

mation is 'measurement quality,' the parameter to be changed may be 'timer period.' Furthermore, in a case where the situation information is 'physical characteristic,' the parameter to be changed may be 'sampling rate.'

TABLE 1

Situation information	Parameter to be changed
Measurement quality	Measurement time (light-emitting time)
Motion	Sampling rate
Measurement result	Measurement trigger
Physical condition	Timer period
User's state	Condition for determining motion

According to an embodiment, the electronic device (e.g., the processor 120 or 500) may reduce power consumption that is generated when biometric information is measured by changing the measurement parameter based on the situation information. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may increase the accuracy of measurement when measuring biometric information by changing the measurement parameter based on the situation information.

For example, the electronic device (e.g., the processor 120 or 500) may determine that it is useless to measure biometric information in a state in which a measurement quality is not good. Accordingly, the electronic device (e.g., the processor 120 or 500) may decrease the light-emitting time of the second sensor in order to reduce power consumption. In another example, when the motion strength of the electronic device that cannot be normally measured is detected while biometric information is being measured, the electronic device (e.g., the processor 120 or 500) may stop the measurement operation in order to reduce power consumption. In yet another example, when the electronic device (e.g., the processor 120 or 500) continually fails to measure biometric information due to the motion thereof, the electronic device may omit the measurement operation until the motion is not generated in order to reduce power consumption. In yet another example, when it is determined that the strength of a motion while biometric information is being measured can be corrected, the electronic device (e.g., the processor 120 or 500) may increase a measurement sampling rate during the measurement operation in order to enhance the accuracy of measurement. In yet another example, when the electronic device (e.g., the processor 120 or 500) detects a physical characteristic (e.g., a dark skin color) of an object to be measured that uses an amount of light more than a reference value, which is emitted from the second sensor, the electronic device may change the measurement period in order to decrease the number of times that biometric information is measured, thereby reducing power consumption. In yet another example, in the case of a user in a motionless state (e.g., a sleep state), the electronic device (e.g., the processor 120 or 500) may omit an operation of determining the motion of the electronic device in order to reduce power consumption.

In operation 1005, the electronic device (e.g., the processor 120 or 500) may measure biometric information based on the changed parameter.

In operation 1007, the electronic device (e.g., the processor 120 or 500) may identify whether the biometric information is completely measured. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may identify whether the biometric information measurement result is successfully obtained.

When the measurement of the biometric information is not completed in operation 1007, the electronic device (e.g., the processor 120 or 500) may measure biometric information. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may measure the biometric information based on the changed parameter in operation 1005.

When the measurement of biometric information is completed in operation 1007, the electronic device (e.g., the processor 120 or 500) may output the measured biometric information in operation 1009. For example, the electronic device (e.g., the processor 120 or 500) may output the measured biometric information through a display or a speaker.

According to various embodiments, at least one of the operations 1001 to 1009 may be omitted. For example, operation 1009 may be omitted.

FIG. 11 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

Referring to FIG. 11, the performance procedure of the method of changing a measurement parameter may be a detailed operation for the operation 1003 illustrated in FIG. 10.

According to an embodiment, the electronic device (e.g., the processor 120 or 500) may determine that it is useless to measure biometric information in a state in which a measurement quality is not good and may decrease the light-emitting time of a second sensor in order to reduce power consumption.

In operation 1101, the electronic device (e.g., the processor 120 or 500) may measure biometric information. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may operate at least one second sensor to emit light for a selected (or specified) period of time and may acquire a signal output from the second sensor.

In operation 1103, the electronic device (e.g., the processor 120 or 500) may identify the measurement quality. According to an embodiment, the measurement quality may be the quality of a signal output from the second sensor. For example, the electronic device (e.g., the processor 120 or 500) may acquire a signal to noise ratio for the signal output from the second sensor as the measurement quality.

In operation 1105, the electronic device (e.g., the processor 120 or 500) may identify whether the measurement quality satisfies a selected (or specified) reference. According to an embodiment, the measurement quality may not satisfy the selected (or specified) reference, which may mean that a signal used for measuring biometric information is not acquired. For example, the measurement quality that fails to satisfy the selected (or specified) reference may indicate that a normal measurement result is not obtained. In another example, the measurement quality that fails to satisfy the selected (or specified) reference may indicate that a measurement result is not obtained for a selected (or specified) measurement time.

When the measurement quality satisfies the selected (or specified) reference in operation 1105, the electronic device (e.g., the processor 120 or 500) may measure biometric information based on a selected (or specified) measurement time in operation 1107. According to various embodiments, the measurement time may be associated with the light-emitting time of the second sensor. According to an embodiment, when the measurement quality satisfies the selected

(or specified) reference, the electronic device (e.g., the processor 120 or 500) may acquire biometric information by emitting light through the second sensor for the selected (or specified) time (e.g., about 10 seconds) every time a biometric information measurement period (e.g., about 1-minute period) arrives.

When the measurement quality does not satisfy the selected (or specified) reference in operation 1105, the electronic device (e.g., the processor 120 or 500) may change the measurement time selected (or specified) based on the quality in operation 1109.

According to various embodiments, the electronic device (e.g., the processor 120 or 500) may change the measurement time in order to reduce power consumption. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may shorten the measurement time in order to reduce power consumption when the measurement result on the biometric information is not obtained for the selected (or specified) measurement time.

According to various embodiments, the electronic device (e.g., the processor 120 or 500) may change the measurement time in order to enhance the accuracy of the measurement result. According to an embodiment, when the measurement result on the biometric information is not obtained for the selected (or specified) measurement time, the electronic device (e.g., the processor 120 or 500) may increase the measurement time in order to sufficiently ensure the information used for measuring the biometric information.

In operation 1111, the electronic device (e.g., the processor 120 or 500) may measure biometric information based on the changed measurement time.

According to various embodiments, at least one of the operations 1101 to 1111 may be omitted.

FIG. 12 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

Referring to FIG. 12, the performance procedure of the method of changing a measurement parameter may be a detailed operation for the operation 1003 illustrated in FIG. 10.

In operation 1201, the electronic device (e.g., the processor 120 or 500) may measure biometric information. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may operate at least one second sensor to emit light for a selected (or specified) period of time and may acquire a signal output from the second sensor.

In operation 1203, the electronic device (e.g., the processor 120 or 500) may identify whether the motion of the electronic device is detected while the biometric information is being measured. According to various embodiments, the electronic device (e.g., the processor 120 or 500) may measure the motion of the electronic device based on at least one first sensor.

When the motion of the electronic device is not detected in operation 1203, the electronic device (e.g., the processor 120 or 500) may measure biometric information in operation 1205. According to various embodiments, the electronic device (e.g., the processor 120 or 500) may acquire information output from the second sensor based on a selected (or specified) sampling rate.

When the motion of the electronic device is detected in operation 1203, the electronic device (e.g., the processor 120 or 500) may identify whether the measured motion satisfies a selected (or specified) reference value in operation 1207.

According to an embodiment, the electronic device (e.g., the processor 120 or 500) may identify whether a low motion strength that does not affect the measurement result is detected. For example, the electronic device (e.g., the processor 120 or 500) may determine the motion strength that does not affect the measurement result to be a motion that satisfies the reference value. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may identify whether a motion strength that may affect the measurement result, but can be corrected, is detected. For example, the electronic device (e.g., the processor 120 or 500) may determine a value larger than the motion strength, which does not affect the measurement result, by a predetermined value or more to be a reference value.

When the motion is identified to satisfy the reference value in operation 1207, the electronic device (e.g., the processor 120 or 500) may change the sampling rate of a signal output from the second sensor in operation 1209. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may increase a signal output from the second sensor by increasing the sampling rate in order to correct the noise according to the motion thereof.

When it is identified in operation 1207 that the motion does not satisfy the reference value, the electronic device (e.g., the processor 120 or 500) may deactivate the second sensor in operation 1211. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may determine that it is impossible to correct the noise according to the motion, and may accordingly stop measuring the biometric information.

According to various embodiments, at least one of the operations 1201 to 1211 may be omitted. For example, the operations 1207 and 1211 may be omitted. In this case, when the motion of the electronic device is detected in operation 1203, the electronic device may change the sampling rate in operation 1209. The electronic device may measure biometric information according to the changed sampling rate in operation 1205.

FIG. 13 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

Referring to FIG. 13, the performance procedure of the method of changing a measurement parameter may be a detailed operation for the operation 1003 illustrated in FIG. 10.

In operation 1301, the electronic device (e.g., the processor 120 or 500) may measure biometric information. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may operate at least one second sensor to emit light for a selected (or specified) period of time and may acquire a signal output from the second sensor.

In operation 1303, the electronic device (e.g., the processor 120 or 500) may identify whether the motion of the electronic device is detected while the biometric information is being measured. According to various embodiments, the electronic device (e.g., the processor 120 or 500) may measure the motion of the electronic device based on at least one first sensor.

In operation 1305, the electronic device (e.g., the processor 120 or 500) may identify whether the measured motion satisfies a selected (or specified) reference. According to an embodiment, the measured motion may satisfy the selected

(or specified) reference, which may indicate that a motion that does not affect the acquisition of biometric information is generated.

When the measured motion satisfies the selected (or specified) reference in operation 1305, the electronic device (e.g., the processor 120 or 500) may measure biometric information in operation 1307. According to various embodiments, the electronic device (e.g., the processor 120 or 500) may acquire information output from the second sensor based on a selected (or specified) measurement time.

When the measured motion does not satisfy the selected (or specified) reference in operation 1305, the electronic device (e.g., the processor 120 or 500) may determine whether information for obtaining a result is collected in operation 1309.

When the information for obtaining a result is collected in operation 1309, the electronic device (e.g., the processor 120 or 500) may perform an operation of determining the motion thereof in operation 1311. According to an embodiment, in order to prevent the measurement operation from being stopped at the time when the measurement is nearly completed, the electronic device (e.g., the processor 120 or 500) may not deactivate the second sensor even though the motion is generated and may perform the operation of determining the motion until the motion satisfies the selected (or specified) reference.

When the information for obtaining a result is not collected in operation 1309, the electronic device (e.g., the processor 120 or 500) may deactivate the second sensor in operation 1313. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may determine that the motion that does not satisfy the selected (or specified) reference is generated in a state in which there is a predetermined period of time before the measurement is completed, and may accordingly stop measuring the biometric information.

According to various embodiments, at least one of the operations 1301 to 1313 may be omitted. For example, the operations 1309 and 1311 may be omitted. In this case, when it is determined in operation 1305 that the measured motion does not satisfy the selected (or specified) reference, the electronic device (e.g., the processor 120 or 500) may deactivate the second sensor in operation 1313. Furthermore, when it is determined in operation 1305 that the measured motion satisfies the selected (or specified) reference, the electronic device (e.g., the processor 120 or 500) may measure biometric information in operation 1307.

FIG. 14 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

Referring to FIG. 14, the performance procedure of the method of changing a measurement parameter may be a detailed operation for the operation 1003 illustrated in FIG. 10.

In operation 1401, the electronic device (e.g., the processor 120 or 500) may measure biometric information. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may operate at least one second sensor to emit light for a selected (or specified) period of time and may acquire a signal output from the second sensor.

In operation 1403, the electronic device (e.g., the processor 120 or 500) may identify whether the motion of the electronic device is detected while the biometric information is being measured. According to various embodiments, the

electronic device (e.g., the processor 120 or 500) may measure the motion of the electronic device based on at least one first sensor.

In operation 1405, the electronic device (e.g., the processor 120 or 500) may identify whether the measured motion satisfies a selected (or specified) reference. According to an embodiment, the measured motion may satisfy the selected (or specified) reference, which may indicate that a motion that does not affect the acquisition of biometric information is generated.

When the measured motion satisfies the selected (or specified) reference in operation 1405, the electronic device (e.g., the processor 120 or 500) may measure biometric information based on the operation of a timer in operation 1407. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may attempt to measure biometric information according to a period having a selected (or specified) period of time.

When the measured motion does not satisfy the selected (or specified) reference in operation 1405, the electronic device (e.g., the processor 120 or 500) may change the measurement trigger based on the motion in operation 1409. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may measure biometric information with a period having a selected (or specified) period of time as a trigger, but may not perform the measurement operation even though the measurement period arrives in a situation in which the motion that satisfies the selected (or specified) reference is not measured. For example, the electronic device (e.g., the processor 120 or 500) may identify whether the motion that satisfies the selected (or specified) reference is measured without performing the biometric information measurement operation, and when the motion that satisfies the selected (or specified) reference is detected, the electronic device may perform the biometric information measurement operation.

In operation 1411, the electronic device (e.g., the processor 120 or 500) may measure biometric information based on the changed measurement trigger. According to various embodiments, at least one of the operations 1401 to 1411 may be omitted.

FIG. 15 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

Referring to FIG. 15, the performance procedure of the method of changing a measurement parameter may be a detailed operation for the operation 1003 illustrated in FIG. 10.

In operation 1501, the electronic device (e.g., the processor 120 or 500) may measure biometric information. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may operate at least one second sensor to emit light for a selected (or specified) period of time and may acquire a signal output from the second sensor.

In operation 1503, the electronic device (e.g., the processor 120 or 500) may identify the previous measurement result while measuring the biometric information. The previous measurement result may be a result based on data acquired by at least one first sensor or at least one second sensor of the electronic device (e.g., the processor 120 or 500).

In operation 1505, the electronic device (e.g., the processor 120 or 500) may identify whether a measurement failure

is continuously generated a selected (or specified) number of times or more based on the previous measurement result.

When it is identified in operation 1505 that the measurement failure is not continuously generated a selected (or specified) number of times or more, the electronic device (e.g., the processor 120 or 500) may perform the biometric information measurement operation based on the operation of a timer in operation 1507. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may attempt to measure biometric information according to a period having a selected (or specified) period of time.

When it is identified in operation 1505 that the measurement failure is continuously generated a selected (or specified) number of times or more, the electronic device (e.g., the processor 120 or 500) may change the measurement trigger based on the motion in operation 1509. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may identify whether the motion that satisfies the selected (or specified) reference is measured and may measure biometric information when the motion that satisfies the selected (or specified) reference is detected, instead of measuring biometric information with a period having a selected (or specified) period of time as a trigger.

In operation 1511, the electronic device (e.g., the processor 120 or 500) may measure biometric information based on the changed measurement trigger. According to various embodiments, at least one of the operations 1501 to 1511 may be omitted.

FIG. 16 is a flowchart illustrating the performance procedure of a method of changing a measurement trigger by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

Referring to FIG. 16, the performance procedure of the method of changing a measurement trigger may be a detailed operation for the operation 1409 illustrated in FIG. 14 or the operation 1509 illustrated in FIG. 15.

In operation 1601, the electronic device (e.g., the processor 120 or 500) may identify the motion of the electronic device. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may measure the motion of the electronic device when the measurement trigger is changed based on the motion. For example, the electronic device (e.g., the processor 120 or 500) may measure the motion of the electronic device based on at least one first sensor.

In operation 1603, the electronic device (e.g., the processor 120 or 500) may identify whether the motion of the electronic device satisfies a condition. For example, the electronic device (e.g., the processor 120 or 500) may determine whether the motion that does not affect the measurement of biometric information is generated.

When the motion that satisfies the condition is not detected in operation 1603, the electronic device (e.g., the processor 120 or 500) may stand by without measuring biometric information in operation 1609. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may stand by without performing the measurement operation until the motion that does not affect the measurement is detected. For example, when the measurement trigger is changed based on the motion, the electronic device (e.g., the processor 120 or 500) may not perform the measurement operation even though a measurement period arrives.

When the motion that satisfies the condition is detected in operation 1603, the electronic device (e.g., the processor 120 or 500) may measure biometric information in operation 1605.

In operation 1607, the electronic device (e.g., the processor 120 or 500) may change the measurement trigger based on a timer after measuring the biometric information based on the motion that satisfies the condition. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may determine that the motion that does not affect the measurement is detected, and may accordingly change the measurement trigger to measure biometric information according to a period having a selected (or specified) period of time. According to various embodiments, at least one of the operations 1601 to 1609 may be omitted.

FIG. 17 is a view illustrating an operation of changing a measurement trigger by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

Referring to FIG. 17, the electronic device (e.g., the processor 120 or 500) may attempt to measure biometric information according to a period having a selected (or specified) period of time (e.g., about 1 minute). For example, the electronic device (e.g., the processor 120 or 500) may measure biometric information with a selected (or specified) period as a measurement trigger.

According to various embodiments, the electronic device (e.g., the processor 120 or 500) may measure biometric information based on at least one second sensor when the selected (or specified) period arrives.

According to various embodiments, the electronic device (e.g., the processor 120 or 500) may change the measurement trigger when the motion of the electronic device does not satisfy a selected (or specified) condition (e.g., when the motion by which biometric information cannot be accurately measured is detected) at the time points t1 and t2 when the selected (or specified) period arrives. For example, the electronic device (e.g., the processor 120 or 500) may not perform the biometric information measurement operation until the motion by which biometric information can be accurately measured is detected. In other words, the electronic device (e.g., the processor 120 or 500) may measure biometric information with the motion satisfying the condition (e.g., the motion by which biometric information can be accurately measured) as the measurement trigger.

According to various embodiments, the electronic device (e.g., the processor 120 or 500) may change the measurement trigger when a measurement failure according to the motion is continuously generated (1701) a selected (or specified) number of times (e.g., about two times) or more.

For example, the electronic device (e.g., the processor 120 or 500) may monitor the motion by changing the measurement trigger based on the detection of the motion after the measurement failure is continuously generated a selected (or specified) number of times or more, and may perform the measurement operation at time t3 when the motion satisfies the selected (or specified) condition without performing the measurement operation (1705) based on the timer for the time interval 1703 for which the motion is detected. The electronic device (e.g., the processor 120 or 500) may change the measurement trigger based on the timer when the electronic device succeeds in acquiring biometric information. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may change the measurement trigger to measure biometric information according to a period 1707 having a selected (or specified) period of time

t3-t4 when the electronic device succeeds in acquiring biometric information by changing the measurement trigger based on the detection of the motion.

FIG. 18 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

Referring to FIG. 18, the performance procedure of the method of changing a measurement parameter may be a detailed operation for the operation 1003 illustrated in FIG. 10.

In operation 1801, the electronic device (e.g., the processor 120 or 500) may measure biometric information. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may operate at least one second sensor to emit light for a selected (or specified) period of time and may acquire a signal output from the second sensor.

In operation 1803, the electronic device (e.g., the processor 120 or 500) may determine a characteristic of a body with which the second sensor makes contact. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may determine a physical characteristic of an object for which biometric information is measured. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may determine the skin color of the body with which the second sensor makes contact as the physical characteristic.

In operation 1805, the electronic device (e.g., the processor 120 or 500) may determine whether the physical characteristic uses an amount of emitted light more than a threshold value. According to various embodiments, the electronic device (e.g., the processor 120 or 500) may measure a melanin index based on the amount of reflected light that is received by a light-receiving part of the second sensor, and may identify the skin color and the amount of emitted light of a light-emitting part that is used during measurement based on the measured melanin index. For example, the amounts of emitted light according to melanin indices may be defined as listed in Table 2 below.

TABLE 2

Melanin Index	Type of skin color	Current intensity
5-10	Very light	31~40%
11-20	Light	41~50%
21-30	Intermediate	51~60%
31-40	Tan	61~70%
41-70	Brown	71~80%
71-99	black	81~100%

For example, a higher melanin index represents a darker skin color, and the electronic device (e.g., the processor 120 or 500) may determine that a larger amount of power is consumed for a darker skin color than for a lighter skin color.

According to various embodiments, the electronic device (e.g., the processor 120 or 500) may store a user's physical characteristic, or receive the same from an external device.

When it is determined in operation 1805 that the physical characteristic does not require an amount of emitted light more than the threshold value, the electronic device (e.g., the processor 120 or 500) may measure biometric information based on a selected (or specified) measurement period in operation 1811.

When it is determined in operation 1805 that the physical characteristic uses an amount of emitted light more than the

threshold value, the electronic device (e.g., the processor 120 or 500) may change the measurement period based on the physical characteristic in operation 1807. For example, the electronic device (e.g., the processor 120 or 500) may increase the measurement period to reduce the number of measurements when an amount of emitted light more than the threshold value is used when biometric information is measured.

When the measurement period is changed in operation 1807, the electronic device (e.g., the processor 120 or 500) may measure biometric information based on the changed measurement period in operation 1809. According to various embodiments, at least one of the operations 1801 to 1811 may be omitted.

FIG. 19 is a flowchart illustrating the performance procedure of a method of changing a measurement parameter by an electronic device (e.g., the electronic device 101, the processor 120 of the electronic device 101, the processor 500, or the electronic device 400) according to various embodiments of the present disclosure.

Referring to FIG. 19, the performance procedure of the method of changing a measurement parameter may be a detailed operation for the operation 1003 illustrated in FIG. 10.

In operation 1901, the electronic device (e.g., the processor 120 or 500) may measure biometric information. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may operate at least one second sensor to emit light for a selected (or specified) period of time and may acquire a signal output from the second sensor.

In operation 1903, the electronic device (e.g., the processor 120 or 500) may identify a user's state. According to an embodiment, the electronic device (e.g., the processor 120 or 500) may identify the user's state, such as a sleep state, an active state, etc.

In operation 1905, the electronic device (e.g., the processor 120 or 500) may determine whether a motion condition needs to be changed. According to various embodiments, when the user is in a sleep state, the motion condition may not be required since it may be determined that there is no motion, or the strength of a motion is low. When the motion condition is not required, it may be determined that the motion condition needs to be changed.

When it is determined in operation 1905 that the motion condition does not need to be changed, the electronic device (e.g., the processor 120 or 500) may measure biometric information based on a selected (or specified) motion condition in operation 1911.

When it is determined in operation 1905 that the motion condition needs to be changed, the electronic device (e.g., the processor 120 or 500) may change the motion condition based on the user's state in operation 1907. For example, the electronic device (e.g., the processor 120 or 500) may omit the condition for determining a motion, or may decrease a desired motion reference value.

When the motion condition is changed in operation 1907, the electronic device (e.g., the processor 120 or 500) may measure biometric information based on the changed motion condition in operation 1909. According to various embodiments, at least one of the operations 1901 to 1911 may be omitted.

FIG. 20 is a view illustrating a screen configuration of an external electronic device (e.g., a smart phone) that is connected to an electronic device (e.g., the electronic device 101 or 400) through wireless or wired communication according to various embodiments of the present disclosure.

Referring to FIG. 20, the electronic device (e.g., the processor 120 or 500) may store biometric information (e.g., heart rate information), which is measured at every selected (or specified) period, in a memory and may display the measurement result 2000 on the screen thereof based on an input.

According to various embodiments, the electronic device (e.g., the processor 120 or 500) may display the measured heart rate information in units of hours 2001, days 2003, and months 2005. According to an embodiment, when displaying the measurement result in units of hours, the electronic device (e.g., the processor 120 or 500) may display the average value of the heart rate information measured at every hour in units of hours. According to another embodiment, when displaying the measurement result in units of days, the electronic device (e.g., the processor 120 or 500) may display the average value of the heart rate information measured at every day in units of days. According to yet another embodiment, when displaying the measurement result in units of months, the electronic device (e.g., the processor 120 or 500) may display the average value of the heart rate information measured at every month in units of months. According to various embodiments, the electronic device (e.g., the processor 120 or 500) may also display the highest and lowest values 2011 and 2013 of the heart rate information that is displayed in units of hours, as well as the average value of the heart rate information.

According to various embodiments, when displaying (2007) the result of the heart rate information, the electronic device (e.g., the processor 120 or 500) may display an indicator together for the time point at which the average value is higher or lower than a predetermined reference. For example, the electronic device (e.g., the processor 120 or 500) may display the indicator using a selected (or specified) color, a selected (or specified) text, a selected (or specified) icon, or the like. In the drawing, an exclamation mark 2009 displayed in the time interval between 3 hour and 4 hour may be an indicator that represents that heart rate information more than the average value is measured.

According to various embodiments, when an input for the indicator is detected, the electronic device (e.g., the processor 120 or 500) may display detailed information relating to the corresponding time interval. For example, the electronic device (e.g., the processor 120 or 500) may display the heart rate information 2021 that corresponds to the time interval relevant to the indicator for which the input is detected and the state information 2023 (e.g., calorie consumption, etc.) of the electronic device or a user for the corresponding time interval together. The drawing illustrates that the user practiced running from 3:15 to 3:20 so that the user's heart rate increased.

A method of operating an electronic device, according to various embodiments, may include: determining whether the electronic device is worn on a part of the user's body using at least one first sensor (e.g., at least one of a proximity sensor, an acceleration sensor, a gyro sensor, and a terrestrial magnetism sensor); acquiring information on the motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time selected (or specified) passes from the time when the electronic device is worn on a part of the user's body; and activating (operating or triggering) at least one second sensor (e.g., a heart rate sensor) based on at least a part of the information on the motion.

According to various embodiments, the second period of time may be shorter than the first period of time.

According to various embodiments, the at least one second sensor may include a biometric sensor.

According to various embodiments, the biometric sensor may include at least one of a heart rate sensor, a blood pressure sensor, an ECG sensor, a skin moisture sensor, and an oxygen saturation sensor.

According to various embodiments, the activating (operating or triggering) may include: acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes; and activating (operating or triggering) the at least one second sensor based on at least a part of the additional information in a case where the information on the motion does not satisfy a selected condition.

According to various embodiments, at least one of the second period of time and the third period of time may include the time point when the first period of time passes.

According to various embodiments, the second period of time may include the time point when the first period of time passes.

According to various embodiments, the activating (operating or triggering) may include: when data is not received from the activated (operated or triggered) second sensor within a selected (or specified) time, activating (operating or triggering) the second sensor again after the selected (specified) time passes.

According to various embodiments, the activating (operating or triggering) may include: acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a period of time that is substantially the same as the second period of time before the selected (or specified) time passes; and activating (operating or triggering) the at least one second sensor again based on at least a part of the additional information.

According to various embodiments, the method of operating the electronic device may further include: acquiring data from at least one of the at least one first sensor and the at least one second sensor; and changing the first period of time and/or the second period of time and/or the third period of time based on at least a part of the acquired data.

According to various embodiments, the method of operating the electronic device may further include: acquiring data from at least one of the at least one first sensor and the at least one second sensor; and changing at least one condition for activating (operating or triggering) the at least one second sensor based on at least a part of the data.

According to various embodiments, the method of operating the electronic device may further include: acquiring data from at least one of the at least one first sensor and the at least one second sensor; and changing at least one of a period of time and a sampling rate for activating (operating or triggering) the at least one second sensor based on at least a part of the data.

A method of operating an electronic device, according to various embodiments, includes: determining whether the electronic device is worn on a part of a user's body using at least one first sensor; activating (operating or triggering) at least one second sensor based on at least a part of the measured motion of the electronic device within a selected (or specified) time range when a selected (or specified) period of time passes from the time when the electronic device is worn on the user's body part; and measuring biometric information based on the at least one second sensor.

According to various embodiments, the at least one first sensor may include at least one of a proximity sensor, a gyro sensor, a terrestrial magnetism sensor, and an acceleration sensor.

According to various embodiments, the method may include: acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a selected (or specified) time from the time when the period of time passes; and activating (operating or triggering) the at least one second sensor based on at least a part of the additional information, without activating (operating or triggering) the second sensor when at least a part of the motion does not satisfy a selected (or specified) condition.

According to various embodiments, the measuring of the biometric information based on the at least one second sensor may include: accumulating the number of failures when the measurement of the biometric information fails; and deactivating the second sensor when the cumulative number of failures satisfies a selected (or specified) condition.

According to various embodiments, the measuring of the biometric information based on the at least one second sensor may include: acquiring data from at least one of the at least one first sensor and the at least one second sensor; and changing at least one condition for activating (operating or triggering) the at least one second sensor based on at least a part of the data.

According to various embodiments, the measuring of the biometric information based on the at least one second sensor may include: acquiring data from at least one of the at least one first sensor and the at least one second sensor; and changing at least one of a period of time or a sampling rate for activating (operating or triggering) the at least one second sensor based on at least a part of the data.

According to various embodiments, the measuring of the biometric information based on the at least one second sensor may include: acquiring data about the user's physical characteristic (e.g., skin color) or state (e.g., a sleep state or an active state); and changing at least one condition for activating (operating or triggering) the at least one second sensor based on at least a part of the data.

According to various embodiments, the acquiring of the additional information on the motion may include changing the selected (or specified) time for acquiring the additional information based on data acquired from at least one of the at least one first sensor and the at least one second sensor.

According to an embodiment, a method of operating an electronic device may include: determining whether the electronic device is close to a user's body; acquiring information on the motion of the electronic device using at least one first sensor within a selected (or specified) time range from the time when a selected (or specified) period passes in a case where the electronic device is close to the user's body part; and acquiring the user's biometric information through at least one second sensor based on at least a part of the information on the motion.

The electronic device and the method, according to the various embodiments of the present disclosure, can selectively operate a biometric sensor based on a selected (or specified) condition, thereby measuring biometric information while reducing power consumption.

Further, the electronic device and the method, according to the various embodiments of the present disclosure, can operate a biometric sensor according to a selected (or specified) period, thereby continually measuring biometric information.

In addition, the electronic device and method, according to the various embodiments of the present disclosure, can measure the motion of the electronic device and can selectively measure a user's biometric information based on the measured motion, thereby improving the accuracy of the measured biometric information.

A terminology "module" used for the present disclosure may mean, for example, a unit including a combination of one or two or more among a hardware, a software, or a firmware. A "module" may be interchangeably used with a terminology such as a unit, a logic, a logical block, a component, or a circuit, etc. A "module" may be a minimum unit of an integrally configured part or a portion thereof. A "module" may be a minimum unit performing one or more functions or a portion thereof. A "module" may be mechanically or electronically implemented. For example, a "module" according to the present disclosure may include at least one of an application-specific IC (ASIC) chip, a field-programmable gate arrays (FPGAs), or a programmable logic device which are known, or to be developed in the future, and performing certain operations.

According to various embodiments, at least a portion of an apparatus (e.g., modules or functions thereof) or a method (e.g., operations) according to the present disclosure may be implemented as an instruction stored in a non-transitory computer-readable storage media, for example, in the form of a programming module. An instruction, when executed by one or more processors (e.g., the processor 120), may configure the one or more processors to perform a function corresponding to the instruction. The non-transitory computer-readable storage media may be, for example, the memory 130. At least a portion of a programming module may be implemented (e.g., executed) by, for example, the processor 120. At least a portion of the programming module may include, for example, a module, a program, a routine, sets of instructions, or a process, etc. for performing one or more functions.

The non-transitory computer-readable storage media may include a hard disk, a magnetic media such as a floppy disk and a magnetic tape, compact disc ROM (CD-ROM), optical media such as DVD, magneto-optical media such as a floptical disk, and a hardware device specially configured for storing and performing a program instruction (e.g., a programming module) such as ROM, RAM, a flash memory, etc. Also, the program instruction may include not only a machine language code generated by a compiler but also a high-level language code executable by a computer using an interpreter, etc. The above-described hardware device may be configured to operate as one or more software modules in order to perform an operation of the present disclosure, and vice versa.

A module or a programming module according to the present disclosure may include at least one of the above-described elements, omit a portion thereof, or further include additional other elements. Operations performed by a module, a programming module, or other elements according to the present disclosure may be executed in a sequential, parallel, or heuristic method. Also, a portion of the operations may be executed in a different sequence, omitted, or other operations may be added.

While the present disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the present disclosure as defined by the appended claims and their equivalents.

45

What is claimed is:

1. An electronic device comprising:

a housing;

a fastener coupled to a part of the housing and detachably
attachable to a part of a user's body;

at least one first sensor;

at least one second sensor;

a memory; and

at least one processor electrically connected to the at least
one first sensor, the at least one second sensor, and the
memory,

wherein the memory stores one or more computer pro-
grams including instructions which, when executed by
the at least one processor, cause the at least one
processor to:

determine whether the electronic device is worn on the
part of the user's body using the at least one first
sensor,

acquire first information on a motion of the electronic
device using at least a part of the at least one first
sensor for a second period of time shortly before
and/or immediately after a first period of time, when
the at least one processor determines that the elec-
tronic device is worn on a part of the user's body,

activate the at least one second sensor based on at least
a part of the first information on the motion,
detect a biometric information of the user using the
activated at least one second sensor,

during detecting the biometric information, acquire
second information on a motion of the electronic
device using at least the part of the at least one first
sensor, and

in response to identifying that the motion of the elec-
tronic device included in the second information is
more than a threshold motion, inactivate the at least
one second sensor.

2. The electronic device of claim 1, wherein the second
period of time is shorter than the first period of time.

3. The electronic device of claim 1, wherein the at least
one second sensor comprises a biometric sensor.

4. The electronic device of claim 3, wherein the biometric
sensor comprises at least one of a heart rate sensor, a blood
pressure sensor, an electrocardiogram sensor, a skin mois-
ture sensor, or an oxygen saturation sensor.

5. The electronic device of claim 1, wherein when the first
information on the motion does not satisfy a selected con-
dition, the instructions which, when executed by the at least
one processor, further cause the at least one processor to:

acquire additional information on the motion of the elec-
tronic device using at least a part of the at least one first
sensor for a third period of time, which is shorter than
the first period of time, immediately after the second
period of time passes; and

activate the at least one second sensor based on at least a
part of the additional information.

6. The electronic device of claim 5, wherein at least one
of the second period of time or the third period of time
comprises a time point when the first period of time passes.

7. The electronic device of claim 1, wherein the second
period of time comprises a time point when the first period
of time passes.

8. The electronic device of claim 1, wherein, when data is
not received from the activated second sensor within a
selected time, the instructions further configure the at least
one processor to activate the second sensor again after the
selected time passes.

46

9. The electronic device of claim 8, wherein the instruc-
tions further configure the at least one processor to:

acquire additional information on the motion of the elec-
tronic device using at least a part of the at least one first
sensor for a period of time that is substantially the same
as the second period of time before the selected time
passes; and

activate the at least one second sensor again based on at
least a part of the additional information.

10. The electronic device of claim 1, wherein the instruc-
tions further configure the at least one processor to:

acquire data from at least one of the at least one first
sensor and the at least one second sensor; and
change at least one of the first period of time, the second
period of time, and the third period of time based on at
least a part of the acquired data.

11. The electronic device of claim 1, wherein the instruc-
tions further configure the at least one processor to:

acquire data from at least one of the at least one first
sensor and the at least one second sensor; and
change at least one condition for activating the at least one
second sensor based on at least a part of the data.

12. The electronic device of claim 1, wherein the instruc-
tions further configure the at least one processor to:

acquire data from at least one of the at least one first
sensor and the at least one second sensor; and
change at least one of a period of time and a sampling rate
for activating the at least one second sensor based on at
least a part of the data.

13. A method comprising:

determining whether an electronic device is worn on a
part of a user's body using at least one first sensor of the
electronic device;

acquiring first information on a motion of the electronic
device using at least a part of the at least one first sensor
for a second period of time shortly before and/or
immediately after a first period of time when it is
determined that the electronic device is worn on a part
of the user's body;

activating at least one second sensor based on at least a
part of the first information on the motion of the
electronic device;

detecting a biometric information of the user using the
activated at least one second sensor;

during detecting the biometric information, acquiring
second information on a motion of the electronic device
using at least the part of the at least one first sensor; and
in response to identifying that the motion of the electronic
device included in the second information is more than
a threshold motion, inactivating the at least one second
sensor.

14. The method of claim 13, wherein the at least one first
sensor comprises at least one of a proximity sensor, a gyro
sensor, a terrestrial magnetism sensor, or an acceleration
sensor.

15. The method of claim 13, further comprising:

acquiring additional information on the motion of the
electronic device using at least a part of the at least one
first sensor for a third period of time, which is shorter
than the first period of time, immediately after the
second period of time passes; and
activating the at least one second sensor based on at least
a part of the additional information.

16. The method of claim 15, wherein at least one of the
second period of time or the third period of time comprises
a time point when the first period of time passes.

47

17. The method of claim 13, further comprising:
 accumulating a number of failures when the detection of
 the biometric information fails; and
 deactivating the at least one second sensor when a cumu-
 lative number of failures satisfies a selected condition. 5

18. The method of claim 13, wherein the detection of the
 biometric information of the user using the activated at least
 one second sensor comprises:

acquiring data about a physical characteristic or state of
 the user; and

changing at least one condition for activating the at least
 one second sensor based on at least a part of the data. 10

19. The method of claim 13, wherein the detection of the
 biometric information of the user using the activated at least
 one second sensor comprises:

acquiring data from at least one of the at least one first
 sensor and the at least one second sensor; and

changing at least one condition for activating the at least
 one second sensor based on at least a part of the data. 15

20. An electronic device comprising:

at least one first sensor; 20

at least one second sensor;

a memory; and

at least one processor electrically connected to the at least
 one first sensor, the at least one second sensor, and the
 memory,

48

wherein the memory stores one or more computer pro-
 grams including instructions which, when executed by
 the at least one processor, cause the at least one
 processor to:

determine whether the electronic device is worn on a
 part of a user's body,

when the electronic device is worn on the part of the
 user's body, determine a motion of the electronic
 device based on first information obtained by the at
 least one first sensor,

determine whether to identify biometric information
 based on the motion of the electronic device,

in response to determining to identify biometric infor-
 mation, identify the biometric information of the
 user through the at least one second sensor,

during identifying the biometric information of the
 user, obtain second information on a motion of the
 electronic device using the at least one first sensor,
 and

in response to identifying that the motion of the elec-
 tronic device included in the second information is
 more than a threshold motion, inactivate the at least
 one second sensor.

* * * * *

EXHIBIT 2



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Choi et al.

(10) **Patent No.:** **US 10,642,359 B2**

(45) **Date of Patent:** **May 5, 2020**

(54) **WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF**

(71) Applicant: **Samsung Electronics Co., Ltd.**,
Suwon-si (KR)

(72) Inventors: **Chang Mok Choi**, Suwon-si (KR);
Sang Joon Kim, Suwon-si (KR);
Seung Keun Yoon, Suwon-si (KR)

(73) Assignee: **Samsung Electronics Co., Ltd.**,
Suwon-si (KR)

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2014, now Pat. No. 10,241,574.

(30) **Foreign Application Priority Data**

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(Continued)

(52) **U.S. Cl.**
CPC **G06F 3/015** (2013.01); **A61B 5/1123**
(2013.01); **A61B 5/221** (2013.01); **A61B**
5/6898 (2013.01);
(Continued)

(58) **Field of Classification Search**

CPC G06F 3/015
See application file for complete search history.

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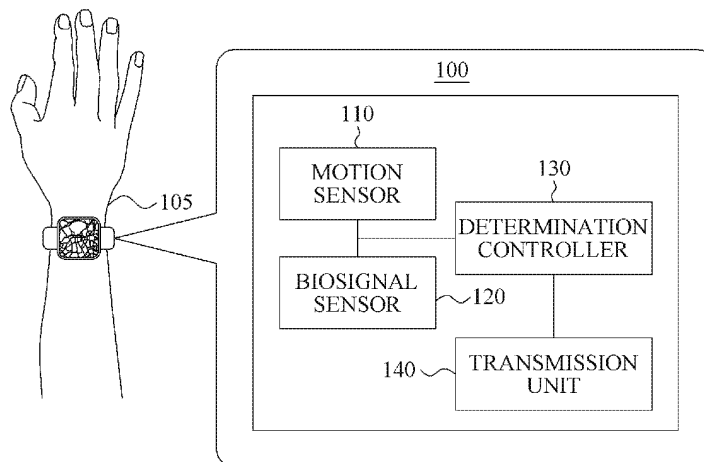
Primary Examiner — Gustavo Polo

(74) *Attorney, Agent, or Firm* — NSIP Law

(57) **ABSTRACT**

A wearable biosignal interface, an apparatus for operating an external device, and an operation method of a wearable biosignal interface are provided. A wearable biosignal interface may include a motion sensor configured to obtain a first signal based on a motion of a target, a biosignal sensor disposed on the target and configured to obtain a second signal from the target, and a determination controller configured to determine a validity of the second signal based on the first signal.

34 Claims, 5 Drawing Sheets



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	CPC	<i>A61B 5/7475</i> (2013.01); <i>G06F 1/163</i>	KR	10-2010-0027375 A 3/2010
		(2013.01); <i>G06F 1/1694</i> (2013.01); <i>G06F</i>	KR	10-2011-0062558 A 6/2011
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		<i>G06F 3/038</i> (2013.01); <i>G06F 3/0346</i>	KR	10-2012-0014233 A 2/2012
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FIG. 1

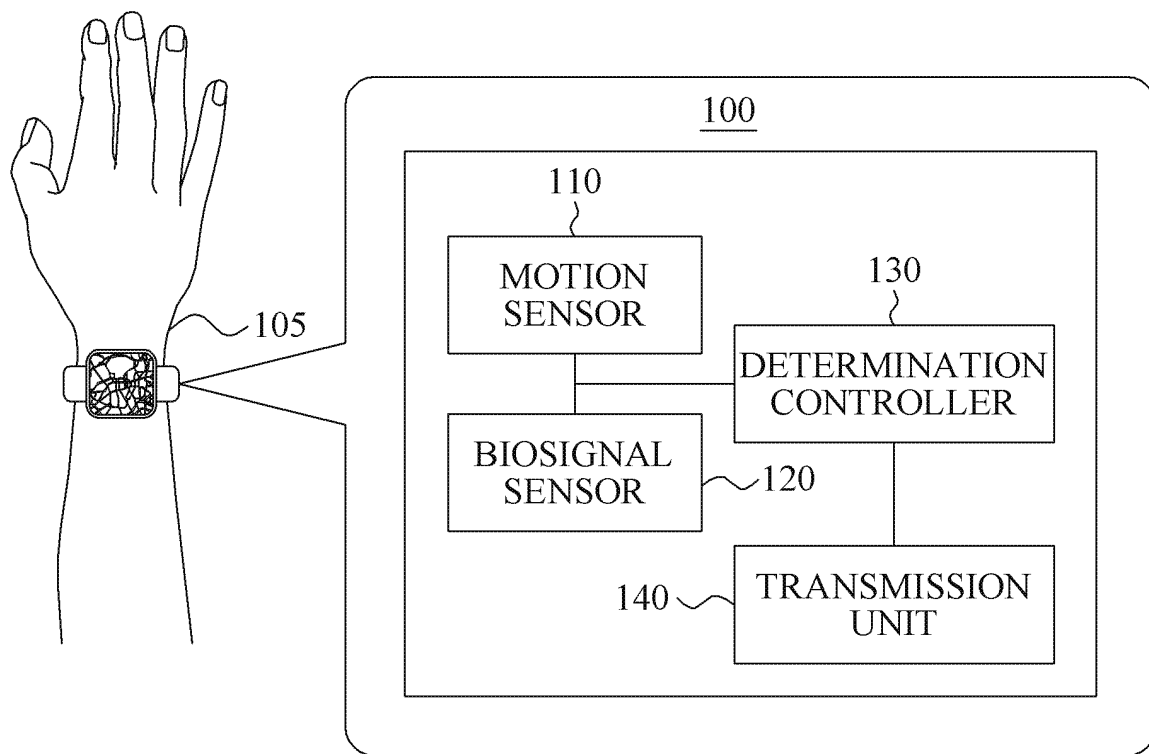


FIG. 2A

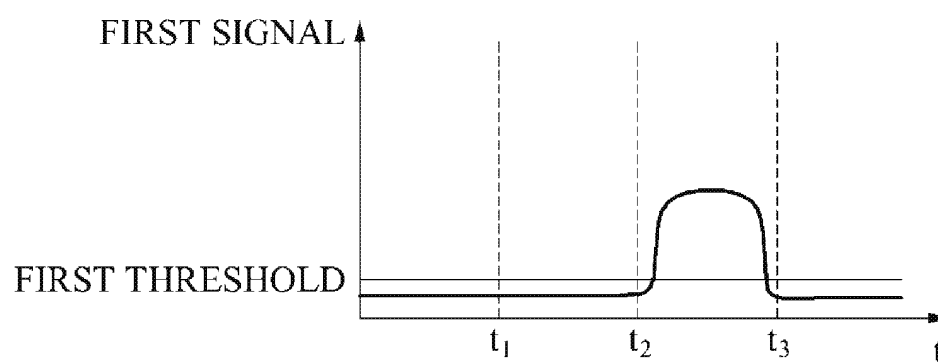


FIG. 2B

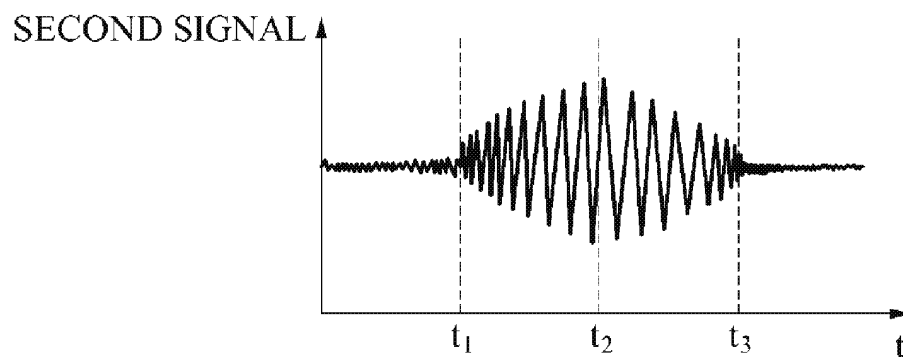


FIG. 2C

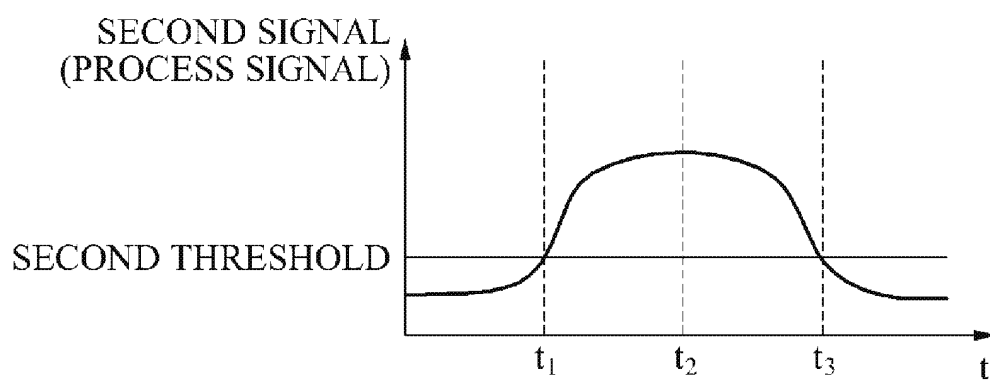


FIG. 2D

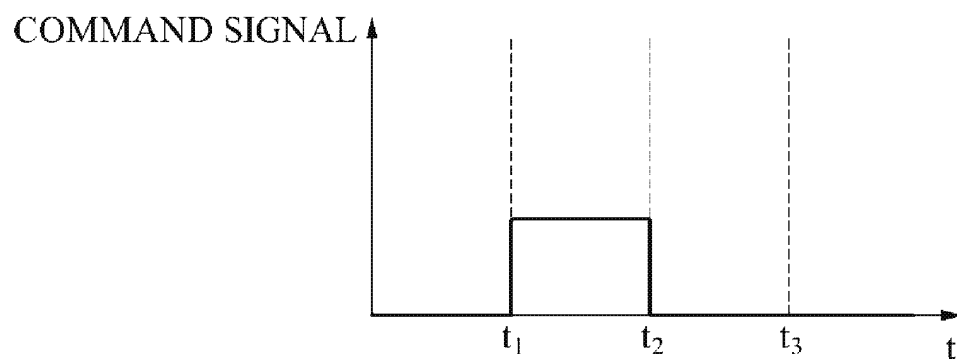


FIG. 3

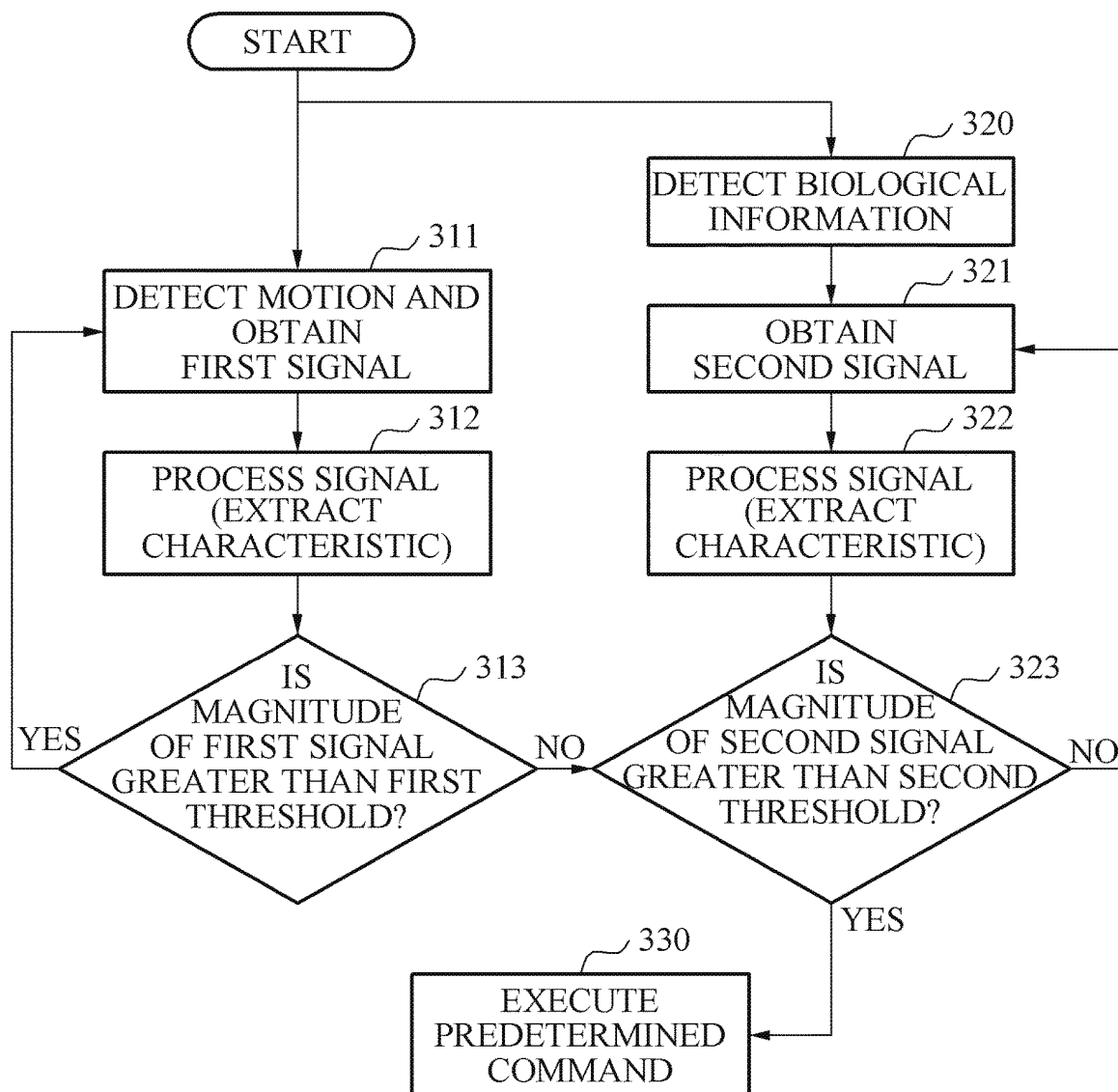
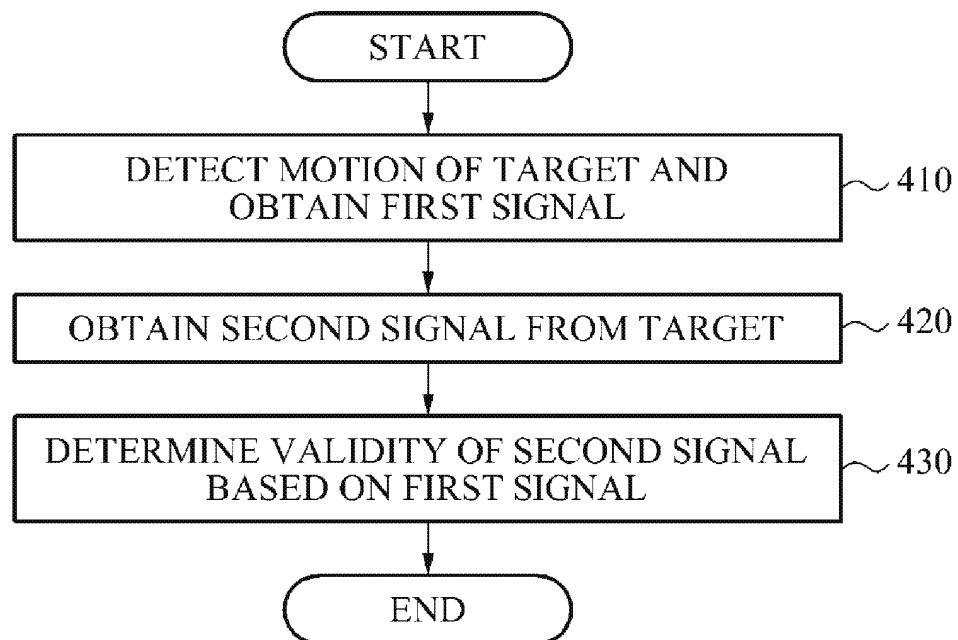


FIG. 4



WEARABLE BIOSIGNAL INTERFACE AND METHOD THEREOF

CROSS-REFERENCE TO RELATED APPLICATION(S)

This application is a Continuation application of U.S. application Ser. No. 14/913,085, filed on Feb. 19, 2016, which is a U.S. National Stage Application of International Application No. PCT/KR2014/004073, filed on May 8, 2014, which claims the benefit under 35 USC 119(a) of Korean Patent Application No. 10-2013-0098476 filed on Aug. 20, 2013, in the Korean Intellectual Property Office, the entire disclosure of which is incorporated herein by reference for all purposes.

BACKGROUND

1. Field

The following description relates to a wearable biosignal interface and an operation method thereof, and to an apparatus for controlling an external device and a method of controlling an external device using a wearable apparatus.

2. Description of Related Art

Recently, wearable-type mobile devices are emerging at an accelerated rate amid the proliferation of smartphones. A wearable-type mobile device may include a separate screen for displaying information obtained by a smartphone through a wireless data network, allowing its user to process data without directly viewing the screen of the smartphone.

A gesture recognized by an accelerometer or a biosignal sensor may be used to operate the wearable mobile device. However, operating the wearable mobile device based only on the gesture may increase a probability of misrecognizing the intention of the user. For example, in a wearable mobile device designed to be operated by recognizing a hand gesture, a movement of the arm connected to the hand may be recognized as a hand gesture, misinterpreting the intention of the user in operating the wearable mobile device.

Accordingly, a wearable mobile device model that recognizes an intention of a user with a greater precision is desirable to improve user experience.

SUMMARY

In one general aspect, there is provided a wearable biosignal interface including a motion sensor configured to obtain a first signal based on a motion of a target, a biosignal sensor disposed on the target and configured to obtain a second signal from the target, and a determination controller configured to determine a validity of the second signal based on the first signal.

The second signal may include biological information obtained by one or more biosensor in contact with skin of the target.

The determination controller may be configured to determine the second signal to be valid in response to a magnitude of the first signal satisfying a first threshold value.

The determination controller may be configured to identify an interval during which a magnitude of the first signal is less than or equal to a first threshold value and to determine a signal, among a plurality of second signals obtained by the biosignal sensor in the identified interval, to be a valid second signal.

The determination controller may be configured to determine the first threshold value based on a magnitude of the second signal.

The determination controller may be configured to determine the second signal to be invalid, irrespective of the first signal, in response to the magnitude of the second signal not satisfying a second threshold value.

In response to a determination that the second signal is valid, the determination controller may be configured to operate in an active mode and to control an external device based on the second signal.

In another general aspect, an operation method of a wearable biosignal interface may involve using a motion sensor to obtain a first signal based on a motion of a target, using a biosignal sensor disposed on the target to obtain a second signal, and determining a validity of the second signal based on the first signal.

The second signal may include biological information obtained by one or more biosensor in contact with skin of the target.

The determining of the validity may involve determining the second signal to be valid in response to a magnitude of the first signal satisfying a first threshold value.

The determining of the validity may involve identifying an interval during which a magnitude of the first signal is less than or equal to a first threshold value, and determining a signal, among second signals obtained by the biosignal sensor in the identified interval, to be a valid second signal.

The determining of the validity may further include determining a first threshold value based on a magnitude of the second signal.

The determining of the validity may involve determining the second signal to be invalid, irrespective of the first signal, in response to a magnitude of the second signal not satisfying a second threshold value.

The general aspect of the method may further involve operating in an active mode and controlling an external device based on the second signal, in response to the second signal being determined to be valid.

In another general aspect, an apparatus including a motion sensor configured to detect movement of a target, a second sensor configured to detect biological information of the target, and a controller configured to determine an interval of stability from the movement of the target detected by the motion sensor, to determine whether a magnitude of a biological signal obtained during the interval is less than or equal to a first threshold value, and to identify a command for controlling another device in response to the magnitude being less than or equal to the first threshold value.

The interval may be determined by extracting a first processed signal from the detected movement of the target, and the biological signal may be obtained by extracting a second processed signal from the biological information obtained by the second sensor, the second sensor comprising one or more biosensor.

The interval may correspond to a time duration in which a magnitude of the first processed signal is less than or equal to a second threshold value.

The general aspect of the apparatus may further include a transmission unit configured to transmit the command to the another device, and the determination controller may be configured to identify the command based on the second processed signal.

In response to a determination that the magnitude of the biological signal is greater than the first threshold value, the

apparatus may be configured not to transmit a command based on the biological information obtained during the interval.

Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram illustrating an example of a wearable biosignal interface.

FIG. 2A through FIG. 2D are diagrams illustrating an example of a method of determining the validity of a signal based on a first signal and a second signal.

FIG. 3 is a diagram illustrating an example of a method of performing a command given to an external device, using a magnitude of a first signal and a magnitude of a second signal.

FIG. 4 is a flowchart illustrating an example of a method of operating a wearable biosignal interface.

Throughout the drawings and the detailed description, unless otherwise described, the same drawing reference numerals will be understood to refer to the same elements, features, and structures. The relative magnitude and depiction of these elements may be exaggerated for clarity, illustration, and convenience.

DETAILED DESCRIPTION

The following detailed description is provided to assist the reader in gaining a comprehensive understanding of the methods, apparatuses, and/or systems described herein. Accordingly, various changes, modifications, and equivalents of the methods, apparatuses, and/or systems described herein will be suggested to those of ordinary skill in the art. The progression of processing steps and/or operations described is an example; however, the sequence of and/or operations is not limited to that set forth herein and may be changed as is known in the art, with the exception of steps and/or operations necessarily occurring in a certain order. Also, description of well-known functions and constructions may be omitted for increased clarity and conciseness.

The features described herein may be embodied in different forms, and are not to be construed as being limited to the examples described herein. Rather, the examples described herein have been provided so that this disclosure will be thorough and complete, and will convey the full scope of the disclosure to one of ordinary skill in the art.

As described herein, a sensor may refer to a sensor or a plurality of sensors that may be attached to a target to be in contact therewith or disposed thereon so as to obtain a signal generated from the target or to detect a motion of the target.

For example, a sensor may include a biosensor that may be attached to a body part of a living being or living organism such as, for example, a wrist or a forearm of a person, in order to obtain a biological or physiological signal. For example, a biosensor may be able to detect signals generated by muscle contraction or muscle relaxation of a body part, such as a wrist or a heart. Herein, such a sensor may refer to as a biosignal sensor. A biosignal sensor may detect biological information from a target and may convert the biological information into a signal waveform to generate a biosignal. Such a biosignal sensor may include a photo sensor, a piezoelectric sensor, a force sensor, an electromyogram (EMG) sensor, and the like.

Herein, a muscle may refer to a striated muscle, a cardiac muscle or a smooth muscle, and may refer to a muscle group, a number of muscle groups in a body region, bundles

of muscle fibers and the like. For instance, an arm may include several different muscle groups, including a tricep, a bicep and a brachioradialis.

A sensor may include an acceleration sensor, a gyro sensor, a geomagnetic sensor, and a Global Positioning System (GPS) sensor. Such a sensor may directly or indirectly monitor a motion performed by the living target and may obtain a signal associated with the motion performed by the living target. Hereinafter, the sensor described above will be referred to as a motion sensor.

To control an external device using a biosignal obtained by a biosignal sensor, an example of a wearable biosignal interface may ignore a biosignal in the event that a magnitude of a signal associated with a motion of a target exceeds a threshold value. Conversely, the wearable biosignal interface may receive the biosignal in the event that the magnitude of the signal associated with the motion of the target is less than or equal to the threshold value.

The wearable biosignal interface may make a determination as to whether a motion is unintentionally or spontaneously made by a user, and may ignore a biosignal obtained as a result of the unintentional or spontaneous motion. Using this approach, the accuracy of determination regarding the intention of the user may be improved.

FIG. 1 is a diagram illustrating an example of a wearable biosignal interface 100.

Referring to FIG. 1, the example of the wearable biosignal interface 100 include a motion sensor 110, a biosignal sensor 120, a determination controller 130, and a transmission unit 140.

The motion sensor 110 may detect a motion of a target 105 and may obtain a first signal. For instance, the first signal may include an acceleration signal, a rotation signal, a bearing signal, and a coordinate signal. These signals may be associated with velocity variance based on a motion of the target 105.

The motion sensor 110 may monitor a motion of the target 105 and may receive the first signal recognizing a movement of the wearable biosignal interface 100 associated with the motion. Also, the motion sensor 110 may identify a gesture performed by the target 105 by obtaining a second signal after obtaining a first signal associated with an identified gesture.

The motion sensor 110 may include, for example, an acceleration sensor. Such a motion sensor 110 may convert an acceleration signal corresponding to the obtained first signal into a velocity component by calculating an integral of the acceleration signal. The motion sensor 110 may determine one or more numerical values indicating a magnitude of a motion of the target 105.

In the event that the target 105 is an arm that is flexed in any direction, the motion sensor 110 may obtain a first signal associated an "arm motion" based on the flexing. Also, the motion sensor 110 may measure a magnitude of the obtained first signal and may measure a magnitude of a velocity of the "arm motion."

The motion sensor 110 may include a gyrosensor. Using a gyrosensor, such a motion sensor 110 may detect a rotation signal corresponding to the first signal.

The target 105 may be an arm. In the event that the arm performs a large rotation motion, the motion sensor 110 may obtain a first signal associated with the "arm rotation" based on the rotating motion of the arm. In addition, the motion sensor 110 may measure a magnitude of the obtained first signal and may measure a magnitude of a rotational velocity of the "arm rotation."

5

The motion sensor **110** may include a geomagnetic sensor or a GPS sensor. Using the geomagnetic sensor or the GPS sensor, such a motion sensor **110** may detect a bearing signal or a coordinate signal corresponding to the first signal.

The motion sensor **110** may be fixed to a form of a bracelet fastened onto the target **105**, along with the biosignal sensor **120**.

The biosignal sensor **120** may be in contact with the target **105** and may obtain a second signal from the target **105**. In the illustrated example, the biosignal sensor **120** directly contact the skin of the target **105**. However, in another example, the biosignal sensor **120** may be disposed on the target **105** indirectly, such as through a clothing that covers the skin. The second signal may be a biosignal associated with biological information such as, for example, muscle contraction and muscle relaxation occurring in the target **105**. The second signal may be a bio-electric/magnetic signal, a bio-impedance signal, and a bio-mechanical signal generated in association with a muscle.

The biosignal sensor **120** may be in contact with the target **105** and may obtain the second signal generated by at least one of muscle contraction and muscle relaxation. In the event that the target **105** is a wrist, the biosignal sensor **120** may detect the second signal generated by the muscle contraction or muscle relaxation of the wrist.

The target **105** may be a wrist, a forearm, a face, a neck, or the like of a living being, and a muscle that generates a second signal may be located within the portion of body.

The biosignal sensor **120** may include at least one biosensor that may detect the second signal from a muscle of the target **105**, and may monitor the contraction or relaxation of the muscle using the at least one biosensor. The biosensor may be a photo sensor, a piezoelectric sensor, a force sensor, an electromyogram (EMG) sensor, and the like.

For example, the biosignal sensor **120** may include the photo sensor. Using the photo sensor, such a biosignal sensor **120** may obtain the second signal associated with a status of the target **105**, based on light scattering.

The biosignal sensor **120** may include a piezoelectric sensor or a force sensor. Using a piezoelectric sensor or a force sensor, such a biosignal sensor **120** may obtain a second signal associated with an extension and/or contraction of a muscle of the target **105**.

The biosignal sensor **120** may include an EMG sensor. Using an EMG sensor, a biosignal sensor may obtain a second signal associated with the muscle contraction or relaxation of a heart muscle of a target **105**.

The biosignal sensor **120** may be fixed to the bracelet fastened onto the target **105**, along with the motion sensor **110**. A sensing surface of the biosignal sensor **120** may be fixed to the bracelet to be in a direct contact with the target **105**. On the sensing surface, the at least one biosensor may be arranged to detect the second signal generated from a target **105**. Using the biosignal sensor **120**, the bracelet may broadly detect a second signal such as the bio-electric/magnetic signal, the bio-impedance signal, and the bio-mechanical signal over an area of the body of the target **105**.

The bracelet may be fastened onto the target **105** by being wrapped around a wrist muscle over the skin, for example. The bracelet may be fastened onto the target **105** by covering the wrist muscle with an area of the bracelet, for example, the sensing surface, in which one or more biosignal sensor **120** is arranged. Using one or more biosensors, the biosignal sensor **120** may collect a second signal from throughout an entire area of the wrist muscle of the target **105**.

The determination controller **130** may determine a validity of the second signal based on the first signal. The

6

determination controller **130** may determine whether the second signal obtained using the biosignal sensor **120** is generated from the target **105** due to an intentional motion or a motion incidentally generated from the target **105** due to an unintentional or spontaneous motion made irrespective of an intention of a user to operate an external device.

The determination controller **130** may determine the second signal to be valid when a magnitude of the first signal satisfies a first threshold value.

The first threshold value may be a reference value used to determine the intentional motion of the target **105**. The first threshold value may be determined as an average value of magnitudes of first signals obtained when the user moves a muscle for an actual operation.

The determination controller **130** may determine the first threshold value based on a magnitude of the second signal to be obtained. The determination controller **130** may determine the first threshold value for an operation involving a large motion of the target **105** to be greater in proportion to the magnitude of the second signal. Accordingly, the second signal generated by an intentional motion may not be ignored due to a low threshold value.

For example, the determination controller **130** may determine the first threshold value for a quick display shift operation of a display apparatus for which a relatively greater second signal is generated to be greater than the first threshold value for an ON/OFF operation of an illumination apparatus for which a relatively smaller second signal is generated.

To determine a validity of a second signal, the determination controller **130** may identify an interval during which the magnitude of the first signal is less than or equal to the first threshold value and may determine a signal, among the second signals obtained during the identified interval, to be a valid second signal. The determination controller **130** may determine, to be valid, the second signal obtained in the interval during which the magnitude of the first signal is less than or equal to the first threshold value and a small motion of the target **105** is performed.

In the event that the magnitude of the second signal does not satisfy a second threshold value, the determination controller **130** may determine the second signal to be invalid, irrespective of the first signal. The second threshold value may be a reference value used to determine a second signal that is valuable as information. The second threshold value may be set as an averaged minimum value obtained based on data obtained by simulations in which users moved their muscles to initiate an actual operation.

In response to the magnitude of the second signal being less than or equal to the determined second threshold value, the determination controller **130** may determine whether the target **105** moves intentionally or unintentionally to be indefinite and invalidate the second signal. Conversely, in response to the magnitude of the second signal being greater than the determined second threshold value, the determination controller **130** may further determine whether the magnitude of the first signal satisfies the first threshold value and may determine the validity of the second signal.

Based on the determination, the determination controller **130** may operate in an active mode in response to the second signal being determined to be valid and may allow an external device to be controlled based on the second signal.

An external device may refer to a device controlled by a command signal, such as a mobile device, a television (TV) screen, a digital versatile disc (DVD) player, a radio, a heating thermostat, a vehicle door, and the like. The com-

mand signal may be generated in association with a valid second signal of the wearable biosignal interface **100**.

In the event that the magnitude of the first signal is less than or equal to the first threshold value and the magnitude of the second signal is greater than the second threshold value, the determination controller **130** may shift a mode of the wearable biosignal interface **100** to the active mode and may allow the external device to be controlled based on the second signal. Thus, the accuracy of determining an intention of a user to operate the device may be improved.

To control the external device after the shift to the active mode, the wearable biosignal interface **100** may transmit the command signal based on the second signal determined to be valid to an appliance or a device in order to enable the external device to perform a control operation, using the transmission unit **140**. For example, the wearable biosignal interface **100** may transmit a command signal to an appliance such as a TV, an air conditioner, a computer, a refrigerator, and the like, or to a living environment controlling device such as a light, a curtain, a window, a door lock device, and the like. The signal transmission may enable the external device to perform a control operation of the appliance or device such as an ON/OFF operation, an open/close operation, and the like, in accordance with the command signal.

In another example, the wearable biosignal interface **100** may use a valid second signal (or a second signal determined to be valid) as a signal of a remote controller of a vehicle to start the vehicle or as an ID signal to make a payment.

In yet another example, the wearable biosignal interface **100** may use the valid second signal for an interaction control with a computer, or a signal to edit a file stored on the computer or to move a page for a presentation.

The wearable biosignal interface **100** may detect and invalidate a biosignal generated by the unintentional motion of the target **105**; thus, the wearable biosignal interface **100** may optimally avoid a control error for the external device.

The wearable biosignal interface **100** may identify a biosignal generated only when the target **105** is under a certain level of stability, and may use the biosignal obtained during the stable condition to control an external device; thus, the wearable biosignal interface **100** may provide a precise determination of the user's intention for operating the external device.

FIG. 2A through FIG. 2D are diagrams illustrating examples of methods of determining a second signal as a valid signal based on a first signal and a second signal.

FIG. 2A is a diagram illustrating an example of a first signal obtained using a motion sensor **110** of FIG. 1 of a wearable biosignal interface **100** of FIG. 1. When a target **105** of FIG. 1 performs a rapid motion, such as a motion of flexing an arm, in a interval from t_2 to t_3 , the motion sensor **110** may obtain the first signal having a magnitude measured to be great in the interval from t_2 to t_3 . The first signal obtained in the interval from t_2 to t_3 may be greater than a first threshold value used as a reference value to determine an intentional motion of the target **105**.

FIG. 2B is a diagram illustrating an example of a second signal obtained by a biosignal sensor **120** of FIG. 1 of a wearable biosignal interface **100** of FIG. 1. As illustrated in FIG. 2B, the obtained second signal may be in a form of an analog sine waveform and may have a greater waveform in a interval from t_1 to t_3 during which muscles of an arm are tense due to the flexing of the arm.

FIG. 2C is a graph illustrating an example in which a determination controller **130** of FIG. 1 of a wearable biosignal interface **100** of FIG. 1 processes magnitudes of the

obtained second signals and compares the magnitudes. The determination controller **130** may plot, based on the second signal of FIG. 2B, a parabola in which an inclination increases drastically after a point t_1 at which a waveform becomes larger and the inclination decreases drastically after a point t_3 at which the waveform becomes smaller.

The second signal in the interval from t_1 to t_3 may be obtained when satisfying a second threshold value used as a reference value to determine a second signal that is valuable as information.

FIG. 2D illustrates an example of a method of determining, by a determination controller **130** of FIG. 1, a validity of a second signal based on a first signal and a second signal.

The determination controller **130** may invalidate the second signal obtained in the interval from t_2 to t_3 in which the first signal exceeding the first threshold value is obtained. Also, the determination controller **130** may invalidate the second signal obtained during an interval before t_1 and after t_3 during which the second signal less than or equal to the second threshold value is obtained.

As shown in FIG. 2D, the determination controller **130** may determine the second signal obtained in of the interval from t_1 to t_2 to be a valid signal and may enable a command signal to be generated to control an external device during the interval.

The wearable biosignal interface **100** may operate, using the second signal, an external device such as a mobile device, a TV screen, a DVD, a radio, a heating thermostat, a vehicle door, and the like. The wearable biosignal interface **100** may allow the operation of the external device, using the second signal, only when the magnitude of the first signal decreases to be less than or equal to the first threshold value due to a small motion performed by a user in order to improve the accuracy of the wearable biosignal interface **100** in determining an intention of a user for operation.

The wearable biosignal interface **100** may also allow the operation of the external device in response of the magnitude of the second signal satisfying the second threshold value, indicating that the second signal includes valuable information.

FIG. 3 is a diagram illustrating an example of a method of performing a command given to an external device, using a magnitude of a first signal and a magnitude of a second signal.

Referring to FIG. 3, in **311**, a motion sensor of a wearable biosignal interface may detect a motion of a target, and the motion sensor may obtain a first signal. For example, the motion sensor may detect and obtain a first signal associated with the flexing of an arm. In this example, the arm is the target from which signals can be detected.

In **312**, the wearable biosignal interface may process the first signal obtained by the motion sensor and may extract a characteristic. For example, wearable biosignal interface may extract the magnitude of the first signal corresponding to a characteristic associated with a speed at which an arm is flexed.

In **313**, the wearable biosignal interface may determine whether the magnitude of the first signal is greater than a first threshold value. For example, the wearable biosignal interface may determine a magnitude of a first signal obtained in response to a user moving a muscle to actually operate an external device to be the first threshold value, and may compare the first threshold value to magnitudes of the obtained first signals.

Also, in **313**, the wearable biosignal interface may identify an interval during which the magnitude of the first signal is less than or equal to the first threshold value.

In response to a determination that the magnitude of the first signal is greater than the first threshold value, for example, in a 'Yes' direction from 313, the wearable biosignal interface may ignore the second signal obtained in the same interval during which the first signal is obtained. Also, the wearable biosignal interface may re-obtain the first signal in 311.

Conversely, in response to a determination that the magnitude of the first signal is less than or equal to the first threshold value, for example, in a 'No' direction of 313, the wearable biosignal interface may proceed to 323, as described later. The wearable biosignal interface may determine whether the second signal is valid based on a result of comparing the magnitude of the second signal obtained in the same interval during which the first signal is obtained to the second threshold value.

If the magnitude of the second signal is less than the second threshold value, in 321, a biosignal sensor of the wearable biosignal interface may detect and may obtain a second signal from the target. For example, the biosignal sensor may obtain the second signal in response to detecting a tenseness of the muscles of the arm due to a flexing of the arm. The second signal may be obtained from the biological information detected by the biosignal sensor in 320.

In 322, the wearable biosignal interface may process the second signal obtained by the biosignal sensor 320 and may extract a characteristic of the second signal. For example, the wearable biosignal interface may extract a magnitude of the second signal corresponding to a characteristic associated with the tension in the flexed arm.

In 323, the wearable biosignal interface may determine whether the magnitude of the second signal is greater than a second threshold value. For example, the wearable biosignal interface may determine a reference value of the magnitude of the second signal, which may be a valuable information for determining the second threshold value, and may compare the second threshold value to magnitudes of the obtained second signals. In 323, the wearable biosignal interface may determine a signal obtained in the identified interval during which the magnitude of the first signal is less than or equal to the first threshold value to be a valid second signal.

In the event that the magnitude of the second signal is less than or equal to the second threshold value, for example, a 'No' direction of 323, the wearable biosignal interface may ignore the obtained second signal irrespective of a determination made in 313. Also, the wearable biosignal interface may re-obtain the second signal in 321.

Conversely, in the event that the magnitude of the second signal is greater than the second threshold value, for example, in a 'Yes' direction of 323, the wearable biosignal interface may ultimately determine the second signal to be valid based on a result of determination in 313 and may proceed to 330 to execute the predetermined command.

Operations described in the foregoing include using a motion sensor to obtain a first signal in 311, and using a biosignal sensor to obtain a second signal in 321. However, it may be possible to perform 321 prior to or concurrent with 311.

In 330, the wearable biosignal interface may execute a command based on the second signal determined to be valid. In 330, the wearable biosignal interface may operate in an active mode and may allow an external device to be controlled based on the second signal. Using the second signal that is determined to be valid, the wearable biosignal inter-

face may operate an external device such as a mobile device, a TV screen, a DVD player, a radio, a heating thermostat, a vehicle door and the like.

Hereinafter, operations of a wearable biosignal interface 100 according to another example will be described with reference to FIG. 4.

FIG. 4 is a flowchart illustrating an example of an operation method of a wearable biosignal interface 100.

In 410, the wearable biosignal interface 100 may detect a motion of a target 105 of FIG. 1 and may obtain a first signal. The first signal may be, for example, an acceleration signal, a rotation signal, a bearing signal, a coordinate signal, and the like, that is associated with a velocity variance based on the motion of the target 105. In 410, the wearable biosignal interface 100 may monitor, using a motion sensor 110 of FIG. 1, the motion of the target 105 and may receive the first signal to perceive a movement involved with the motion. For example, the wearable biosignal interface 100 may identify a gesture performed by the target 105 in response to obtaining a biosignal, and may obtain the first signal associated with the identified gesture.

For example, the motion sensor 110 may include an acceleration sensor. By applying integral calculus, the motion sensor 110 may convert an acceleration signal obtained by the acceleration sensor as the first signal into its velocity component. Based on the calculation, the wearable biosignal interface 100 may numerically indicate a magnitude of a motion of the target.

For example, in the event that the target 105 is an arm that is flexed, the wearable biosignal interface 100 may obtain the first signal associated with an "arm motion" based on the flexing of the arm. Also, the wearable biosignal interface 100 may measure the magnitude of the obtained first signal and may measure a velocity magnitude of the "arm motion."

The motion sensor 110 may include a gyrosensor. For example, using the gyrosensor, the wearable biosignal interface 100 may measure a rotation signal corresponding to the first signal.

In one example, target 105 may be an arm, and the arm may be performing a large rotation. The wearable biosignal interface 100 may obtain the first signal associated with the arm rotation based on the detected movement of the arm. Also, the wearable biosignal interface 100 may measure the magnitude of the obtained first signal and/or may measure the magnitude of the angular velocity of the arm rotation.

The motion sensor 110 may include a geomagnetic sensor or a GPS sensor. Using the geomagnetic sensor or the GPS sensor, the motion sensor 110 may measure a bearing signal or a coordinate signal corresponding to the first signal.

According to an example, the motion sensor 110 may be fixed to a form of a bracelet fastened onto the target 105, along with a biosignal sensor 120.

The wearable biosignal interface 100 may be in contact with the target 105 and may obtain a second signal from the target 105. In this example, the second signal may be a biosignal associated with biological information such as muscle contraction and muscle relaxation that is occurring in the target 105. The second signal may be a bio-electric/magnetic signal, a bio-impedance signal, and/or a bio-mechanical signal generated in association with a muscle movement or muscle tone.

In 420, the wearable biosignal interface 100 may be in contact with the target 105 through the biosignal sensor 120 and may obtain the second signal generated by at least one of muscle contraction and muscle relaxation. In an example in which the target 105 is a wrist, the wearable biosignal

11

interface **100** may detect the second signal generated by the muscle contraction or muscle relaxation of the wrist.

The target **105** may be, for example, a wrist, a forearm, a face, a neck and the like of a living being. The body part may include a muscle that generates the second signal.

The biosignal sensor **120** may include at least one biosensor detecting the second signal from a muscle of the target **105** and may monitor contraction or relaxation of the muscle. The biosensor may be a photo sensor, a piezoelectric sensor, a force sensor, an EMG sensor, and the like by using the at least one biosensor.

For example, the biosignal sensor **120** may include the photo sensor. Using the photo sensor, the wearable biosignal interface **100** may obtain the second signal associated with a state of the target **105**, based on light scattering.

The biosignal sensor **120** may include a piezoelectric sensor or a force sensor. Using the piezoelectric sensor or the force sensor, the wearable biosignal interface **100** may obtain the second signal associated with an extension and/or contraction state of the target **105**.

The biosignal sensor **120** may include an EMG sensor. Using the EMG sensor, the wearable biosignal interface **100** may obtain the second signal associated with a muscle contraction or relaxation of the target **105**.

The biosignal sensor **120** may be affixed to a bracelet fastened onto the target **105**, along with a motion sensor **110**, as illustrated in FIG. 1. In this example, a sensing surface of the biosignal sensor **120** may be affixed to the bracelet to be in a direct contact with the skin of the target **105**. On the sensing surface, one or more biosensor may be arranged to detect a second signal generated from the target **105**. For example, the bracelet may detect, using the biosignal sensor **120**, various forms of second signals such as a bio-electric/magnetic signal, a bio-impedance signal, and a bio-mechanical signal associated with the target **105**.

The bracelet may be fastened onto the target **105**, wrapping around the skin that covers a wrist muscle. For example, the bracelet may be fastened onto the target **105** by covering the wrist muscle with a specialized area such as the sensing surface of the bracelet on which the one or more biosignal sensor **120** is disposed. Using the one or more biosensor, the biosignal sensor **120** may collect the second signal throughout an entire area of the wrist muscle of the target **105**.

In **430**, the wearable biosignal interface **100** may determine a validity of the second signal based on the first signal. In **430**, the wearable biosignal interface **100** may determine whether the second signal is generated by an intentional motion of the target **105** or incidentally generated by an unintentional or spontaneous motion of the target **105**.

The wearable biosignal interface **100** may determine the second signal to be valid in the event that the magnitude of the first signal satisfies a first threshold value.

Here, the first threshold value may be a reference value used to determine the intentional motion of the target **105**. The first threshold value may be determined as an average value of magnitudes of the first signals obtained while a user moves a muscle for an actual operation.

The wearable biosignal interface **100** may determine the first threshold value based on a magnitude of the second signal to be obtained. For example, the wearable biosignal interface **100** may determine the first threshold value for operation involving a large motion of the target **105** to be greater in proportion to the magnitude of the second signal; thus, the second signal generated by the intentional motion of the target **105** may not be ignored due to a threshold value determined to be small.

12

The wearable biosignal interface **100** may determine the first threshold value for a quick display shift operation of a display apparatus for which a relatively greater second signal is generated to be greater than the first threshold value for an ON/OFF operation of an illumination apparatus for which a relatively smaller second signal is generated.

To determine a validity of the second signal, the wearable biosignal interface **100** may identify interval during which the magnitude of the first signal is less than or equal to the first threshold value and may determine a signal, among the second signals, obtained in the identified interval to be a valid second signal. The wearable biosignal interface **100** may determine the second signal obtained in the interval during which the magnitude of the first signal is less than or equal to the first threshold value and a small motion of the target **105** is determined to be valid.

The wearable biosignal interface **100** may determine the second signal to be invalid, irrespective of the first signal, in the event that the magnitude of the obtained second signal does not satisfy a second threshold value. The second threshold value may be a reference value used to determine the second signal that is valuable as information. The second threshold value may be determined as an averaged minimum value obtained a user moves a muscle for an actual operation. For instance, the second threshold value may be determined from historic values obtained based on many actual operations performed by a target.

In the event that the magnitude of the obtained second signal is less than or equal to the second threshold value, the wearable biosignal interface **100** may determine whether the target **105** moved intentionally for operation to be indefinite and accordingly, invalidate the second signal. Conversely, in the event that the magnitude of the second signal is greater than the second threshold value, the wearable biosignal interface **100** may further determine whether the magnitude of the first signal satisfies the first threshold value and may determine the validity of the second signal.

Based on a result of the determination, in the event that the second signal is determined to be valid, the wearable biosignal interface **100** may operate in an active mode and may allow an external device to be controlled by the second signal. The external device may be a general term for devices controlled by a command signal, such as a mobile device, a TV screen, a DVD player, a radio, a heating thermostat, a vehicle door, and the like. Using the wearable biosignal interface **100**, the command signal may be generated based on the valid second signal.

For example, in the event that the magnitude of the first signal is less than or equal to the first threshold value, and the magnitude of the second signal exceeds the second threshold value, the wearable biosignal interface **100** may shift a mode of the wearable biosignal interface **100** to the active mode and may allow the external device to be controlled based on the second signal; thus, the accuracy in determining an intention of a user may be improved.

To control the external device after the shift to the active mode, the wearable biosignal interface **100** may transmit the command signal based on the valid second signal to an appliance such as a TV, an air conditioner, a computer, a refrigerator, and the like, or to a living environment controlling device such as a light, a curtain, a window, a door lock device, and the like. The transmission of the command signal enables the external device to perform a control operation such as an ON/OFF operation and an open/close operation, requested by the command signal.

The wearable biosignal interface **100** may use the valid second signal as a signal for a remote controller of a vehicle to start the vehicle, or as an identification (ID) signal to make a payment.

The wearable biosignal interface **100** may use the valid second signal as a signal for interactive control of a computer, or to edit a file stored on the computer or move a page for a presentation.

The wearable biosignal interface **100** may invalidate a biosignal generated by the unintentional or spontaneous motion of the target **105** to optimally avoid a control error for the external device.

The wearable biosignal interface **100** may identify a biosignal exclusively generated during the state in which the target **105** is in a certain level of stability and may use the biosignal to control the external device. By using the biosignals generated during a stable state, it is possible to determine an intention of a user with improved precision.

Program instructions to perform a method described herein, or one or more operations thereof, may be recorded, stored, or fixed in one or more computer-readable storage media. The program instructions may be implemented by a computer. For example, the computer may cause a processor to execute the program instructions. The media may include, alone or in combination with the program instructions, data files, data structures, and the like. Examples of non-transitory computer-readable storage media include magnetic media, such as hard disks, floppy disks, and magnetic tape; optical media such as CD ROM discs and DVDs; magneto-optical media, such as optical discs; and hardware devices that are specially configured to store and perform program instructions, such as read-only memory (ROM), random access memory (RAM), flash memory, and the like. Examples of program instructions include machine code, such as produced by a compiler, and files including higher level code that may be executed by the computer using an interpreter. The program instructions, that is, software, may be distributed over network coupled computer systems so that the software is stored and executed in a distributed fashion. For example, the software and data may be stored by one or more computer readable storage mediums. Also, functional programs, codes, and code segments that accomplish the examples disclosed herein can be easily construed by programmers skilled in the art to which the examples pertain based on and using the flow diagrams and block diagrams of the figures and their corresponding descriptions as provided herein. Also, the described unit to perform an operation or a method may be hardware, software, or some combination of hardware and software. For example, the unit may be a software package running on a computer or the computer on which that software is running.

While this disclosure includes specific examples, it will be apparent to one of ordinary skill in the art that various changes in form and details may be made in these examples without departing from the spirit and scope of the claims and their equivalents. The examples described herein are to be considered in a descriptive sense only, and not for purposes of limitation. Descriptions of features or aspects in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if the described techniques are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined in a different manner and/or replaced or supplemented by other components or their equivalents. Therefore, the scope of the disclosure is defined not by the detailed description, but by the claims and their equivalents, and all variations within the

scope of the claims and their equivalents are to be construed as being included in the disclosure.

The invention claimed is:

1. A wearable device comprising:

a motion sensor;

a biological sensor; and

a controller configured to:

obtain a movement signal of an object using the motion sensor, and a biological signal of the object using the biological sensor;

determine, based on an analysis of the movement signal, whether to perform a specified function of the wearable device using the biological signal, and

perform the specified function using a select portion of the biological signal obtained during a select period, of the movement signal, determined to correspond to a specified condition.

2. The wearable device of claim **1**,

wherein the specified condition comprises the movement signal being determined to be less than or equal to a first threshold value.

3. The wearable device of claim **2**, wherein the controller is configured to determine the first threshold value based on a magnitude of the biological signal.

4. The wearable device of claim **1**, wherein the controller is configured to determine the biological signal to be invalid, irrespective of the movement signal, in response to the magnitude of the biological signal not satisfying a second threshold value.

5. The wearable device of claim **1**, wherein, in response to a determination that the biological signal is valid, the controller is configured to operate in an active mode and to control an external device based on the biological signal.

6. The wearable device of claim **1**, further comprising a transmission unit, wherein the controller is configured to transmit an instruction associated with an external electronic device to the external electronic device using the transmission unit as at least part of the specified function.

7. A wearable device comprising:

a motion sensor;

a biological sensor; and

a controller configured to:

obtain a movement signal of an object external to the wearable device using the motion sensor;

obtain a biological signal of the object using the biological sensor;

identify a period, of the movement signal, the movement signal is determined to be less than a specified condition; and

perform a specified function using a select portion of the biological signal corresponding to the period.

8. The wearable device of claim **7**, wherein the biological sensor comprises an optical sensor configured to obtain an optical signal corresponding the biological signal.

9. The wearable device of claim **7**, wherein the controller is configured to adjust, prior to the identifying of the period, the specified condition based at least in part on a magnitude of the biological signal.

10. The wearable device of claim **7**, wherein the controller is configured to:

in response to a magnitude of the biological signal being determined to be within a first specified range, determine the biological signal to be a valid signal; and

in response to the magnitude of the biological signal being determined to be within a second specified range, determine the biological signal to be an invalid signal.

15

11. The wearable device of claim 7, wherein, for the performing of the specified function, the controller is configured to generate an instruction using the select portion of the biological signal.

12. The wearable device of claim 11, wherein, for the performing of the specified function, the controller is configured to control an external electronic device based at least in part on the instruction as at least part of the specified function.

13. The wearable device of claim 12, further comprising a transmission unit, wherein the controller is configured to transmit the instruction to the external electronic device using the transmission unit.

14. A wearable device comprising:

a motion sensor;

a biological sensor; and

a controller configured to:

obtain a movement signal of an object external to the wearable device using the motion sensor;

obtain a biological signal of the object using the biological sensor;

determine a select period, of the movement signal, that meets a specified condition;

determine, as a valid signal, at least a portion of the biometric signal that is determined to correspond to the determined select period; and

perform a specified function using the valid signal.

15. The wearable device of claim 14, wherein the biological sensor comprises an optical sensor configured to obtain an optical signal as the biological signal.

16. The wearable device of claim 14, wherein the controller is configured to adjust the specified condition based at least in part on a magnitude of the biological signal prior to the determining of the select period.

17. The wearable device of claim 14, wherein, for the performing of the specified function, the controller is configured to generate an instruction using at least a portion of the valid signal.

18. The wearable device of claim 17, wherein, for the performing of the specified function, the controller is configured to control an external electronic device based at least in part on the instruction as at least part of the specified function.

19. The wearable device of claim 18, further comprising a transmission unit, wherein the controller is configured to transmit the instruction to the external device using the transmission unit.

20. The wearable device of claim 1, wherein

the controller is further configured to validate the biological signal, and

for the performing of the specified function, the controller is configured to perform the specified function using a portion of the validated biological signal.

21. A wearable device system including one or more wearable devices, the wearable device system comprising:

a motion sensor of the one or more wearable devices;

a biological sensor of the one or more wearable devices; and

a controller of the one or more wearable devices, the controller configured to:

obtain, based on motion information from the motion sensor, a movement signal of an object;

obtain, based on biological information from the biological sensor, a biological signal of the object;

determine whether a period of the movement signal exists that meets a specified motion type indicating condition;

16

determine whether any portions the biological signal correspond to the period and meet a specified value condition; and

perform, in response to at least one of the one or more portions of the biological signal being determined to correspond to the period, a specified function of at least one of the one or more wearable devices and/or an external device respectively dependent on the movement signal.

22. The wearable device system of claim 21, wherein the determination of whether the period of the movement signal exists includes determining what period of the movement signal meets the specified motion type indication condition that is predetermined to be one or more measures of relatedness of obtained motions to an incidental motion, corresponding to a non-spontaneous or non-intentional motion of the object, and/or an intentional motion of the object.

23. The wearable device system of claim 22, the determination of whether the period of the movement signal exists that meets the specified motion type indicating condition includes determining whether a measured rotational velocity magnitude is below a predetermined velocity magnitude, where the measured rotational velocity magnitude being below the predetermined velocity magnitude is representative of the movement signal being not the incidental motion and/or being the intentional motion of the object.

24. The wearable device system of claim 22, the determination of whether the period of the movement signal exists that meets the specified motion type indicating condition includes determining whether a measured velocity variance magnitude is below a predetermined velocity variance magnitude, where the measured rotational velocity variance magnitude being below the predetermined velocity variance magnitude is representative of the movement signal being not the incidental motion and/or being the intentional motion of the object.

25. The wearable device system of claim 24, wherein the predetermined velocity variance magnitude is a predetermined acceleration-based threshold, a predetermined rotation-based threshold, a predetermined bearing signal-based threshold, or a predetermined coordinate signal-based threshold.

26. The wearable device system of claim 21, wherein the specified value condition is a measure of valuableness of a measured biological signal for intentional controlling of one or more functions of an external device, the measure being a determination that a magnitude of at least one portion of the biological signal, corresponding to the select period, is greater than a predetermined threshold.

27. The wearable device system of claim 26, wherein the magnitude of the at least one portion of the biological signal is a measure of a relaxation and/or contraction of one or more muscles of the object.

28. The wearable device system of claim 27, wherein the biosensor includes one or more biosensors, and the muscles include one or more of one or more wrist muscles, one or more hand muscles, one or more forearm muscles, one or more face muscles, and/or one or more neck muscles of the object.

29. The wearable device system of claim 28, further comprising determining which one or more of different interaction controls are represented the movement signal, with the performing of the specified function being a performance of the specified function of the determined one or more of the different interaction controls represented by the movement signal.

17

30. The wearable device system of claim 29, wherein the specified motion type indicating condition is a predetermined condition that, when met, the one or more wearable devices are controlled according to a determined gesture of the object, represented by the movement signal, and the biological signal of the object. 5

31. The wearable device system of claim 30, wherein the determination of whether the period of the movement signal exists includes determining what period of the movement signal meets the specified motion type indication condition that is predetermined to be one or more measures of relatedness of obtained motions to an incidental motion, corresponding to a non-spontaneous or non-intentional motion of the object, and/or an intentional motion of the object, and wherein the determining of what period of the movement signal meets the specified motion type indication condition includes determining whether a measured rotational velocity magnitude is below a predetermined velocity magnitude, where the measured rotational velocity magnitude being below the predetermined velocity magnitude is representative of the movement signal being not the incidental motion and/or being the intentional motion of the object, and/or wherein the determining of what period of the movement signal meets the specified motion type indication con-

18

dition includes determining whether a measured velocity variance magnitude is below a predetermined velocity variance magnitude, where the measured rotational velocity variance magnitude being below the predetermined velocity variance magnitude is representative of the movement signal being not the incidental motion and % or being the intentional motion of the object.

32. The wearable device system of claim 31, wherein the predetermined velocity variance magnitude is a predetermined acceleration-based threshold, a predetermined rotation-based threshold, a predetermined bearing signal-based threshold, or a predetermined coordinate signal-based threshold.

33. The wearable device system of claim 21, wherein the specified motion type indicating condition is a predetermined condition that, when met, the one or more wearable devices are controlled to shift modes of operation from an inactive mode of operation to an active mode of operation.

34. The wearable device system of claim 21, wherein the biological signal is a bio-electric/magnetic signal, a bioimpedance signal, and/or a bio-mechanical signal generated in association with skeletal muscle movement signal or skeletal muscle tone of the object.

* * * * *

EXHIBIT 3

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON D.C.**

**Before the Honorable Doris Johnson Hines
Administrative Law Judge**

In the Matter of

CERTAIN WEARABLE DEVICES

Inv. No. 337-TA-1478

**RESPONDENTS' INVALIDITY AND INELIGIBILITY CONTENTIONS FOR U.S.
PATENT NOS. 10,642,359, 10,231,675, 10,945,677, AND 10,978,789**

I. INTRODUCTION

Pursuant to the Procedural Schedule (Order No. 6), Respondents Ouraring Inc. and Ōura Health Oy (collectively, “Oura” or “Respondents”), by and through counsel, hereby provide these invalidity and ineligibility contentions. Complainants Samsung Electronics Co., Ltd. and Samsung Electronics America, Inc. (collectively, “Samsung” or “Complainants”) currently asserts the following patents in this Investigation: (1) U.S. Patent Nos. 10,642,359 (“the ’359 Patent”); (2) 10,231,675 (“the ’675 Patent”); (3) 10,945,677 (“the ’677 Patent”); and (4) 10,978,789 (“the ’789 Patent”). These invalidity and ineligibility contentions concern the ’359, ’675, ’677, and ’789 Patents and are, with respect to those patents, responsive to Interrogatory No. 8 of Samsung’s First Set of Interrogatories to Oura.

II. GENERAL RESERVATIONS

Respondents reserve the right to revise or supplement these contentions in light of party and third-party discovery, Samsung’s infringement contentions, any claim construction order issued by the ALJ, review and analysis by expert witnesses, and further investigation and discovery regarding the defenses asserted by Respondents. Respondents expressly reserve the right to amend these contentions after review of Samsung’s infringement contentions, after issuance of the claim construction order, should Samsung provide any information that it failed to provide in its disclosures, if Samsung amends its disclosures in any way, or when Samsung provides further discovery. Further, because discovery is ongoing, Respondents reserve the right to revise, amend, and/or supplement the information provided herein, including identifying, charting, and relying on additional references. Further, Respondents reserve the right to revise their ultimate contentions concerning the invalidity of the asserted claims, which may change depending upon further and ongoing investigation, the construction of the asserted claims and/or positions that Samsung or expert witnesses may take concerning claim construction, infringement, eligibility, and/or

indefiniteness of a qualitative term where “intrinsic evidence provided guidance as to the scope of the claims,” emphasizing that “the written description is key to determining whether a term of degree is indefinite”).

While “absolute precision is unattainable” in patented claims, the definiteness requirement “mandates clarity.” *Nautilus, Inc. v. Biosig Instruments, Inc.*, 572 U.S. 898, 910 (2014). A court will not find a patented claim indefinite unless the claim interpreted in light of the specification and the prosecution history fails to “inform those skilled in the art about the scope of the invention with reasonable certainty.” *Id.* at 901.

The following claims are indefinite (under 35 U.S.C. § 112, second paragraph) because the ’675 Patent fail to sufficiently disclose them as required by 35 U.S.C. § 112:

- “A first period of time,” “a second period of time,” and “a third period of time, which is shorter than a first period of time.” (Claims 1, 15, 16, and 20). These claim terms are indefinite because the claim language is irresolvably ambiguous and the specification of the ’675 Patent does not provide sufficient clarity for a POSITA to understand the metes and bounds of a first, second, or third “period of time” as recited by the claims. “A period of time” is a term for which the specification supplies no benchmarks, examples, or measurement techniques. Accordingly, a POSITA would not understand with reasonable certainty the claim scope of a first, second or third “period of time.”
- “At least a part of the at least one first sensor” and “at least a part of the first information.” (Claims 1 and 15). These claim terms are indefinite because the claim language is irresolvably ambiguous and the specification of the ’675 Patent does not provide sufficient clarity for a POSITA to understand the metes and bounds of

a “at least a part of the at least one first sensor” and “at least a part of the first information.” The specification does not describe how to partition a first sensor or examples of partitioning a first sensor to arrive at “at least a part of the at least one first sensor”. “[A]t least a part of the first information” is a term for which the specification supplies no benchmarks, examples, or measurement techniques. Accordingly, a POSITA would not understand with reasonable certainty the claim scope of a “at least a part of the at least one first sensor” and “at least a part of the first information.”

- “acquiring first information on a motion . . . for a second period of time shortly before and/or immediately after a first period” and “immediately after the second period of time passes.” (Claims 13 and 15). These claim terms are indefinite because the claim language is irresolvably ambiguous and the specification of the ’675 Patent does not provide sufficient clarity for a POSITA to understand the metes and bounds of a “shortly before” and “immediately after” as recited by the claims. The ’675 specification supplies no benchmarks, examples, or measurement techniques that sufficiently describe “shortly before and/or immediately after” and “immediately after the second period of time passes.” Accordingly, a POSITA would not understand with reasonable certainty the claim scope of “shortly before and/or immediately after” and “immediately after the second period of time passes.” Further, a POSITA would not understand with reasonable certainty how to determine when the motion sensor is obtaining first information during the second period as compared to when the motion sensor was obtaining the information to determine that a user is wearing the device.

- “during the detecting the biometric information, acquiring second information on a motion” (Claim 13) and “determine whether the electronic device is worn on a part of a user's body, “determine a motion of the electronic device based on first information obtained by the at least one first sensor,” and “obtain second information on a motion of the electronic device using the at least one first sensor” (Claim 20). These claim limitations are indefinite because the claim language is irresolvably ambiguous. A POSITA would not understand with reasonable certainty the claim scope how the device distinguishes between the information to determine a user is wearing the device, the first information used to activate the biological sensor, and the second information used to deactivate the device.
- “acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes.” (Claim 15). This claim limitation is indefinite because the claim language is irresolvably ambiguous. A POSITA would not understand with reasonable certainty the claim scope as to how does a POSITA determine what the different time periods when the same sensor is making the same measurement.
- “at least one of the second period of time or the third period of time comprises a time point when the first period of time passes.” (Claim 16) This claim limitation is indefinite because the claim language is irresolvably ambiguous. The claim language can be read that the “second period of time or the third period of time comprises a time point” when time passes in the first period (the claim requires the second and third period to include a portion of the first period). On the other hand,

the claim language could also be read that the “second period of time or the third period of time comprises a time point” that includes a point when the first time period has passed (in which case the second and third period would need to include a time point outside the first period of time). Accordingly, a POSITA would not understand with reasonable certainty the claim scope of this limitation.

- “[A] selected condition,” and “changing at least one condition for activating the at least one second sensor.” (Claims 17, 18 and 19). These claim terms are indefinite because the claim language is irresolvably ambiguous and the specification of the ’675 Patent does not provide sufficient clarity for a POSITA to understand the metes and bounds of a “condition” as recited by the claims. The specification of the ’675 Patent does not disclose benchmarks, examples, or measurement techniques that sufficiently describe a “condition.” Accordingly, a POSITA would not understand with reasonable certainty the claim scope of “a selected condition,” and “changing at least one condition for activating the at least one second sensor.”
- “[A] physical characteristic or state of the user.” (Claim 18) This claim term is indefinite because the claim language is irresolvably ambiguous and the specification of the ’675 Patent does not provide sufficient clarity for a POSITA to understand the metes and bounds of a “physical characteristic or state of the user” as recited by the claims. The specification of the ’675 Patent does not disclose benchmarks, examples, or measurement techniques that sufficiently describe a “condition.” Accordingly, a POSITA would not understand with reasonable certainty the claim scope of “physical characteristic or state of the user.”

Elecs. Co. LTD. et al v. Ouraring, Inc., IPR2024-01078, Final Written Decision, at 21-23 (PTAB Jan. 27, 2026).

The claims that lack support as identified above are entitled to, at the earliest, a priority date of the actual filing date of the application that issued as the asserted patent. Under Samsung's infringement theories by the products Samsung accuses of infringement and relies on for technical domestic industry were known, used, and sold before their priority date and invalidate those claims. Samsung's infringement and technical domestic industry charts (including any amendment or supplement thereto) in each of the above as to the '789 patent in this investigation and any other proceeding are herein incorporated by reference as if fully set forth herein. For the same reason, members of the Cho family, including Cho, as issued on July 28, 2020, are prior art under pre-AIA Section 102(b) and/pr AIA Section 102. The Cho family does not disclose these limitations for the same reasons. However, it would have been obvious to combine their teachings with the references discussed above (*e.g.*, Michiwaki, Williams, etc.) to render these claims obvious.

Dated: March 16, 2026

Respectfully submitted,

/s/ Jasjit S. Vidwan

Jasjit S. Vidwan
Saqib J. Siddiqui
Michael L. Lindinger
James A. Fussell, III
Bryan C. Nese
Tiffany A. Miller
Clark S. Bakewell
Alison T. Gelsleichter
Tariq Javed
Amanda E. Stephenson
Seth W. Bruneel
Courtney Krawice
Séké Godo
Angeline Premraj
Paul Choi
So Ra Ko

Andrew Huntsinger
Jaspreet Momi
Jennifer Weinberg
Sydney N. Royer
MAYER BROWN LLP
1999 K Street, NW
Washington, DC 20006-1101
Tel.: +1 202.263.3000

Robert G. Pluta
Nicholas H. Ciulla
MAYER BROWN LLP
71 S. Wacker Drive
Chicago, IL 60606
Tel.: +1 312.782.0600

*Counsel for Respondents Ouraring Inc. and Oura
Health Oy*

Oura-ITC@mayerbrown.com

CERTIFICATE OF SERVICE

I hereby certify that a copy of the foregoing **RESPONDENTS' INVALIDITY AND INELIGIBILITY CONTENTIONS FOR U.S. PATENT NOS. 10,642,359, 10,231,675, 10,945,677, AND 10,978,789** was served upon the following as indicated below on March 16, 2026:

Counsel for Complainant Samsung Electronics Co., Ltd and Samsung Electronics America, Inc. D. Sean Trainor O'MELVENY & MYERS LLP 1625 Eye Street, NW Washington, D.C. 20006-4061 Telephone: (202) 383-5300	Via email: dstrainor@omm.com
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/s/ Jasjit S. Vidwan
Jasjit S. Vidwan

*Counsel for Respondents Ouraring Inc.
and Oura Health Oy*

EXHIBIT 4

NEWTON's TELECOM DICTIONARY

27th Expanded and Updated Edition

Harry Newton

Steve Schoen, Contributing Editor

Gail Saari, Artistic Director



New York

NEWTON's TELECOM DICTIONARY

27th Expanded and Updated Edition

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email: Harry@HarryNewton.com

book site: www.NewtonsTelecomDictionary.com



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Steve Schoen, Contributing Editor
Gail Saari, Artistic Director

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signals. The timing elements at the sending and receiving terminal define where one character ends and another begins. There are no start or stop elements in this form of transmission. For a more detailed explanation, see BSC.

binary to decimal conversion Conversion from base 2 to base 10. See Binary.

binary transfer See Binary File Transfer and MIME.

BIND Berkeley Internet Name Daemon. BIND is the software that allows us to type site names like *www.yahoo.com* instead of a string of numbers. BIND is the implementation of a DNS server originally developed and distributed by the University of California at Berkeley. Many Internet Hosts run BIND, and it is the ancestor of many commercial BIND implementations.

bind A request to activate a session between two logical units (LUs). See BIND.

binder A way to separate groups of 25 pairs in a twisted-pair cable with more than 25. Color plastic ribbon binds separate each group of 25. The first group (1-25) is white/blue, the second (26-50) is white/orange, the third (51-75) is white/green, the fourth (76-100) is white/brown, etc. Binders are helically applied colored thread, yarn, or plastic ribbon. They're used to confine, separate and identify groups of fibers or wires in a cable. Binders are usually used for holding assembled cable components in place.

binder group A typical phone line—business or residence, analog or digital—is a pair of wires. When the phone company installs cabling along the street—up in the air or underground—it installs a cable containing many pairs of wires. The pairs are color coded so you can find which is which. Often there are so many of them you have to do something better—like take a bunch of cable pairs and wind something (or bind something) around them. That's called a binder group. The binder is a spirally wound colored thread or plastic ribbon used to separate and identify cable pairs by means of color coding. The groups are composed of insulated twisted copper pairs that are also twisted within each binder. Typically, they are wrapped in 25-pair bundles. Normal telephone color-coding provides for only 25 pairs of wire, so binder groups allow multiple pairs of the same color wire to be in one cable. A 50-pair cable has blue and orange binder groups; a 75-pair cable has blue, orange, and green groups. Since several wire pairs have the same color markings, installers must be careful when stripping cable insulation so they do not destroy the binder threads. For example, pairs 1-25 might be in one binder group and pairs 26-50 in another. In xDSL, one often hears discussions of signal interference between adjacent pairs within a binder group. The best of all worlds is to keep a data pair separated from another data pair by assigning it to an adjacent binder group. If the data pairs are too close to each other they create what telephone companies call "disturbances" (i.e. crosstalk).

bindery A Novell NetWare database containing definitions for entities such as users, groups, and workgroups. The bindery contains three components: objects, properties, and property data sets. Objects represent any physical or logical entity, including users, user groups, file servers, print servers, or any other entity given a name. Properties are the characteristics of each bindery object, including passwords, account restrictions, account balances, internetwork addresses, list of authorized clients, workgroups, and group members. Property data sets are values assigned to entities' bindery properties.

binding A Windows definition. Binding is a process that establishes the communication channel between a protocol driver and a network adapter driver.

binding post 1. BP. A screw with a small nut. You take your phone wires and join them together on a binding post by wrapping them together around the screw and then tightening the nut on them. You'll find binding posts on huge wall-mounted things called terminal boxes. We have several in the basement of the building in Manhattan, New York, in which I live. Binding posts are numbered on the terminal box and those numbers are entered into the phone company's cable management system. That way, when the technician comes to check out a trouble report, he can quickly find the offending pairs. These days, other more modern termination devices (such as Krone Boxes, 110-connecting blocks, and even 66-blocks) are replacing binding posts. Our building's binding posts are probably 60 years old. In telephone parlance, the binding post is neither a rate demarcation point or a network interface. The binding

post is part of the telephone company's plant and cannot be accessed by the telephone subscriber. The only exception I can think of is the wholesale (CLEC) scenario. The issue of who would have access to the binding post would depend on the agreement signed by the ILEC and CLEC. The reason for limiting access to binding posts is simple: If retail customers have access to binding posts for do-it-yourself tinkering, they could accidentally or maliciously cause trouble. In a shared scenario (for example, in an apartment building), a subscriber could connect his inside wire to his neighbor's binding post and listen to his neighbor's phone calls and make long-distance calls on his neighbor's dime. Or he could clumsily damage LEC facilities. See also House Box.

2. A post that witches were attached to just before they were burned. Rude switchboard operators and ACDs with long queues should suffer this fate.

BinHex When sending files which aren't plain ASCII across a network—dial-up, leased line or the Internet—you basically have two options. First, you can attach them as a binary file (i.e. non-ASCII file). Or second, you can encode them into ASCII characters and send the file as part of your message. The first method is preferable. But you can typically only send binary files from one account to another on the same network, or between two networks that have agreed between themselves to a method of transferring files. But that's a rarity. And it certainly doesn't work in and around the Internet. Thus something called MIME was created. MIME stands for Multipurpose Internet Mail Extensions. It is the name for encoding binary files into ASCII characters for transfer across the Internet and on-line services. Under MIME, there are a number of methods of encoding binary files, one of which is called BinHex, which is a popular encoding algorithm that uses Run-Length Encoding (RLE). To send a binary file, you encode it using BinHex encoding software. You then include the ASCII encoded file in your message. The recipient of your message then decodes the ASCII using BinHex decoding software, which may or may not be built into his electronic mail or browser package.

binomial distribution The binomial is a two-parameter distribution, the parameters being *n*, the number of trials, and *p*, the probability of a particular outcome of a single trial.

bio break Meeting-speak for "bathroom break." "Let's take a short bio break before moving on to action plans."

bio-surveillance Monitoring hospital admissions to detect increases in similar symptoms that could indicate a biological attack.

biobology The art of scrutinizing and interpreting ambiguous, lo-res imagery. Practiced by spy agencies and, more recently, neuroscientists studying brain scans.

biogorrrhea Pointless or excessive filler in a weblog. As one Net diarist wrote: "I was accused recently of biogorrrhea. Well, yes, that's fair. But see, here's the thing. I'm leaving town for a month..." Courtesy, Gareth Branwyn, *Wired Magazine*.

biohacker A hobbyist who tinkers with DNA and other aspects of genetics. Also: bio-hacker. Some speculate a "biohacker"—the equivalent of a computer hacker seeking thrills rather than impact—could be behind the anthrax letters. "Hacker" is usually considered a bad word.

bioinformatics Bioinformatics is the process of using computer systems for the purposes of acquisition, storage and analysis of biological data. Computer databases and algorithms are used to speed up biological research and to characterize the molecular components of living things. Bioinformatics is notably being used in the Human Genome Project, the effort to identify the 80,000 genes in human DNA. Bioinformatics includes a wide spectrum of technologies including computer architectures, knowledge management, networking and collaboration tools as well as traditional life-science equipment needed to handle biological samples.

biomass addition New growth of vegetation. See vegetation management.

biometrics Biometrics is the study of unique and measurable physical (or biological) characteristics such as fingerprints, retinal pattern, or handwriting, biometric authentication devices capture, encrypt and use these unique and measurable characteristics as the basis for confirming identity and determining whether to grant or deny access to physical or logical assets. When used for au-

thentication, personal biometrics are used much like a door key, however that key cannot be duplicated or lost. The basic principles of biometric verification have been understood and practiced for thousands of years. References occur as early as ancient Egypt, where individuals were identified via unique "biometrics" such as scars or a combination of measured features such as complexion, eye color, height and so on. Biometrics as we know it today really began in the late nineteenth century when criminology researchers attempted to tie physical features and characteristics with criminal tendencies. The results were not conclusive but the idea of measuring individual physical characteristics became accepted, leading the way for fingerprinting to become the international methodology among police forces for identity verification. Today it is taken for granted that fingerprints (and foot and handprints) accurately identify not just criminals, but newborns, minors (as a means of thwarting abduction), military and police personnel, and a host of others. Once technology simplified the collection, codification and encryption of biometric data, the stage has become set for biometrics to become the security standard of the electronic world, replacing passwords, bits of paper, etc. See also biometric access control.

biometric access control Any means of controlling access through human measurements, such as fingerprinting, handwriting analysis, retinal scanning, hand geometry and voiceprinting. See Biometrics.

biometric device A device used in authenticating access to a system. A biometric device authenticates a user by measuring some hard-to-forge physical characteristic, such as a fingerprint or retinal scan. You see a lot of biometric devices in Hollywood thrillers, including *Mission Impossible*, a movie which was released in the summer of 1996.

bionic code A problem-solving routine for human behavior as it is exercised in the realm of networks and cyberspace. The first bionic codes were developed by Ebon Fisher based on a series of his theatrical experiments involving communication systems amongst audience members. Fisher's bionic codes have been formalized as a series of diagrams and statements which "float" in the infosphere in a variety of media.

BIOS Basic Input/Output System of desktop computers. The BIOS contains the buffers for sending information from a program to the actual hardware device the information should go to. Every personal computer has a system basic input/output system, or BIOS, which is what takes control of your computer the moment you turn it on. The screen that you first see when you turn on your computer is called the power on self test screen, better known as the POST screen. If you purchased your computer from one of the major computer manufacturers, this screen is often hidden by the manufacturer's logo. To get rid of this logo from the screen, press the ESC button on your keyboard; you'll then see what is going on in the background. At this stage in the system boot, the BIOS is probing the hardware to test the system memory and other device connections. Once the POST is completed, the BIOS proceeds to look for a device to boot from. Once it finds your hard drive, it will begin to load Windows. The BIOS also acts as a main system component control panel, where low-level settings for all of your hardware devices are made. The device boot order, port addresses, and feature settings such as plug and play are all found in the BIOS setup screens. For example, if you want to change the order of the drives that your computer checks to boot from, then you will want to modify the device boot order. I have to modify this setting almost every time that I install Windows because I want my computer to boot off the CD-ROM to launch the Windows XP setup application instead booting off of the operating system on my hard drive. BIOSs on each and every PC may be made by different companies or accessed in different ways. Nevertheless, the most common way to access the setup screen is to press F2 or the Delete key when the POST screen is displayed. Some computers even tell you which key to push to enter setup, as my laptop does. If your PC doesn't allow you to access the setup screen in this way, consult your computer documentation or contact your computer manufacturer for instructions. See also extensible firmware interface.

BIOS Enumerator A Windows term. Bios enumerator is responsible, in a Plug and Play system, for identifying all hardware devices on the motherboard of the computer. The BIOS supports an API that allows all Plug and Play computers to be queried in a common manner. See BIOS.

biosensors Devices such as fingerprint readers and signature recognition systems.

BIP 1. Billing Interconnection Percentage. A calculation of who owns how much of a route when there are multiple providers between two points. Used to allocate revenue in MPB arrangements.

2. See Broadband Initiatives Program.

BIP-8 Bit Interleaved Parity 8. A method of error monitoring used in SONET optical fiber transmission systems. A SONET frame of data comprises a large number of bits organized into bytes, or 8-bit values. The bytes, of course, are interleaved into a byte stream. The BIP-8 method looks across all matching bit positions for those distinct bytes in a frame, and calculates parity, which should be even, rather than odd. If an odd parity is calculated at the receiving end, a bit error is indicated. The number of such errors in frames over a period of time constitutes the Bit Error Rate (BER). The specific period of time involved is known as a "sliding time window," as it varies according to the maximum detection time, which is sensitive to the transmission rate, or level of the Optical Carrier (e.g. OC-1, OC-3, OC-24, OC-48, OC-192). The higher the OC-N level, the higher the rate of transmission, and the shorter the sliding time window. In other words, the faster the rate of transmission, the more quickly the error detection process must take place in order to establish the BER and to correct the problem.

BIP-N Bit Interleaved Parity N. A method of error monitoring. With even parity, an N bit code is generated by the transmitting equipment over a specified portion of the signal in such a manner that the first bit of the code provides even parity over the first bit of all N-bit sequences within the specified portion, etc. Even parity is generated by setting the BIP-N bits so that there are an even number of 1s in each of all N-bit sequences including the BIP-N. See BIP-8 for a concrete example.

bipolar 1. The predominant signaling method used for digital transmission services, such as DDS and T-1. The signal carrying the binary value alternates between positive and negative. Zero and one values are represented by the signal amplitude at either polarity, while no-value "spaces" are at zero amplitude.

2. Bipolar Signal: A signal that can take on two polarities, of which neither is zero.

bipolar coding The T carrier line coding system that inverts the polarity of alternate "one" bits.

bipolar signal Any mechanism that converts digital data to digital signals on a wire through the use of two non-zero voltage levels—positive and negative—to encode binary 1s, and a zero-voltage level to encode binary 0s. The term "bipolar" literally means "two poles." The prefix "bi" in "bipolar" comes from a Latin word meaning "two." The term "bipolar" refers to the aforementioned fact that two different voltage polarities (positive and negative) are used to encode 1s; the first occurrence of a 1 in a bitstream is encoded as a positive voltage on the wire; the second occurrence of a 1 in the same bitstream is encoded as a negative voltage; the third occurrence of a 1 in the bitstream is encoded as a positive voltage; and so on. Bipolar encoding mechanisms have an advantage over unipolar encoding mechanisms in that the greater variability in voltage levels and polarities that is produced by the former makes it easier for the receiver to figure out where each bit begins and ends, especially when there is an unvarying sequence of voltage pulses (representing a string of identical bits). Multiple bipolar encoding techniques have been developed, including, but not limited to, alternate mark inversion, B8ZS, and HDB3. You can find explanations of these terms in this dictionary. Also see unipolar encoding.

bipolar violation The presence of two consecutive "one" bits of the same polarity on the T carrier line. See also Bipolar Coding.

Bipolar with 8 Zero Substitution See B8ZS.

bird A satellite.

bird dog device An electronic tracking device used by law enforcement to track the physical location of a subject.

birdie 1. A birdie is a lightweight device that you blow through underground cement pipes through which you want to pull cable. Here's how it typically works: First, you use a bore to make an underground hole. Then you fill that hole with hollow concrete cement pipes joined together to form one long un-

EXHIBIT 5

Dictionary of Computer Vision and Image Processing

R.B. Fisher

University of Edinburgh

K. Dawson-Howe

Trinity College Dublin

A. Fitzgibbon

Oxford University

C. Robertson

Heriot-Watt University

E. Trucco

Heriot-Watt University



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binocular: A system that has two cameras looking at the same scene simultaneously usually from a similar viewpoint. See also stereo vision.

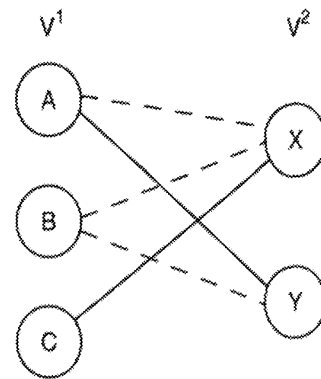
binocular stereo: A method of deriving depth information from a pair of calibrated cameras set at some distance apart and pointing in approximately the same direction. Depth information comes from the parallax between the two images and relies on being able to derive the same feature in both images.

binocular tracking: A method that tracks objects or features in 3D using binocular stereo.

biometrics: The science of discriminating individuals from accurate measurement of their physical features. Example biometric measurements are retinal lines, finger lengths, fingerprints, voice characteristics and facial features.

bipartite matching: Graph matching technique often applied in model based vision to match observations with models or stereo to solve the correspondence problem. Assume a set V of nodes partitioned into two non-intersecting subsets V^1 and V^2 . In other words, $V = V^1 \cup V^2$ and $V^1 \cap V^2 = \emptyset$. The only arcs E in the graph lie between the two subsets, i.e., $E \subset \{V^1 \times V^2\} \cup \{V^2 \times V^1\}$. This is the bipartite graph. The bipartite matching problem is to find a maximal matching

in the bipartite graph, in other words, a maximal set of nodes from the two subsets connected by arcs such that each node is connected by exactly one arc. One maximal matching in the graph below with sets $V^1 = \{A, B, C\}$ and $V^2 = \{X, Y\}$ pairs (A, Y) and (C, X) . The selected arcs are solid, and other arcs are dashed.



bit map: An image with one bit per pixel.

bit-plane encoding: An image compression technique where the image is broken into bit planes and run length coding is applied to each plane. To get the bit planes of an 8-bit gray scale image, the picture has a boolean AND operator applied with the binary value corresponding to the desired plane. For example, ANDing the image with 00010000 gives the fifth bit plane.

bitangent: See curve bitangent.

bitshift operator: The bitshift operator shifts the binary representation of each pixel to the left or right by a set number of bit positions. Shifting

EXHIBIT 6

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

**Before the Honorable Doris Johnson Hines
Administrative Law Judge**

In the Matter of

CERTAIN WEARABLE DEVICES

Investigation No. 337-TA-1478

**DECLARATION OF DR. DARRIN YOUNG
REGARDING CLAIM CONSTRUCTION**

1. I, Darrin Young, Ph.D., declare as follows:

I. INTRODUCTION

2. My name is Darrin Young. I have been retained by Complainants Samsung Electronics Co., Ltd. and Samsung Electronics America, Inc. (“Samsung” or “Complainants”) as an independent expert consultant in connection with the above-captioned Investigation No. 337-TA-1478 regarding Certain Wearable Devices, to provide my analyses and opinions on certain technical aspects of this dispute related to U.S. Patent Nos. 10,231,675 (the “’675 Patent”) and 10,642,359 (the “’359 Patent”).

3. This declaration contains statements of my opinions formed to date, and the bases and rationale for these opinions. I may offer additional opinions based on further review of materials in this case, including opinions and/or testimony of other expert witnesses.

4. For my efforts in preparing this declaration, I have been compensated at my usual and customary rate for this type of consulting activity. My compensation is in no way contingent on the results of these or any other proceedings related to the ’675 and ’359 Patents.

A. Qualifications / Professional Background

5. My qualifications generally are set forth in my Curriculum Vitae, which is attached as **Exhibit 7**. Exhibit 7 also includes a list of the publications I have authored and a list of the other cases in which I have testified during the last five years.

6. I am a Utah Science Technology and Research Initiative (USTAR) Associate Professor and the chair of the solid-state/micro-electro-mechanical systems/circuit area of the Electrical and Computer Engineering Department of the University of Utah.

7. I earned a bachelor of science degree in Electrical Engineering and Computer Sciences from the University of California at Berkeley in 1991, a master of science in Electrical Engineering and Computer Sciences from the University of California at Berkeley in 1993, and a

Ph.D. in Electrical Engineering and Computer Sciences from the University of California at Berkeley in 1999.

8. I joined the University of Utah in 2009. Prior to joining the University of Utah, I was an Assistant and Associate Professor in the Electrical Engineering and Computer Science Department at Case Western Reserve University for ten years. Prior to joining the Case Western Reserve University, I was a participating researcher at the Lawrence Livermore National Laboratory in Livermore, California for two years, where I worked on the design, fabrication, and testing of MEMS-based 3D high-Q coil inductors. My research has focused on micro-electro-mechanical systems (“MEMS”) design and fabrication, integrated electronic interface circuit design, wireless power transfer, wireless sensing technology, and microsystem packaging technology for a wide range of applications, including wireless sensing, implantable and wearable biomedical sensing, RF communication, industrial sensing, personal navigation, and environmental monitoring.

9. I have extensive experience in academia. I have been teaching and conducting research for over 26 years. I teach both undergraduate and graduate level electrical engineering courses, ranging from an Introduction to Circuits and Instruments to detailed lectures on Applied Electronics for various sensing and communication applications, including biomedical applications and industrial sensing applications. I have taught numerous graduate-level courses and invited lectures that bear on wireless sensing design, including Radio-Frequency Integrated Circuits Design, MEMS for Sensing and Communication, Electrical Interface for MEMS, and Analog Integrated Circuit Design.

10. Since 2009, I have been the Director of the Wireless Microsystem Laboratory at the University of Utah, where I conduct and oversee research on the design and development of

wireless sensing systems for biomedical applications, personal navigation, and environmental sensing applications.

11. In my research lab I have designed, built, and tested a number of wireless sensing systems. Those systems include wearable sensors to detect physiological signals such as blood pressure waveforms, heart rate, pulse rate, and pulse wave velocity, implantable sensors for detecting blood pressure waveforms, ECG, glucose level, and micro-implantable coils array for wireless power transfer to the brain, flexible tactile sensors array for personal navigation applications under GPS-denied environment, and MEMS-based wireless strain sensing module that can detect surface conditions of metallic rotating machinery elements.

12. I have served as a technical program committee member, track chair, and session chair for a number of international conferences. I served as an editor for the IEEE Transactions on Electron Devices with a focus on sensors and actuators between 2015 and 2024. I was an associate editor of the IEEE Journal of Solid-State Circuits from 2006 to 2011. I served as the chair of the Micro-Electro-Mechanical Systems committee under the IEEE Electron Devices Society between 2010 and 2014. I also served on the IEEE Sensors Council as an administrative committee member as well as a representative from the IEEE Solid-State Circuits Society between 2002 and 2016.

13. I am an author of over 100 peer-reviewed publications including refereed journals and conferences, in the field of sensors, actuators, MEMS, wireless microsystems design, packaging, and interface circuits design. I am an invited author of multiple book chapters in the field of MEMS. I have also given numerous presentations and speeches around the world on similar topics including sensors, actuators, MEMS, wireless microsystems design technology, and related applications. I am the inventor of two patents, both in the field of sensors.

14. I have been retained by counsel for Complainants to provide my expert opinion in connection with the above-captioned proceeding as set forth herein.

B. Materials Considered

15. The analysis that I provide in this Declaration is based on my education, research, and experience, as well as the documents I have considered. In forming my opinions, I have read and considered the '675 and '359 Patents, their prosecution histories, the parties' proposed constructions, the extrinsic evidence identified, and any other documents referenced in this declaration.

II. LEVEL OF ORDINARY SKILL IN THE ART

16. I have been advised that to determine the appropriate level of a person having ordinary skill in the art, the following factors may be considered: (1) the types of problems encountered by those working in the field and prior art solutions thereto; (2) the sophistication of the technology in question, and the rapidity with which innovations occur in the field; (3) the educational level of active workers in the field; and (4) the educational level of the inventor.

17. In my opinion, a person of ordinary skill in the art ("POSITA") at the time of the patents' priority date would have possessed at least a four-year degree in electrical engineering, mechanical engineering, biomedical engineering, optical engineering, or related field of study, or equivalent experience, and at least two years' experience in academia or industry studying or developing sensor systems. A higher level of education or skill might make up for less experience, and vice versa. The '359 Patent's earliest claimed filing date is May 8, 2014 but it claims priority to an earlier Korean application filed on August 20, 2013. The '675 Patent's earliest claimed filing date is June 3, 2016, but it claims priority to an earlier Korean application filed on August 12, 2015.

18. Although my qualifications and experience exceed those of the hypothetical person having ordinary skill in the art defined above, my analysis and opinions regarding the '675 and '359 Patents have been based on the perspective of a POSITA as of the relevant priority dates identified above.

III. LEGAL STANDARDS FOR CLAIM CONSTRUCTION

19. I understand that the purpose of claim construction is to determine what a person of ordinary skill in the art would have understood the claim terms to mean. Claim terms are generally given their ordinary and customary meaning, which is the meaning that the term would have to a person of ordinary skill in the art in question as of the effective filing date.

20. I understand that the person of ordinary skill in the art is deemed to read the claim term not only in the context of the particular claim in which the disputed term appears, but in the context of the entire patent, including the specification. I understand that the patent specification, under the legal principles, has been described as the single best guide to the meaning of a claim term, and is thus highly relevant to the interpretation of claim terms. And I understand for terms that do not have a customary meaning within the art, the specification usually supplies the best context of understanding the meaning of those terms.

21. I further understand that other claims of the patent in question, both asserted and unasserted, can be valuable sources of information as to the meaning of a claim term. Because the claim terms are normally used consistently throughout the patent, the usage of a term in one claim can often illuminate the meaning of the same term in other claims. Differences among claims can also be a useful guide in understanding the meaning of particular claim terms.

22. I understand that particular embodiments appearing in the written description do not limit claim language that has broader effect, and that the scope of the claims is not necessarily limited to inventions that look like the ones shown in the figures and described in the specification.

However, I also understand that the patentee is required to define precisely what he claims his invention to be, and the claims must be construed in a manner consistent with the specification.

23. I am informed that a term must be interpreted with a full understanding of what the inventors actually invented and intended to include within the scope of the claim as set forth in the patent itself. Thus, claim terms should not be broadly construed to encompass subject matter that, while technically within the broadest reading of the term, is not supported when the claims are viewed in light of the invention described in the specification.

24. I understand that the prosecution history can further inform the meaning of the claim language by demonstrating how the inventors understood the invention and whether the inventors limited the invention in the course of prosecution, making the claim scope narrower than it otherwise would be. Extrinsic evidence, such as dictionaries, may also be consulted in construing the claim terms.

25. I have been instructed by counsel to apply the “*Phillips*” claim construction standard for purposes of interpreting the claims in this proceeding, to the extent they require an explicit construction. The description of the legal principles set forth above thus provides my understanding of the “*Phillips*” standard as provided to me by counsel.

26. I understand that a patent must be precise enough to afford clear notice of what is claimed and apprise the public of what subject matter is still open to them in a manner that avoids uncertainty. I understand that a claim term is indefinite if, when the term is viewed in light of the specification and prosecution history, it fails to inform a POSITA about the scope of the invention with reasonable certainty. The definiteness requirement must take into account the knowledge of a POSITA and inherent limitations of language; reasonable certainty in light of the subject matter, and not absolute precision, is required.

IV. SUMMARY OF PATENTS

27. The '675 Patent, titled "Method for Detecting Biometric Information and Electronic Device Using Same," was filed on June 3, 2016, and claims priority to Korean Application No. 10-2015-0113700 filed on August 12, 2015. I understand that Claims 13-20 of the '675 Patent are at issue in this Investigation.

28. The '359 Patent, titled "Wearable Biosignal Interface and Method Thereof," was filed on February 5, 2019, and claims priority to Application No. 14 / 913,085 filed on May 8, 2014 as Application No. PCT / KR2014 / 004073. However, the '359 Patent also claims priority to Korean Application No. 10-2013-0098476 filed on August 20, 2013, the earliest priority date claimed for the '359 Patent. I understand that Claims 1-2, 4, 7-8, 10, 14-15, 20 of the '359 Patent are at issue in this Investigation.

29. The '675 Patent and '359 Patent are each directed to wearable devices with two sensors: a motion sensor (sometimes called the "first" sensor) and a biological or biometric sensor (sometimes called the "second" sensor). The motion sensor measures movement and triggers certain actions, like (1) activating or deactivating the biological/biometric sensor (e.g., '675 Patent at Claim 13), or (2) deciding whether a biological signal measured by the second sensor is valid (e.g., '359 Patent at Claim 14) based on the amount of measured movement.

V. OPINIONS ON CLAIM CONSTRUCTION

A. U.S. Patent No. 10,231,675 (“’675 Patent”)

1. **“acquiring first information on a motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user's body” (’675 Patent, Claim 13)**

Samsung	Oura
Within this phrase, “second period of time shortly before and/or immediately after a first period of time when it is determined that the electronic device is worn on a part of the user's body” means: “second period of time which begins: (i) before the first period of time has ended when it is determined that the electronic device is worn, or (ii) immediately after the first period of time has ended when it is determined that the electronic device is worn”	Indefinite

30. I agree with Samsung’s proposed construction and disagree with Oura’s proposal that the term is indefinite. The phrase “second period of time shortly before and/or immediately after a first period of time” carries a definite meaning. As I detail below, the specification supplies objective boundaries through multiple figures and embodiments that afford a POSITA reasonable certainty about this term’s scope.

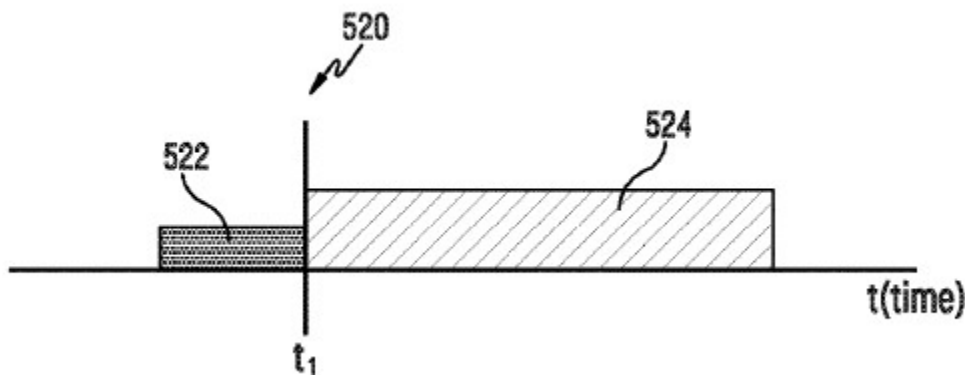
31. Claim 13 recites that the “second period” can be either “[1] shortly before and/or [2] immediately after a first period.” Claim 13 also makes clear that “first period” is directed at wear detection, as it recites: “first period of time when it is determined that the electronic device is worn on a part of the user's body.” During the “second period,” in turn, motion is used to decide whether to activate or inactivate the “second sensor.” For example, the next step in Claim 13, after the disputed term, recites, “activating at least one second sensor based on at least a part of the first information on the motion of the electronic device.” These two periods and their relationship, including what it means for “second period” to be “immediately after” or “shortly before,” are discussed in the specification.

32. In fact, the specification discloses an embodiment that closely tracks Claim 13:

[D]etermine whether the electronic device is worn on (or close to) a part of the user's body using the at least one first sensor; acquire information on the motion of the electronic device using at least a part of the at least one first sensor for a second period of time shortly before and/or immediately after a first period of time selected (or specified) passes from the time when the electronic device is worn on (or close to) a part of the user's body; and activate (operate or trigger) the at least one second sensor based on at least a part of the information on the motion.

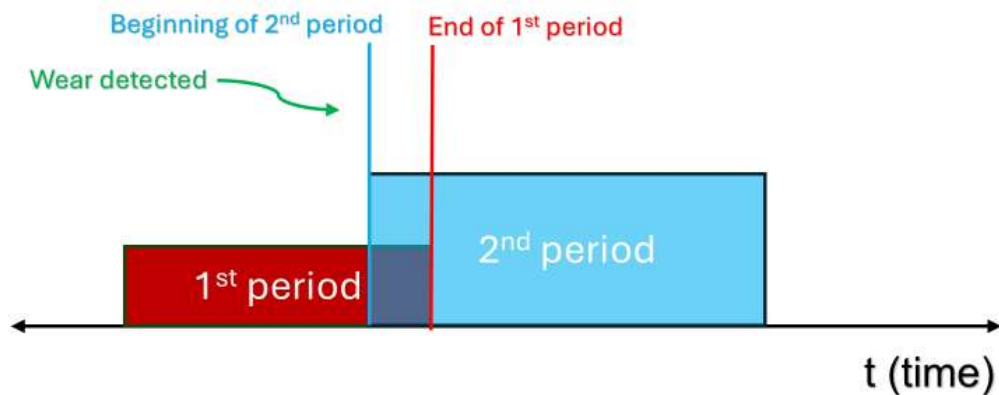
Id. at 20:66-21:8 (emphasis added), Fig. 6-7.

33. For the situation where the “second period” is “immediately after” the “first period” ends, Figure 5B (below) of the '675 Patent illustrates sequential time periods 522 (which could be the “first period of time” or at least part of it) and 524 (which could be the “second period of time”). The boundary between these periods (reference numeral 520) marks the point when “the operation of the first timer ends.” '675 Patent at 20:13-15. So when the first period ends, the second period commences immediately thereafter. The specification confirms this relationship, demonstrating that the periods follow one another consecutively without gap. *See* '675 Patent at Fig. 5B.



34. For the situation where the “second period” begins “shortly before” the “first period” ends, the specification teaches that “the second period of time may be set to include the

time point when the first period of time passes.” ’675 Patent at 21:55–57. In this situation, wear detection—i.e., the determination that the electronic device is worn on a part of the user’s body—serves as the triggering event, and it can occur before the first period ends. When that triggering event takes place before the first period concludes, the second period begins “shortly before” the end of the first period, causing the two periods to overlap as illustrated below. In this situation, because the device already completed wear detection, the “second period” (where motion is used to determine whether to activate another sensor) can begin right away, before waiting for the “first period” to end as in the “immediately after” scenario.



35. Thus, Samsung’s proposed construction—distinguishing between the two cases (before the first period ends, or immediately after it ends) with wear detection as the triggering event—captures what a POSITA would understand this term to mean in view of the specification.

2. “acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor for a third period of time, which is shorter than the first period of time, immediately after the second period of time passes; and activating the at least one second sensor based on at least a part of the additional information” (’675 Patent, Claim 15)

Samsung	Oura
Within this phrase, “third period of time” means: “period of time that immediately follows the second period of time and during which additional information regarding the motion of the electronic device is collected by at least a part of the at least one first sensor and used to determine whether to activate the at least one second sensor”	Indefinite

36. I agree with Samsung’s proposed construction, specifically of the phrase “third period of time” within this larger disputed term and disagree with Oura’s indefiniteness proposal, which may be premised on a misunderstanding of what the “third period of time” is. This term carries a definite meaning.

37. In Claims 13 and 15 of the ’675 Patent, the same first sensor measures motion across all of the recited time periods:

13. A method comprising:

...

determining whether an electronic device is worn on a part of a user's body **using at least one first sensor** of the electronic device;

acquiring **first information on a motion** of the electronic device **using at least a part of the at least one first sensor for a second period of time** shortly before and/or immediately after a **first period of time when it is determined that the electronic device is worn** on a part of the user's body;

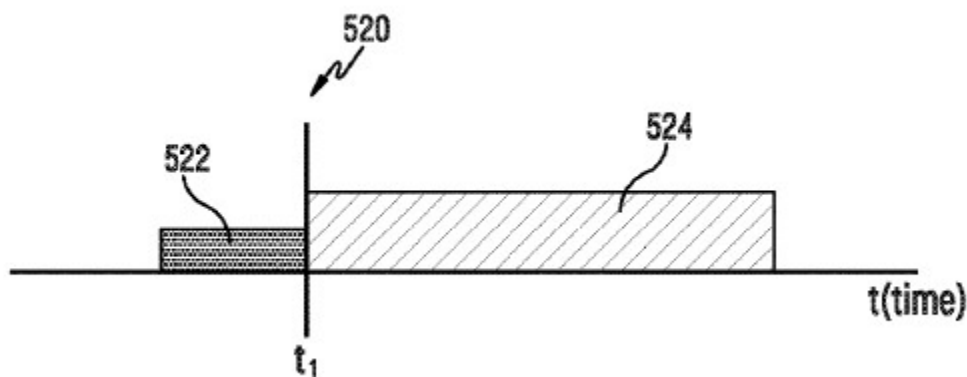
...

15. The method of claim 13, further comprising

acquiring additional information on the motion of the electronic device **using at least a part of the at least one first sensor for a third period of time**, which is shorter than the first period of time, immediately after the second period of time passes; and

...

38. In the context of Figure 5B, the '675 Patent's specification illustrates and confirms that multiple sequential time periods (522 and 524) may both sequentially measure motion. See '675 Patent at 19:46-64, 20:14-29, Fig. 5B.



39. What distinguishes each period is not what the sensor measures, as the “first sensor” measures motion in both periods, but rather the *purpose* that the measured motion serves:

13. A method comprising:

...

activating at least one second sensor based on at least a part of the first information on the motion of the electronic device;

detecting a biometric information of the user using the activated at least one second sensor;

during detecting the biometric information, acquiring second information on a motion of the electronic device using at least the part of the at least one first sensor; and

in response to identifying that the motion of the electronic device included in the second information is more than a threshold motion, inactivating the at least one second sensor.

15. The method of claim 13, further comprising

acquiring additional information on the motion of the electronic device using at least a part of the at least one first sensor ***for a third period of time***, which is shorter than the first period of time, immediately after the second period of time passes; and

activating the at least one second sensor based on at least a part of the additional information.

40. During Claim 13's “first period of time,” the measured motion is used to determine wear detection.

41. During Claim 13’s “second period of time,” the device uses the measured motion to decide whether to activate and then inactivate the second sensor. More specifically, in the second period, the device acquires first information on the electronic device’s motion and, drawing on that information, determines whether to activate the second sensor for measuring biometric information—and later whether to inactivate the second sensor if the motion level exceeds a threshold.

42. During Claim 15’s “third period of time,” the device uses the measured motion to decide whether to *reactivate* the second sensor after the device inactivates it in the last step of Claim 13. The “third period of time” thus commences “immediately after” (as recited in the claim) the “second period of time” and serves the purpose of gathering additional motion data to assess whether conditions warrant reactivating the second sensor.

43. The specification reinforces this reading with embodiments that closely mirror Claim 15’s language:

[A]cquire additional information on the motion of the electronic device using at least a part of the at least one first sensor ***for a third period of time***, which is shorter than the first period of time, ***immediately after the second period of time passes; and activate (operate or trigger) the at least one second sensor based on at least a part of the additional information***, without activating (operating or triggering) the second sensor when the information on the motion does not satisfy a selected (or specified) condition.

See ’675 Patent at 21:42–51, *see id.* at 42:8-17 (similar); *see also id.* at 19:46-64 (disclosing that sequential time periods may both operate a biological sensor). Thus, the specification describes a process in which, after the device inactivates the second sensor, the device continues monitoring motion to evaluate whether it should reactivate the second sensor—precisely the function that the “third period of time” serves as recited in Claim 15. *Id.*

44. Accordingly, a POSITA would understand the “third period of time” as Samsung proposes: “period of time that immediately follows the second period of time and during which additional information regarding the motion of the electronic device is collected by at least a part of the at least one first sensor and used to determine whether to activate the at least one second sensor.”

B. U.S. Patent No. 10,642,359 (“359 Patent”)

1. “biometric signal” (’359 Patent, Claim 14)

Samsung	Oura
Correcting “the biometric signal” to be “the biological signal” and applying the plain and ordinary meaning (i.e., a signal that is associated with biological information of the user).	Correcting “the biometric signal” to be “a biometric signal” and applying the plain and ordinary meaning (i.e., a signal relating to the Identification of an Individual based on physical characteristics, such as fingerprints, finger lengths, facial features, retinal patterns, voice recognition, and weight).

45. I agree with Samsung that in Claim 14, the term “biometric signal” refers to another claim term, “biological signal” recited earlier in the claim. “[B]iological signal” should be construed according to its plain and ordinary meaning. I disagree with Oura’s proposed construction as detailed below.

46. The intrinsic evidence clarifies that “the biometric signal” is “the biological signal.”

47. Claim 14 recites, in relevant part:

obtain a movement signal of an object external to the wearable device using the motion sensor;

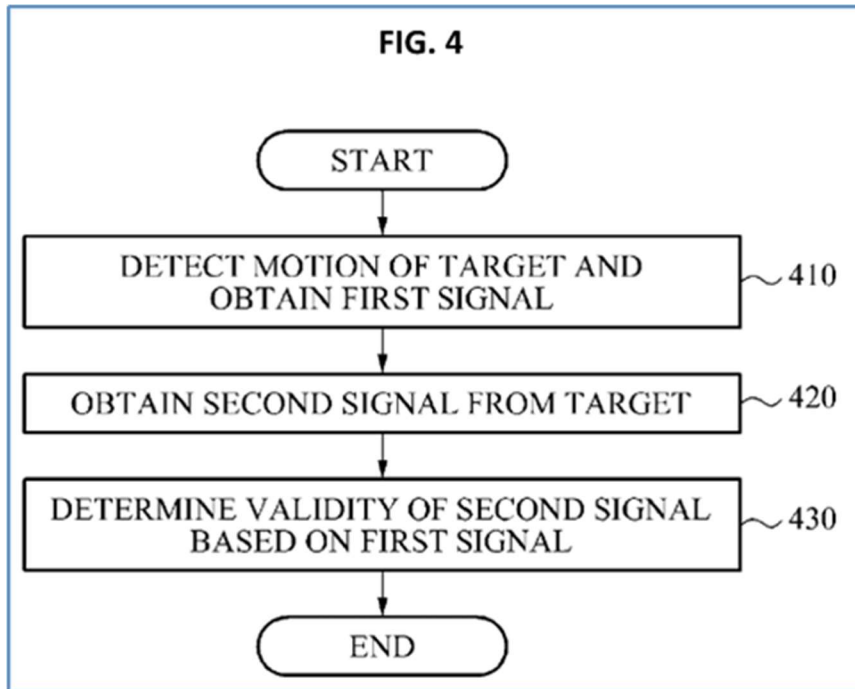
obtain a biological signal of the object using the biological sensor;

determine a select period, of the movement signal, that meets a specified condition; determine, as a valid signal, at least a portion of the biometric signal that is determined to correspond to the determined select period; and

perform a specified function using the valid signal.

48. I understand that where the definite article “the” is used in a patent claim, the following noun is intended to refer back to a substantially similar term previously introduced using an indefinite article like “a.” The term “biometric signal” in Claim 14 of the ’359 Patent draws its antecedent basis from the “biological signal” recited earlier in the same claim. Claim 14 recites only two types of signals before introducing the term “biometric signal:” a “movement signal” and a “biological signal.” The “movement signal” cannot constitute the “biometric signal” because the “movement signal” identifies a “select period” during which a portion of the “biometric signal” qualifies as valid. Put differently, the “movement signal” *filters* the “biometric signal,” deciding which portions of that signal are valid or not. The “movement signal” therefore itself cannot be a “biometric signal.” The “biometric signal” therefore necessarily refers to the “biological signal.”

49. Figure 4 of the ’359 Patent specification (reproduced below) corroborates this correspondence between “biometric signal” and “biological signal” as it recites a process that parallels the steps in Claim 14. At step 410, the device measures a movement signal using a motion sensor; at step 420, the device measures a biological signal (“generated by at least one of” “muscle contraction and muscle relaxation”) using a biosensor; and at step 430, the device assesses the validity of the biological signal based on the measured movement. ’359 Patent at 10:9-22, 10:63-11:2, 11:45-54 (explaining each step of Fig. 4 in detail). Thus, “biometric signal” in Claim 14 maps to the “biological signal” measured at step 420.



50. Claim 7 of the '359 Patent further bolsters this reading. Claim 7 employs the term “biological signal” in the same manner as Claim 14 employs “biometric signal,” confirming that the two terms denote the same signal. As charted below, Claims 7 and 14 track the same steps directed to: (1) measuring movement and biological signals; (2) identifying a period in which the movement signal satisfies some condition; and (3) performing a specified function based on a portion of the biological signal corresponding to the identified period.

Claim 7	Claim 14 (disputed)
“obtain a movement signal of an object external to the wearable device using the motion sensor; obtain a biological signal of the object using the <i>biological sensor</i> ;”	“obtain a movement signal of an object external to the wearable device using the motion sensor; obtain a <i>biological signal</i> of the object using the biological sensor;”
“identify a period, of the movement signal, the movement signal is determined to be less than a specified condition; and”	“determine a select period, of the movement signal, that meets a specified condition;”

“perform a specified function using a select portion of the biological signal corresponding to the period.”	“determine, as a valid signal, at least a portion of the biometric signal that is determined to correspond to the determined select period; and perform a specified function using the valid signal.”
--	---

51. Claim 7 identifies a “select portion” of the “biological signal” corresponding to a “select period” when the “movement signal” meets a “specified condition.” Claim 7 also performs a “specified function” using this “select portion” of the “biological signal.” The only difference between Claim 14 and Claim 7 is that Claim 14 refers to the “select portion” of the biometric/biological signal as the “valid signal.” This “valid signal” is equivalent to the “select portion” because it is a “portion of the biometric signal” corresponding to the “select period” identified based on the “movement signal.”

52. The specification also provides context about the plain and ordinary meaning of “biological signal.” Specifically, the specification discloses “biological signal[s] or physiological signal[s]” obtained by a “biosignal sensor” that relate to biological information, such as “muscle contraction or muscle relaxation of a body part, such as a wrist or a heart.” *See, e.g., id.* at 3:52-64. And consistent with Samsung’s proposed construction, the specification explains that a biological signal is “**associated with biological information** such as, for example, muscle contraction and muscle relaxation.” ’359 Patent at 5:14-17.

53. Oura’s construction—proposing that “biometric signal” means a signal used for “identification of an individual based on physical characteristics, such as fingerprints, finger lengths, facial features, retinal patterns, voice recognition, and weight”—finds no support in the specification and should be rejected. The ’359 Patent never describes using any signal to identify an individual based on physical characteristics. Instead, the specification refers only to “biological

signal or physiological signal[s]” relating to health information such as “muscle contraction or muscle relaxation.” *See* ’359 Patent at 3:52–64.

54. Oura’s construction appears to rely on certain, selected dictionary definitions, which improperly narrow “biometric signal” to a meaning that lacks any support in the specification and contradicts how the term operates throughout the claims and specification of the ’359 Patent. Additionally, contrary extrinsic definitions of “biometric signal” are consistent with the ’359 Patent specification and Samsung’s construction. For example, the ’675 Patent describes “biometric information” as biological information “includ[ing] blood pressure, a heart rate, an electrocardiogram, skin moisture, saturation of peripheral oxygen, and the like.” ’675 Patent at 1:29-35.

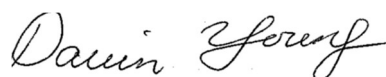
55. Accordingly, a POSITA would understand that a “biological signal” is “a signal associated with biological information” consistent with Samsung’s construction.

VI. OTHER COMMENTS

56. I reserve the right to revise my opinions based on any expert opinions that Respondents may later present, information I may receive in the future, or additional work I may perform. With that reservation, based on the analysis I have conducted and for the reasons set forth above, I stand by the conclusions and opinions in this Declaration.

57. I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true, and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under 18 U.S.C. § 1001.

Dated: April 30, 2026



Dr. Darrin Young

EXHIBIT 7

Curriculum Vitae

Darrin J. Young

Email: darrin_young@yahoo.com

Phone: 510-282-3638

Education:

University of California at Berkeley

Ph.D. Electrical Engineering and Computer Sciences, May 1999.

University of California at Berkeley

M.S. Electrical Engineering and Computer Sciences, May 1993.

University of California at Berkeley

B.S. Electrical Engineering and Computer Sciences, May 1991.

Research Interests:

Integrated microsystem technologies (based on MEMS device design, fabrication, and integrated circuit design) for industrial sensing, wireless sensing, biomedical implant, wearable health monitoring, environmental sensing, navigation, power and energy scavenging, and RF communication applications.

Professional Experiences:

2009-Present

USTART Associate Professor, Electrical and Computer Engineering Department

The University of Utah, Salt Lake City, Utah.

2013-2020

Associate Chair, Electrical and Computer Engineering Department

The University of Utah, Salt Lake City, Utah.

2005-2009

Associate Professor, Electrical Engineering and Computer Science Department

Case Western Reserve University, Cleveland, Ohio.

1999-2005

Assistant Professor, Electrical Engineering and Computer Science Department

Case Western Reserve University, Cleveland, Ohio.

1991-1999

Research Assistant, Electrical Engineering and Computer Sciences Department
University of California, Berkeley, California.

1997-1999

Participating Researcher, Lawrence Livermore National Laboratory, Livermore,
California.

1997 -1998

Analog RF IC Design Intern, Rockwell Semiconductor Systems, Newport Beach,
California.

1991-1993

Intern, Hewlett-Packard Laboratories, Palo Alto, California.

Honors:

- Recognition and Appreciation of Valued Services and Contributions as IEEE EDS Microelectromechanical Systems Chair, December 2011
- Nomination for 2003 Carl F. Wittke Award at Case Western Reserve University for Excellence in Undergraduate Teaching
- Bachelor of Science Degree with Honors in EECS at U.C. Berkeley, 1991
- Scholastic All American Collegiate Award, 1988

Professional Service:

- Administrative Committee of IEEE Sensors Council, 2002-2016
- Technical Program Session Chair, The 12th International Conference on Solid-State Sensors, Actuators, and Microsystems, Boston, MA, 2003
- MEMS Plenary Session Chair, Emerging Information Technology Conference, Princeton University, Princeton, NJ, 2003
- MEMS Session Chair, Emerging Information Technology Conference, Princeton University, Princeton, NJ, 2004
- Advisor Board, Emerging Information Technology Conference, Princeton University, Princeton, NJ, 2004

- Technical Program Committee for IEEE Sensors Conference, Vienna, Austria, 2004
- Technical Program Committee for International Electron Device Meeting, 2004
- Technical Program Session Chair, International Electron Device Meeting, 2004
- Technical Program Committee for the 1st International Workshop on Networked Sensing Systems, University of Tokyo, Tokyo, Japan, 2004
- Organizing Committee for the 7th International Conference on Solid-State and Integrated-Circuit Technology, Beijing, China, 2004
- Technical Program Session Chair, The 13th International Conference on Solid-State Sensors, Actuators, and Microsystems, Seoul, Korea, 2005
- Conference MEMS Session Chair, Emerging Information Technology Conference, National Taiwan University, Taiwan, R.O.C., 2005
- Advisor Board, Emerging Information Technology Conference, National Taiwan University, Taiwan, R.O.C., 2005
- Technical Program Committee for IEEE Sensors Conference, Irvine, California, USA, 2005
- Technical Program Session Chair, IEEE Sensors Conference, Irvine, California, USA, 2005
- Technical Program Committee for International Electron Device Meeting, Washington, D.C. 2005
- Technical Program Session Chair, International Electron Device Meeting, Washington, D.C. 2005
- Technical Program Committee for the 2nd International Workshop on Networked Sensing Systems, San Diego, California, USA, 2005
- Technical Program Session Chair, International Workshop on Networked Sensing Systems, San Diego, California, USA, 2005
- Technical Program Committee for IEEE International Symposium on Circuits and Systems, Island of Kos, Greece, 2006
- Technical Program Committee for IEEE International Conference on Nano/Micro Engineered and Molecular Systems, Zhuhai, China, 2006

- Program Vice-Chair for the 3rd International Conference on Networked Sensing Systems, Chicago, Illinois, USA, 2006
- Technical Program Committee for IEEE Sensors Conference, Korea, 2006
- Conference MEMS Session Chair, Emerging Information Technology Conference, University of Dallas, Dallas, Texas, 2006
- Advisor Board, Emerging Information Technology Conference, University of Dallas, Dallas, Texas, 2006
- Technical MEMS Program Committee of Electronic Device Society, 2006
- Technical Program Committee for MEMS/MOEMS Technologies and Applications Conference as part of the Photonics Asia Symposium, Beijing, China, 2007
- **Associated Editor for the IEEE Journal of Solid-State Circuits, 2006 to 2011**
- IEEE Electron Devices Society MEMS Committee Member, January 2006 to December 2009
- **Chair of the IEEE Electron Devices Society MEMS Committee, January 2010 to 2014**
- Technical Program Committee Member and Conference Publicity Co-Chair for the 4th International Conference on Networked Sensing Systems, Braunschweig, Germany, 2007
- Steering Committee and Technical Program Committee Member the 5th International Conference on Networked Sensing Systems, Kanazawa, Japan, 2008
- Organizing Committee Member and Session Chair for the 9th International Conference on Solid-State and Integrated-Circuit Technology, Beijing, China, 2008
- Steering Committee Member for 2008 Emerging Information Technology Conference
- Technical Program Committee Member and MEMS-Track Co-Chair for the 21st Annual IEEE International SOC Conference, 2008
- Technical Program Committee Member and Industry-Track Co-Chair for the 6th International Conference on Networked Sensing Systems, Pittsburgh, Pennsylvania, 2009
- Technical Program Committee Member for the 22nd Annual IEEE International SOC Conference, 2009

- Medical IC Session Chair, Emerging Information Technology Conference, M.I.T., Cambridge, MA, August 2009
- Technical Program Session Chair, The 5th Annual IEEE International Conference On Nano/Micro Engineered and Molecular Systems, Xiamen, China, 2010
- Steering Committee and Technical Program Committee Member the 7th International Conference on Networked Sensing Systems, Kassel, Germany 2010
- Technical Program Committee Member for the 10th International Conference on Solid-State and Integrated-Circuit Technology, Shanghai, China, 2010
- Technical Program Committee Member of the 22nd Annual IEEE International SOC Conference, 2010
- Technical Program Committee Member and Session Chair of the 24th IEEE International Conference on Micro Electro Mechanical Systems, Cancun, Mexico, 2011
- Technical Program Committee Member and Session Chair of the 25th IEEE International Conference on Micro Electro Mechanical Systems, Paris, France, 2012
- Technical Program Committee Member of the 8th International Conference on Networked Sensing Systems, Penghu, Taiwan, 2011
- Technical Program Session Chair for the 8th International Conference on Networked Sensing Systems, Penghu, Taiwan, 2011
- Technical Program Committee Member of the 9th International Conference on Networked Sensing Systems, Germany, 2012
- Guest Editor for the Special Issue of ACM Journal of Emerging Technologies in Computing Systems (JETC) – Special Issue on Implantable Electronics, Volume 8 Issue 2, June 2012
- Member of Editorial Board of the Journal of Sensor and Actuator Networks, 2012
- Technical Program Committee Member of the 24th Annual IEEE International SOC Conference, 2012
- Technical Session Chair for the 6th Asia-Pacific Conference on Transducers and Micro/Nano Technologies, Nanjing, China, July 2012
- Technical Track Chair and Session Chair of the 34th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, 2012

- Technical Program Committee Member and Session Chair of the International Conference on BioElectronics, BioSensors, BioMedical Devices, BioMEMS/NEMS and Applications, Singapore, November 2012
- IEEE Sensors Council Meritorious Award Committee Member, 2012-2013
- Technical Program Committee Member of the 8th IEEE International Conference on Nano/Micro Engineered and Molecular Systems in 2013
- Chair of IEEE Sensors Council Meritorious Award Committee, 2014
- Technical Program Committee Member of the 9th IEEE International Conference on Nano/Micro Engineering and Molecular Systems, Hawaii, USA, 2014
- Technical Program Committee Member of the 9th IEEE International Conference on Intelligent Sensors, Sensor Networks and information Processing, Singapore, 2014
- Technical Program Committee Member and Session Chair of the International Conference on BioElectronics, BioSensors, BioMedical Devices, BioMEMS/NEMS and Applications, Shanghai China, November 2014
- Technical Program Committee and Track Chair for Circuits and Systems for Biomedical Applications of the IEEE 58th International Midwest Symposium on Circuits and Systems, 2015
- Special Sessions Co-Chair, IEEE Sensors Conference, 2015
- Chair of IEEE Sensors Council Meritorious Award Committee, 2015
- Technical Session Chair of the 29th IEEE International Conference on Micro Electro Mechanical Systems, Shanghai, China, January 2016
- IEEE Sensors Council Technical Achievement Award Committee Member, 2017-2019
- Member of IEEE Sensors Conference Venue Selection Committee, 2018
- IEEE Sensors Council Meritorious Award Committee Member, 2020 and 2021
- Technical Session Chair of the IEEE Sensors Conference, Dallas, Texas, 2022
- Technical Session Chair of the IEEE PowerMEMS Conference, Salt Lake City, Utah, 2022

- Member of Awards Committee of the IEEE PowerMEMS Conference, Salt Lake City, Utah, 2022
- Technical Track Co-Chair of the IEEE Sensors Conference, Kobe, Japan, 2024
- **Editor for the IEEE Transactions on Electron Devices, 2015 to 2024**
- NSF Review Panel for Major Research Instrumentation (MRI) Program, 2004
- NSF Review Panel for Sensor Networks, RF Communications, and VLSI Program, 2004
- NSF Review Panel for SBIR Program, 2003 and 2004
- NSF Review Panel for Sensor Networks, 2005
- NSF CAREER Panel, 2005 and 2008
- NSF Review Panel for Biomedical Engineering / Research to Aid Persons with Disabilities (BME/RAPD) Program, 2006
- NSF Review Panel, 2006
- NSF Review Panel, 2007
- NSF MNS Horizon 2040 Workshop-Health Care Panelist, June 2009
- NSF Biomedical Engineering CAREER Panel, October 2009
- NSF Biomedical Engineering Review Panel, May 2010
- NSF Biomedical Engineering Review Panel, January 2011
- NSF Biomedical Engineering Review Panel, May 2011
- NSF SitS Review Panel, June 2020
- Review for San Francisco State University Faculty Small Grant for Research and Creative Activities, May 2013
- Journal Review for IEEE Journal of MicroElectroMechanical Systems
- Journal Review for IEEE Transactions on Electron Devices
- Journal Review for IEEE Electron Devices Letter

- Journal Review for IEEE Sensors Journal
- Journal Review for Journal of Micromechanics and Microengineering
- Journal Review for IEEE Transaction on Biomedical Engineering
- Journal Review for IEEE Transaction on Biomedical Circuits and Systems
- Journal Review for Sensors & Actuators: A. Physical
- Journal Review for IEEE Transactions on Microwave Theory and Techniques
- Journal Review for Journal of Micromechanics
- Journal Review for Journal of Microelectronics Engineering
- Journal Review for Biosensors
- Journal Review for Journal of Sensors and Sensor Systems
- Journal Review for Microsystems & Nanoengineering

Expert Witness Service During Last Five Years

Momentum Dynamics Corp. (2021-2022)
 Expert witness for Momentum Dynamics Corp.
 WiTricity v. Momentum Dynamics Corp.
 Inter Partes Review

DexCom, Inc. (2022)
 Expert witness for DexCom, Inc.
 DexCom v. Abbott Diabetes Care, Inc.
 Patent Infringement Litigation

Echologics, Inc (2022-2023)
 Expert witness for Echologics, Inc,
 Echologics v. Orbis Intelligent Systems, Inc.
 Patent Infringement Litigation

Masimo Corporation (2023-2024)
 Expert witness for Masimo Corporation
 Apple, Inc. v. Masimo Corporation
 Patent Infringement Litigation

Nintendo Corporation (2024-2025)
Expert witness for Nintendo Corporation
Nintendo Co. v. American GNC Corporation
Inter Partes Review

Bragi GmbH (2024)
Expert witness for Bragi GmbH
K/S HIMPP v. Bragi GmbH
Inter Partes Review

Mueller Systems, LLC (2024)
Expert witness for Mueller Systems, LLC
SIPCO, LLC v. Mueller Systems, LLC
Patent Infringement Litigation Within License Dispute

SMAXTEC INC (2024-2025)
Expert witness for SMAXTEC INC. AND SMAXTEC ANIMAL CARE GMBH
SMAXTEC INC v. ST REPRODUCTIVE TECHNOLOGIES, LLC
Inter Partes Review

Publications:

Journals

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2. J. Du, W. H. Ko, D. J. Young, "Single Crystal Silicon MEMS Fabrication Based On Smart-Cut Technique," *Sensors and Actuators A*, vol. 112, issue 1, pp. 116-121, 2004.
3. M. Suster, W. H. Ko, D. J. Young, "An Optically-Powered Wireless Telemetry Module for High-Temperature MEMS Sensing and Communication," *IEEE Journal of Microelectromechanical Systems*, June, pp. 536-541, 2004.
4. D. J. Young, J. Du, C. A. Zorman, W. H. Ko, "High-Temperature Single Crystal 3C-SiC Capacitive Pressure Sensor," *IEEE Sensors Journal, Special Issue on Microsensors and Microactuators*, August, pp. 464-470, 2004.
5. R. Wang, W. H. Ko, and D. J. Young, "SiC-Carbide-MESFET-Based High-Temperature MEMS Sensing and Data Telemetry Module," *IEEE Sensors Journal*, December, pp. 1389-1394, 2005.

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20. D. J. Young, M. Z. Zurcher, M. Semann, C. A. Megerian, W. H. Ko, "MEMS Capacitive Accelerometer-Based Middle Ear Microphone," *IEEE Transactions On Biomedical Engineering*, Vol. 59, No. 12, pp. 3283-3292, December 2012.
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22. R. Surapaneni, Q. Guo, Y. Xie, D. J. Young and C. H. Mastrangelo, "A three-axis high-resolution capacitive tactile imager system based on floating comb electrodes," *Journal of Micromechanics and Microengineering*, Vol. 23, No. 7, 075004, 2013.
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25. Y.-P. Hsu and D. J. Young, "Skin-Coupled Personal Wearable Ambulatory Pulse Wave Velocity Monitoring System Using Microelectromechanical Sensors," *IEEE Sensors Journal, Special Issue*, Vol. 14, No. 10, pp. 3490-3497, 2014.
26. D. J. Young, P. Cong, M. A. Suster, M. Damaser, "Implantable Wireless Battery Recharging System for Bladder Pressure Chronic Monitoring," *Lab on a Chip*, DOI: 10.1039/C5LC00821B, 2015, 15, 4338-4347.
27. Q. Guo, W. Deng, O. Bebek, C. Cavusoglu, C. Mastrangelo, and D. Young, "Personal Inertial Navigation System Assisted by MEMS Ground Reaction Sensor Array and

Interface ASIC for GPS-Denied Environment,” *IEEE Journal of Solid-State Circuits*, Vol. 53, No. 11, pp. 3039-3049 (Special Issue), November 2018.

28. H. Basaeri, Y. Yu, D. Young, and S. Roundy, “A MEMS-Scale Ultrasonic Power Receiver for Biomedical Implants,” *IEEE Sensors Letters*, 2019.

29. D. Young and P. Cong, “Wireless Implantable Sensors: From Lab to Technology Breakthrough Ambitions,” *Sensors and Actuators A*, 2019.

30. H. Bsaeri, Y. Yu, D. Young, and S. Roundy “Acoustic Power Transfer for Biomedical Implants Using Piezoelectric Receivers: Effects of Misalignment and Misorientation,” *Journal of Micromechanics and Microengineering* (Special Issue), 2019.

31. Y. Yu, T. Nguyen, P. Tathireddy, S. Roundy, and D. J. Young, “Wireless Inductive Sensing and Robust Packaging for Future Implantable Hydrogel-based Glucose Monitoring Applications,” *IEEE Sensors Journal*, 2020.

32. W. Luo, V. Sharma, and D. J. Young, “A Paper-Based Flexible Tactile Sensor Array for Low-Cost Wearable Human Health Monitoring,” *IEEE Journal of Microelectromechanical Systems*, Vol. 29, No. 5, pp. 825-831, 2020.

33. Y. Yu, T. Nguyen, P. Tathireddy, S. Roundy, and D. J. Young, “A Wireless Battery-less and Moisture-Resistant Packaged Glucose Sensing System Employing Hydrogel-based Inductive Sensing Technique and Low-power ASIC for Long-term Glucose Monitoring,” *Sensors and Actuators A*, July, 2022.

34. J. M. Arteaga, J. Sanchez, F. Elsakloul, M. Marin, C. Zesiger, N. Pucci, G. J. Norton, D. J. Young, D. E. Boyle, E. M. Yeatman, P. D. Hallett, S. Roundy, and P. D. Mitcheson, “High Frequency Inductive Power Transfer Through the Soil for Agricultural Applications”, *IEEE Transactions on Power Electronics*, 2023. DOI: 10.1109/TPEL.2023.3305642

35. W. Luo, V. Sharma, and D. J. Young, “A sock-embedded wireless sensing system employing paper-substrate-based MEMS tactile sensors array and low power ASIC for accurate and comfortable AFib monitoring,” *IEEE Sensors Journal*, Volume; 23, Issue: 24, December, 2023. DOI: 10.1109/JSEN.2023.3329100

36. J. Sanchez, J. M. Arteaga, C. Zesiger, P. Mitcheson, D. Young, S. Roundy, “Inductive Power Transfer Coil Misalignment Perception and Correction for Wirelessly Recharging Underground Sensors,” *Sensors* 2025, 25(2), 309; <https://doi.org/10.3390/s25020309>

37. S. Ding, S. Roundy, R. Goel, C. Zesiger, and D. J. Young, “Wireless Battery-Free Active and Passive Underground Soil Moisture Sensing System Design and Field Characterization for Future Agricultural,” Submitted to *IEEE Sensors Journal*, 2025.

Peer Reviewed Conference Papers

1. D. J. Young, B. E. Boser, "A Micromachined Variable Capacitor for Monolithic Low-Noise VCOs," *Technical Digest, IEEE Solid-State Sensor and Actuator Workshop*, Hilton Head Island, SC. June 1996, pp. 86-89.
2. D. J. Young, B. E. Boser, "A Micromachine-Based RF Low-Noise Voltage-Controlled Oscillator," *Technical Digest, IEEE Custom Integrated Circuits Conference*, Santa Clara, CA. May 1997, pp. 431-434.
3. D. J. Young, V. Malba, J. J. Ou, A. F. Bernhardt, B. E. Boser, "Monolithic High-Performance Three-Dimensional Coil Inductors for Wireless Communication Applications," *Technical Digest, International Electron Devices Meeting*, Washington, D.C. December 1997, pp. 67-70.
4. V. Malba, D. J. Young, J. J. Ou, A. F. Bernhardt, B. E. Boser, "High-performance RF coil inductors on silicon," *48th IEEE Electronic Components & Technology Conference*, 1998, pp. 252-255.
5. D. J. Young, V. Malba, J. J. Ou, A. F. Bernhardt, B. E. Boser, "A Low-Noise RF Voltage-Controlled Oscillator Using On-chip Three-Dimensional Coil Inductor and Micromachined Variable Capacitor," *Technical Digest, IEEE Solid-State Sensor and Actuator Workshop*, Hilton Head Island, SC. June 1998. pp. 128-131.
6. D. J. Young, J. L. Tham, B. E. Boser, "A Micromachine-Based Low Phase-Noise GHz Voltage-Controlled Oscillator for Wireless Communication," *Technical Digest, The 10th International Conference on Solid-State Sensors and Actuators (Transducers '99)*, Sendai, Japan, June 1999. pp. 1386-1389.
7. D. J. Young, "Micromachining for RF Communications," *MRS Bulletin on Microelectromechanical Systems: Technology and Applications*, Volume 26, No. 4, pp.331-332, April 2001.
8. M. Suster, D. J. Young and W. H. Ko, "Micro-Watt Wireless System for MEMS Sensors Network," *IEEE 2001 Microelectromechanical Systems Conference*, Berkeley, CA. August 2001, pp. 25-28.
9. M. Suster, D. J. Young, W. H. Ko, "Micro-Power Wireless Transmitter for High-Temperature MEMS Sensing and Communication Applications," *Technical Digest, The 15th IEEE International Conference on Micro Electro Mechanical Systems*, Las Vegas, Nevada, January, 2002, pp. 641-644.
10. M. Suster, D. J. Young, and W. H. Ko, "Micro-Power Wireless Transmitter with Printed Battery for MEMS Sensing and Communication Applications," *Technical Digest, Solid-State Sensor, Actuator, and Microsystems Workshop*, Hilton Head Island, South Carolina, June 2002. pp. 333-336.

11. J. Du, D. J. Young, and W. H. Ko, "Single Crystal Silicon MEMS Fabrication Technology Using Proton-Implantation Smart-Cut Technique," Technical Digest, *IEEE Sensors 2002*, Orlando, Florida, June 2002, pp. 585-588.
12. D. J. Young, "Micromachined RF Voltage-Controlled Oscillator with Phase Noise Characterization (invited)," Technical Digest, *The 9th IEEE International Conference on Electronics, Circuits and Systems*, Dubrovnik, Croatia, September 2002. pp. 295-298.
13. M. Suster, W. H. Ko, D. J. Young, "Optically-Powered Wireless Transmitter For High-Temperature MEMS Sensing and Communication," *Technical Digest, The 12th International Conference on Solid-State Sensors and Actuators (Transducers'2003)*, Boston, June 2003. pp. 1703-1706.
14. J. Du, D. J. Young, C. A. Zorman, W. H. Ko, "Single Crystal SiC Capacitive Pressure Sensor at 400 °C," *Technical Digest, International Electron Devices Meeting*, Washington, D.C. December 2003, pp. 783-786.
15. J. Guo, H. Kuo, D. J. Young, and W. H. Ko, "Buckled Beam Linear Output Capacitive Strain Sensor," *Technical Digest of Solid-State Sensor, Actuator, and Microsystems Workshop*, Hilton Head Island, South Carolina, June 2004. pp. 344-347.
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17. N. Chaimanonart, W. H. Ko, and D. J. Young, "Remote RF Powering System for MEMS Industrial Strain Sensing Applications," *IEEE Sensors Conference*, Vienna, Austria, October, 2004. pp. 1522-1525.
18. P. Cong, W. H. Ko, and D. J. Young, "Single Crystal 6-H Silicon Carbide Smart-Cut MEMS Fabrication Technology," *IEEE Sensors Conference*, Vienna, Austria, October, 2004. pp. 1161-1164.
19. P. Cong, D. J. Young, and W. H. Ko, "Novel Long-Term Implantable Blood Pressure Monitoring System," *IEEE Sensors Conference*, Vienna, Austria, October, 2004. pp. 1359-1362.
20. M. Suster, J. Guo, N. Chaimanonart, W. H. Ko, and D. J. Young, "High Performance Strain Sensing Microsystem," *IEEE Custom Integrated Circuits Conference*, Orlando, Florida, October 2004, pp. 693-696.
21. D. J. Young and W. H. Ko, "A High-Performance Strain Sensing Microsystem with Remote RF Power Capability (invited)," *the 7th International Conference on Solid-State and Integrated-Circuit Technology*, Beijing, China, 2004.
22. M. Suster, N. Chaimanonart, J. Guo, W. H. Ko, and D. J. Young, "Remote-Powered High-Performance Strain Sensing Microsystem," *the 18th IEEE International Conference*

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26. N. Chaimanonart, K. Olszens, M. Zimmerman, W. Ko, D. J. Young, "Implantable RF Power Converter for Small Animal In Vivo Biological Monitoring," *the 27th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC-2005)*, Shanghai, China, September 2005, pp. 1559-1562.

27. J. Guo, M. Suster, D. J. Young, W. H. Ko, "High-gain Mechanically Amplified Capacitive Strain Sensor," technical proceeding of the *IEEE Sensors Conference*, Irvine, CA, October 2005, pp. 464-467.

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Invited Book Chapters

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2. D. J. Young and H. Kim, “Microelectromechanical Systems (MEMS),” *Chapter 18 of Guide to State-of-the-Art Electronic Devices*, Jonh Wiley & Sons Ltd, 2013.
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Patents

US 8,634,924 “MEMS Implanted Acoustic Sensor”

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Presentations

“Wireless Sensing Technology with Example Applications” presented at ECE Department Graduate Seminar, University of Utah, February 4, 2026.

“Microsystem Research Overview and Graduate Programs in ECE Department at University of Utah” presented at School of Electronics and Information Engineering, Soochow University, March 27, 2018.

“About Graduate School” presented at the ECE Department Junior Seminar, University of Utah, Salt Lake City, Utah, March 3, 2017

“Smart and Robust Sensing Systems Enabled by MEMS Technology and Low Power Electronics”, presented at the graduate seminar, Electrical and Computer Engineering Department, University of Utah, Salt Lake City, Utah, August 29, 2016.

“MEMS-Based Personal Navigation System Under GPS-Denied Environment”, presented at the Mechanical Engineering Department Seminar, National Taiwan University, Taiwan, R.O.C., June, 6, 2016.

“Smart and Robust Sensing Systems Enabled by MEMS Technology and Low Power Electronics”, presented at Shanghai Institute of Microsystem and Information Technology, Chinese Academy of Sciences, May 20, 2016.

“Bionic Human”, panel discussion of Leo After Hours at the Leonardo Museum, Salt Lake City, Utah, March 10, 2016.

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“Skin-Coupled Wearable Ambulatory Health Monitoring System Using MEMS Sensors,” presented as an invited talk at the International conference on BioSensors, BioElectronics, BioMedical Devcies, BioMEMS/NEMS and Applications 2014 & The 5th Symposium on Green Microsystem Technology, Shanghai, China, November 17, 2014,

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“High Performance Microsystems Enabled by MEMS and Integrated Circuits,” presented at the Electrical and Computer Engineering Department Graduate Seminar, Univeristy of Utah, September 16, 2013.

“Biomedical Implant System Design Enabled by MEMS and CMOS IC,” presented at the International Conference on BioElectronics, BioSensors, BioMedical Devices, BioMEMS/NEMS and Applications, Singapore, November 20, 2012.

“Wireless Microsystem Technologies and Future Direction”, presented at National Taiwan University, October 31, 2012.

“MEMS-Based Biomedical Implant Research and Commercialization Opportunities”, presented at the 6th Asia-Pacific Conference on Transducers and Micro/Nano Technologies, Nanjing, China, July 10, 2012.

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“Wireless MEMS-Based Strain Sensing Microsystem” presented at the Department of Information and Electronic Engineering, Zhejiang University, Hangzhou, China, 9-22-2009.

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“Wireless Microsystem Research”, presented at EECS Department Seminar at Case Western Reserve University, Cleveland, Ohio, 1-20-2009.

“Wireless Microsystem Technology for Biomedical Applications”, presented at the ECE Department Seminar at University of Utah, Salt Lake City, Utah, 12-1-2008.

“Wireless Implantable EMG Sensor for Powered Prosthesis Control”, presented in *the 9th International Conference on Solid-State and Integrated-Circuit Technology*, Beijing, China, 10-22-2008.

“CMOS Cross-Coupled VCO Architecture Comparison at 2GHz and 16GHz”, presented in *the 9th International Conference on Solid-State and Integrated-Circuit Technology*, Beijing, China, 10-22-2008.

“Wireless Microsystem Technology for Biomedical Applications”, presented at State Key Laboratory of Precision Measuring Technology & Instruments, Tianjin University, China, 10-20-2008

“Wireless Microsystem Technology for Biomedical Interface”, presented at the Institute of Electronics, Chinese Academy of Sciences, Beijing, China, 10-16-2008.

“Wireless Microsystem Technology for Biomedical Applications”, presented at the ECE Department graduate seminar, University of Utah, September 23, 2008.

“MEMS Middle Ear Microphone for Fully Implantable Cochlear Prosthesis”, presented at National Chiao Tung University, Taiwan, R.O.C., 6-26-2008.

“Wireless Implantable Less-Invasive Blood Pressure Sensing Microsystem for Small Laboratory Animal Real-Time Monitoring”, presented at the Asia-Pacific Conference on Transducers and Micro-Nano Technology, Taiwan, R.O.C., 6-25-2008.

“Wireless Less-Invasive Blood Pressure Sensing Microsystem for Small Laboratory Animal *In Vivo* Real-Time Monitoring”, presented at the 5th International Conference on Networked Sensing System (INSS 2008), Kanazawa, Japan, 6-18-2008.

“MEMS Middle Ear Microphone for Fully Implantable Cochlear Prosthesis”, presented at Tsinghua University, Beijing, China, 2-26-2008.

“Wireless Less-Invasive Blood Pressure Sensing Microsystem for Small Laboratory Animal *In Vivo* Real-Time Monitoring”, presented at the Institute of Microelectronics, Beijing University, Beijing, China, 2-25-2008.

“Wireless Less-Invasive Blood Pressure Sensing Microsystem for Small Laboratory Animal *In Vivo* Real-Time Monitoring”, presented at Southeast University, Nanjing, China, 2-18-2008.

“Wireless Less-Invasive Long-Term Implantable Blood Pressure Sensing System for Small Animal Real-Time Monitoring”, presented at the 2007 International Electron Devices and Materials Symposia (IEDMS), Taiwan, R.O.C., 11-29-2007.

“Wireless Implantable EMG Sensing Microsystem”, presented at the 2007 Biomedical Engineering Society Annual Fall Meeting, Los Angeles, California, 9-29-2007.

“MEMS Accelerometer with Low Noise Interface Circuits as Middle Ear Microphone for Fully Implantable Cochlear Prosthesis”, presented at the College of Mechanical Engineering, Dalian Technical University, Dalian, China, 10-27-2006.

“Microsystem Technology for Fully Implantable Cochlear Prosthesis”, presented at the 6th Emerging Information Technology Conference, University of Texas at Dallas, Dallas, Texas, 8-11-2006.

“Wireless Strain Sensors”, presented at The Timken Company, Canton, Ohio, 2-28-2006.

“Wireless Strain Sensing Microsystem Technology”, presented at Southeast University, Nanjing, China, 9-6-2005.

“Microsystem Technology for Totally Implantable Cochlear Prosthesis”, presented at National Cheng Kung University, Tainan, Taiwan, R.O.C., 8-18-2005.

“Wireless Strain Sensing Microsystem”, presented as a plenary session talk at the 5th Emerging Information Technology Conference, National Taiwan University, Taipei, Taiwan, R.O.C., 8-18-2005.

“Microsystem Technology for Totally Implantable Cochlear Prosthesis”, presented at Neural Prosthesis Seminar at Case Western Reserve University, Cleveland, Ohio, 7-15-2005

“Strain Sensing Microsystem with Remote RF Powering and Data Telemetry”, presented at ECE Department Seminar, Carnegie Mellon University, Pittsburgh, Pennsylvania, 3-18-2005.

“High-Temperature Wireless Sensing Technology and Its Future Directions”, presented at Beijing University, Beijing, China, 10-21-2004.

“A High-Performance Strain Sensing Microsystem with Remote RF Power Capability,” presented at the 7th International Conference on Solid-State and Integrated-Circuit Technology (ICSICT-2004), Beijing, China, 10-19-2004.

“High-Performance MEMS Capacitive Strain Sensing Microsystem”, presented at Army Research Office MEMS Meeting, Arlington VA. 6-24-2004.

“Single Crystal SiC Capacitive Pressure Sensor at 400 °C”, presented at International Electron Devices Meeting, Washington D.C., 12-10-2003.

“3D-MEMS Approaches to Varactors”, presented at IEEE International Microwave Symposium, Philadelphia, Pennsylvania, 6-9-2003.

“MEMS for RF Applications”, presented at MEMGen Corporation, Burbank, CA, 11-20-2002.

“RF MEMS and Its Applications”, presented at Institute of Microelectronics, TsingHua University, Beijing, China, 7-30-2002.

“Micromachined RF Low Phase Noise Voltage-Controlled Oscillator”, presented at Shanghai Institute of Microsystem and Information Technology, Shanghai, China, 7-15-2002.

“MEMS Technologies for Radio Frequency (RF) and Optical Applications” Short Course Presented at Adriatic Research Institute, Berkeley, California, 4-26-2002.

“Micromachined Low Phase Noise Voltage-Controlled Oscillator”, EE Department Seminar, University of Southern California, Los Angeles, California, 3-19-2002.

“MEMS for RF Communications”, Tutorial Session Presented at Material Research Society (MRS) Fall Meeting, Boston, Massachusetts, 11-26-2001.

“Micromachined Low Phase Noise RF Voltage-Controlled Oscillator”, EECS Department Seminar, University of Michigan, Ann Arbor, Michigan, 11-5-2001.

“MEMS for Wireless Communication”, presented at National Taiwan University, Taipei, Taiwan. R. O. C. on 10-29-2001.

“Key Technology for MEMS”, Short Course Presented at Precision Instrument Development Center, National Science Council, The Executive Yuan, Taiwan, R. O. C., 10-22-2001 to 10-25-2001.

“MEMS & Communications - A Complimentary System” presented at Glennan Microsystem Initiative Meeting, Independence, Ohio on 6- 9-2000.

“RF Voltage-Controlled Oscillator (VCO) with MEMS Capacitors and Inductors” presented at Naval Research Lab workshop on Military Application of RF MEMS, Washington, D.C. on 2-8-2000.

“MEMS for Wireless Communications” presented at Symposium of Micromachine and Microtechnology, Bio-Medical, and Industrial Prospects at Case Western Reserve University, Cleveland, Ohio on 11-12-1999.

“A Low-Noise RF VCO Using On-Chip High-Q 3-D Coil Inductor & Micromachined Variable Capacitor” presented at Solid-State Circuit Technology Committee Workshop, Arlington, Virginia on 11-15-1999.

Research Grants and Contracts:

- “Robotic Platform For Precision Irrigation Management Using Passive Zero Maintenance, Long-Life, Buried Sensors” (Co-PI)
USDA, September 1, 2024 to August 30, 2027
- “SitS NSF-UKRI: Wireless In-Situ Soil Sensing Network for Future Sustainable Agriculture” (PI)
NSF, January 1, 2020 to December 31, 2024
- “Acoustic Power and Communication Platform for a Deeply Implanted Intraluminal Glucose Sensor” (Co-PI)
NSF, September 2014 to August 2018
- “Implantable Particle-Based Wireless Neurorecording Probes for Minimally-Intrusive 3D Mapping of Brain Signals” (Co-PI)
NSF, September 2015 to August 2017
- “Soldier Navigation via High-Resolution-Gait-Corrected IMUs” (Co-PI)
DARPA, July 2008 to December 2013
- “Robust Design of High Performance MEMS Resonators” (Co-PI)
NSF, September 2008 to August 2011
- “Wireless Implantable Pressure Monitor for Clinical Applications” (Co-PI)
U.S. Department of Veteran Affairs, September 2007 to August 2010
- “Intelligent Sensor Network for In-Vivo Real Time Biological Dynamic Systems” (PI)
NSF, September 2003 to August 2007
- “RF Data Transmitter and Receiver Modules for MEMS Strain Sensors” (PI)
Army Research Office, June 2002 to May 2007
- “Wireless EMG Sensing Microsystem” (PI)

Veteran Affairs Medical Center, Cleveland, Ohio, September 2005 to August 2006

- “Harsh Environment MEMS Sensing and Communications” (PI)
NASA, October 1999 to October 2002
- “MEMS Tunable Antennas for Power-Efficient Wireless Communications” (PI)
NASA, October 2001 to March 2003
- “Nerve Reshaping” (Co-PI)
NIH, July 2005 to June 2010
- “Micro Sensor for Cochlear and Middle Ear Implant Systems” (Co-PI)
NIH, December 2003 to November 2005
- “Microfabricated Strain Sensor Module with Interface Circuits” (Co-PI)
Army Research Office, June 2002 to May 2007
- “Fast Bonding Techniques for Attaching Strain Sensors to Steel Substrates” (Co-PI)
Army Research Office, June 2002 to May 2007
- “Multi-Range Pressure Sensor Arrays For Harsh Environment” (Co-PI)
NASA, October 1999 to October 2002

Graduate Research Students Advising:

- Weijie Luo
Ph.D. Dissertation Title: “A Wireless Wearable Atrial Fibrillation Monitoring System Employing A Paper-Based MEMS Tactile Sensors Array And A Low-Power ASIC”
Completion Date: December 2021
- Yuechuan Yu
Ph.D. Dissertation Title: “A Wireless and Battery-less Hermetically Packaged Glucose Sensor Employing Hydrogel-Based Inductive Sensing Scheme and Low-Power ASIC for Future Long-Term Implantable Glucose Monitoring”
Completion Date: March 2020
- Qingbo Guo
Ph.D. Dissertation Title: “Personal Inertial Navigation System Assisted By MEMS Wearable Ground Reaction Sensor Array and Sensing Interface ASIC”
Completion Date: December 2018
- Xing Chen (Post-Doc, 2015-2016)
Packaging Technology for Biomedical Implant Applications

- Michael Suster
Ph.D. Dissertation Title: “Wireless Strain Sensing Microsystem with External RF Powering and Two-Channel Data Telemetry Capability”
Completion Date: July 2011
- Nattapon Chaimanonart
Ph.D. Dissertation Title: “Wireless Implantable Multi-channel Bio-sensing Network with Adaptive RF Power Control for Untethered Laboratory Animal Real-time Monitoring”
Completion Date: July 2009
- Peng Cong
Ph.D. Dissertation Title: “Wireless Less-invasive Implantable Blood Pressure Sensing System for Small Animal Real-time Monitoring”
Completion Date: December 2008
- Kalyan Ladhane
M.S. Project Title: “SiC MEMS”
Completion Date: December 2013
- Bradley Farnsworth
M.S. Thesis Title: “Wireless Implantable EMG Sensor”
Completion Date: May 2010
- Mark Zimmerman
M.S. Thesis Title: “*In Vivo* RF Powering for Advanced Biological Research”
Completion Date: June 2008
- Engin Pehlivanoglu
M.S. Thesis Title: “Silicon Carbide MEMS Oscillator”
Completion Date: January 2008
- Mark Zurcher
M.S. Thesis Title: “Development of a MEMS Middle Ear Acoustic Sensor for a Fully Implantable Cochlear Prosthesis”
Completion Date: November 2007
- Matthew Cross
M.S. Thesis Title: “Architectural Comparison Study for RF and Microwave CMOS Voltage-Controlled Oscillators”
Completion Date: August 2007
- Cheng-Kuan Lu
M.S. Thesis Title: “Wireless MEMS Accelerometer for Real-time Small Laboratory Animal Activity Monitoring”
Completion Date: July 2007

- Steven Mallin
M.S. Thesis Title: “Phase Noise Processes and Suppression Techniques in RF LC Voltage-Controlled Oscillators”
Completion Date: May 2006
- Nattapon Chaimanonart
M.S. Thesis Title: “Remote Radio-Frequency Powering System for Wireless MEMS Strain Sensor”
Completion Date: January 2006
- Jiangang Du
M.S. Thesis Title: “Single Crystal Silicon Carbide Capacitive Pressure Transducer”
Completion Date: May 2005
- Run Wang
M.S. Thesis Title: “Silicon-Carbide-MESFET-Based 400 °C MEMS Sensing and Data Telemetry”
Completion Date: August 2004
- Michael Suster
M.S. Thesis Title: “High-Temperature MEMS Wireless Sensing and Telemetry”
Completion Date: August 2004
- Brian Quach
M.S. Thesis Title: “Micromachined Touch Mode Tunable Capacitor Design and Fabrication for RF Communication Applications”
Completion Date: August 2003

Teaching and Courses Development:

- ECE 4900 Senior Clinic Project (Undergraduate Course)
- ECE 6720 Analog Integrated Circuit Design (Graduate Course)
- ECE 6722 Analog Integrated Circuit Testing (Graduate Course)
- ECE 6960 Electrical Interface for MEMS (Graduate Course)
- ECE 6730 Radio Frequency Integrated Circuits Design (Graduate Course)
- ECE 2280 Fundamentals of Engineering Electronics (Undergraduate Course)
- ECE 3110 Electronic Circuits-II (Undergraduate Course)
- EECS 426 MEMS for Sensing and Communication (Graduate Course)

- EECS 600 Radio-Frequency Integrated Circuits Design (Graduate Course)
- EBME 418 Applied Electronics for Biomedical Engineering (Selected Lectures, Graduate Course in BME)
- EECS 344 Microelectronic Design and Analysis (Undergraduate Course)
- EECS 245 Electronic Circuits (Undergraduate Course)
- ENRG 210 Introduction to Circuits and Instruments (Undergraduate Course)

CERTIFICATE OF SERVICE

Investigation No. 337-TA-1478

The undersigned certifies that on May 1, 2026, true and correct copies of the foregoing document were served upon the following parties as indicated below:

The Honorable Lisa R. Barton Secretary to the Commission U.S. INTERNATIONAL TRADE COMMISSION 500 E Street, SW, Room 112 Washington, DC 20436	☐ Via Hand Delivery ☐ Via Overnight Delivery ☐ Via First Class Mail ☒ Via Electronic Filing (EDIS) ☐ Via Electronic Mail
The Honorable Doris Johnson Hines Administrative Law Judge U.S. INTERNATIONAL TRADE COMMISSION 500 E Street, S.W. Washington, D.C. 20436 Email: JohnsonHines1468@usitc.gov	☐ Via Hand Delivery ☐ Via Overnight Delivery ☐ Via First Class Mail ☒ Via Electronic Filing (EDIS) ☒ Via Electronic Mail
<i>Counsel for Respondents Ouraring Inc. and Oura Health Oy</i> Jasjit S. Vidwan James A. Fussell, III Saqib J. Siddiqui Bryan C. Nese MAYER BROWN LLP 199 K. Street NW Washington, DC 20006 Tel: 202-263-5282 Robert G. Pluta Nicholas H. Ciulla MAYER BROWN LLP 71 S. Wacker Drive Chicago, IL 60606 Tel: 312-782-0600 Email: Oura-ITC@mayerbrown.com	☐ Via Hand Delivery ☐ Via Overnight Delivery ☐ Via First Class Mail ☐ Via Electronic Filing (EDIS) ☒ Via Electronic Mail

/s/ Brian A. Treggs

Brian A. Treggs
Senior IPT Specialist
O'Melveny & Myers LLP
400 South Hope Street
Los Angeles, CA 90071